# Bahir Dar University

## Bahir Dar Institute of Technology

## Faculty of Computing

## Requirement Analysis Document (RAD)

for

## Industrial project on Digital Land Citizen Portal with Ownership Verification and Land Marketplace

Submitted to the faculty of computing in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science

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**Bahir Dar University, Bahir Dar Institute of Technology**

# Declaration

We hereby declare that this project is our original work and has not been submitted for a degree or diploma in any other university or institution. All sources of information and materials used in this project have been properly cited and acknowledged.

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**Faculty:** Computing

**Program**: Computer Science

**Project Title**: Digital Land Citizen Portal with Ownership Verification and Land Marketplace

This is to certify that I have read this project and that in my supervision and the students' performance, it is fully adequate, in scope and quality, as a project for the degree of Bachelor of Science.

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It is approved that this project has been written in compliance with the formatting rules laid down by the faculty.

# Roles and Responsibilities of the Group Members

| List of Tasks | List members | | |
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| Requirement Gathering | √ | √ | √ |
| Requirement Analysis | √ | √ | √ |
| Use Case modeling | √ | √ | √ |
| Sequence and Activity modeling | √ | √ | √ |
| UI/UX Design | √ | √ | √ |
| Persistent and Class modeling | √ | √ | √ |
| Component and Deploymentmodeling | √ | √ | √ |

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We would like to express our sincere gratitude to all those who contributed to the development and completion of this Digital Land Citizen Portal with Ownership Verification and Land Marketplace System project.

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We are particularly grateful to the **land administration office managers and staff** in both urban and rural areas who patiently explained their daily workflows, challenges, and citizen interactions. Their candid discussions about technological limitations, user literacy concerns, and operational realities helped us design a system that is both innovative and practical for Ethiopia's context.

# List of acronyms

| List of acronyms | Description |
|---|---|
| API | Application Programming Interface |
| BR | Business Rule |
| CPU | Central Processing Unit |
| CRC | Class-Responsibility-Collaboration |
| CSS | Cascading Style Sheet |
| DBMS | Database Management System |
| EQUILEX | Equity-Lex(Fairness or equality in law) |
| E-R | Entity-Relationship |
| FAQ | Frequently Asked Questions |
| FK | Foreign Key |
| FREQ | Functional Requirement |
| GB | Giga Byte |
| GHz | Giga Hertz |
| GUI | Graphical User Interface |
| HTML | Hypertext Markup Language |
| HTTPS | Hypertext Transfer Protocol |
| HTTPS | Hypertext Transfer Protocol Secured |
| ORM | Object Relational Model |
| PDA | Personal Digital Assistant |
| PK | Primary Key |
| RAM | Random Access Memory |
| SQL | Structured Query Language |
| SSL | Secure Sockets Layer |
| TLS | Transport Layer Security |
| UI/UX | User Interface/User Experience |
| UML | Unified Modeling Language |

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## Abstract

This document presents the design and development of a Digital Land Registry and Ownership Verification System for Ethiopia, aimed at transforming the country's land administration from manual, paper-based processes to an efficient, transparent, and accessible digital platform. The current system requires citizens to physically visit land administration offices, resulting in long processing times, inefficiencies, lack of transparency, and high susceptibility to fraud and document loss. This project addresses these challenges by developing a secure, integrated online portal that enables citizens to manage land records, verify ownership, initiate transactions, and complete payments entirely through digital means. The proposed system consists of a Citizen Portal accessible via web and mobile applications, featuring user registration and authentication, real-time land ownership verification, digital service request submission, and integrated payment processing with Ethiopian financial systems

The system maintains a centralized digital land registry, provides a secure marketplace for property transactions, and ensures end-to-end workflow automation with real-time status tracking and notifications.

# Chapter One: Introduction

## 1.1 Background

Land is the most fundamental asset for economic development, social stability, and wealth generation in any society. Efficient, transparent, and accessible land administration is therefore a cornerstone of good governance and economic progress. Traditionally, land administration systems in many regions have been characterized by paper-based processes, fragmented workflows across multiple government offices, and manual record-keeping. This legacy system creates significant barriers for citizens seeking services like title verification, transaction processing, or dispute resolution, leading to delays, potential errors, and a lack of transparency.

The global digital transformation offers a powerful solution. The integration of Information and Communication Technology (ICT) into public service delivery known as e-Governance has proven to enhance efficiency, reduce corruption, and empower citizens. A centralized Digital Land Citizen Portal represents the next evolutionary step, moving from disjointed, office-bound procedures to a unified, online platform. Such a portal can integrate all land-related services, provide real-time access to records, and even facilitate secure market transactions, thereby bridging the critical gap between the government and the citizenry.

**For Citizens:**

The platform empowers citizens to:

- Receive tailored recommendations for land parcels and property investments.
- Browse detailed land listings and ownership histories freely.
- Make informed choices based on verified titles, land dimensions, and historical data. This approach saves time, reduces the risk of fraud, and minimizes the financial burden of traditional land acquisition.

**For Landowners:**

The platform provides property owners with a centralized space to:

- Showcase their verified land titles, survey maps, and property details.
- Expand their reach to a broader audience of potential buyers and investors.

**Business Process to Be Automated**

The core business process automated by the platform involves:

1. **Citizen Registration and Profile Creation:** Enabling users to create accounts and specify their property interests or land requirements.
2. **Ownership Verification and Land Browsing:** Providing automated verification of land titles through secure records and allowing free browsing of detailed property profiles.
3. **Communication and Negotiation Tools:** Facilitating direct communication between buyers and sellers to discuss terms, negotiate prices, or arrange site visits.
4. **Marketplace Transactions:** Supporting secure online payments for land purchases, documentation fees, and ownership transfer services.

## 1.2 Statement of the problem

**Land Administration and Marketplace**

In Ethiopia's rapidly growing cities, individuals seeking to acquire or verify land often encounter numerous barriers when navigating the current administration system. One of the primary issues is the uncertainty surrounding the authenticity of land titles and whether a specific plot is free from existing legal encumbrances or overlapping claims. Additionally, prospective buyers often struggle with selecting the right property for their needs due to a lack of accessible, reliable information on land history, zoning regulations, and verified ownership status.

Even when a land transaction is initiated, challenges persist. Citizens frequently face bureaucratic delays with land bureaus, as manual record-keeping and institutional inefficiencies can lead to prolonged processing times and dissatisfaction. A major concern is that the current manual system is vulnerable to data manipulation, where some actors may prioritize transactions based on informal incentives rather than legal merit, potentially compromising the integrity and security of land tenure.

Another significant issue is the absence of consistent practices for land price negotiation and transparent transaction standards. In many instances, the lack of a formal digital record allows for "double-selling" or fraudulent listings, leaving citizens without legal protection or the financial recovery they need. Logistical problems also arise regarding the verification of property boundaries. Due to the high cost of formal surveying and the difficulty of accessing official land maps in Addis Ababa and other major cities, many transactions occur in informal spaces without proper site validation, which can compromise the long-term legality of the investment.

Additionally, buyers often select land based on unregulated middlemen or word-of-mouth recommendations, which can prevent legitimate sellers and developers from gaining fair visibility in the marketplace. This creates an imbalance, especially for first-time homeowners or small-scale investors, who find it difficult to navigate the market safely unless they have high-level connections within administrative offices.

Due to diverse challenges, there is a clear need for an online platform that can address these issues. Such a platform would provide a streamlined, transparent way for citizens to verify land ownership, ensure that legal standards are maintained in property transfers, and offer an efficient, professional environment for secure land marketplace interactions.

## 1.3 Objectives

### *1.3.1 General Objective*

To design ,develop and implement Digital Land Citizen Portal with Ownership Verification and Land Marketplace

### *1.3.2 Specific Objective*

The following specific objectives outline the for the development and implementation of the platform:

- To analyze the limitations, stakeholder pain points, and procedural workflows of the existing paper-based land administration system.
- To design a robust system architecture for a portal that integrates core functionalities: user registration, digital application submission, online ownership verification, case tracking, and document management..
- To incorporate a secure land marketplace module within the portal to enable legitimate listings, inquiries, and transparent transaction processes between verified users.
- To develop a functional prototype demonstrating key user interfaces for citizens and administrative staff, along with core backend workflows.
- To evaluate the proposed system's potential impact on reducing service delivery time, minimizing errors, improving data security, and increasing overall citizen satisfaction with land services.

## 1.4 Methodology

This Project uses a mixed-methods approach to examine urban land administration, ownership systems, and land marketplaces in Bahir Dar City, Ethiopia. The city is purposively selected due to its rapid urban growth and increasing land-related challenges**.**

#### *1.4.1 Requirement gathering Methods*

During the requirement gathering process, we visited both rural and urban land administration offices. In the rural land administration office, we met the manager, who explained how land buying and selling are carried out. He mentioned that using an online system is very difficult because the process is complex and many people do not have enough knowledge or experience with technology. After that, we visited the urban land administration office. At first, we were not able to meet the required person and had to return for one week. Later, when we met the official, he explained that although there is a land administration system, it is not effective. The system is not online, land registration is done manually, and buying and selling of land are not handled digitally. Because of this, people find it difficult to access land information and to find land or land marketplaces that meet their need.

**Interviews:**

- Conducted with land administration officials in both rural and urban offices.
- Key purpose: to understand land registration, buying and selling processes, challenges, and existing systems

**Questionnaires/Surveys:**

- Distributed to urban and rural land users (residents, leaseholders).
- Key purpose: to collect data on experiences with land access, tenure security, and awareness of land systems.

**Observation:**

- Visiting rural and urban land offices to see the workflow, documentation process, and system usage.
- Key purpose: to understand the real-world process and challenges in land administration.

**Focus Group Discussions (FGDs) :**

- Conducted with community members in rural and urban areas.
- Key purpose: to capture local experiences, informal practices, and opinions on land marketplaces.

**Document Review / Secondary Data Collection:**

- Reviewing land laws, urban lease proclamations, policy documents, municipal reports, and previous studies.
- Key purpose: to gather official rules, regulations, and historical data for analysis.

**Stakeholder Consultation**

- Meetings with urban planners, municipal authorities, and other experts.
- Key purpose: to understand institutional requirements and policy perspectives.

### *1.4.2 Analysis and design Methodology*

The **Object-Oriented (OO) analysis and design methodology** was selected for developing the Digital Land Registry and Ownership Verification System based on the following critical factors:

- **Modular System Architecture:** The methodology supports creating independent yet interconnected modules (Citizen Portal, Ownership Verification, Payment Integration, Admin Interface) that align perfectly with Ethiopia's phased digital transformation approach.

- **Real-World Entity Representation:** Natural modeling of Ethiopian land administration entities such as Ownership Certificate, Payment Transaction, and Service Request as discrete objects with clear Amharic-contextual attributes.

- **Enhanced Security Through Encapsulation:** Critical for protecting sensitive Ethiopian land records, citizen identification data (Kebele ID), and financial transactions with Ethiopian payment systems (TeleBirr, CBE Birr).

- **Stakeholder-Centric Communication:** UML diagrams bridge understanding between technical teams, land administration officials at ክልል (regional) and ወረዳ (district) levels, and citizens with varying technical literacy.

- **Scalability for National Deployment:** Supports gradual rollout across Ethiopia's 11 regions and 2 city administrations while maintaining data integrity and system performance.

**Tools for Analysis and Design**

**Project Management & Collaboration:**

- **ClickUp**: For managing development sprints, tracking integration timelines with Ethiopian banks, and coordinating with regional land bureau stakeholders.

**Design:**

- **Figma**: Used for designing Amharic-first UI/UX interfaces with right-to-left text support, culturally appropriate icons, and accessibility features for rural users.

**Development:**

- **React.js**: For building the Citizen Portal web application with Amharic/Oromiffa/Tigrinya language support and optimal performance on limited bandwidth.
- **React Native:** For cross-platform mobile applications ensuring accessibility for Ethiopia's predominantly Android mobile user base (75%+ market share).
- **Tailwind CSS**: Utilized as the CSS framework to streamline and standardize the platform's design.

**Database:**

- **PostgreSQL**: Primary database for structured land parcel data with geospatial capabilities for Ethiopia's cadastral mapping requirements.
- **Blockchain**: as an immutable ledger for ownership verification. This ensures that land titles and transaction histories are tamper-proof, transparent, and cryptographically secured against fraud.

**Object Relational Mapping:**

- **Prisma**: Implemented for ORM, enabling efficient database operations and seamless integration with the backend.

**API Development:**

- **Postman**: Utilized for testing API endpoints to ensure reliable and efficient data communication between the frontend and backend systems.

**Ethiopian Payment System Integration:**

- **TeleBirr API Integration**: For Ethiopia's largest mobile money platform (12M+ users).
- **CBE Birr API**: Integration with Commercial Bank of Ethiopia's digital payment system.
- **EthSwitch Gateway**: For card-based transactions across Ethiopian financial institutions.

**Authentication & Ethiopian Context Security:**

- **JWT with Ethiopian Digital ID Context:** Secure session management supporting future integration with Ethiopia's Fayda National ID system.
- **Kebele ID Validation:** Specialized validation for Ethiopia's administrative ID structure.

**Notification System for Ethiopian Context:**

- **Amharic Email Templates**: Culturally appropriate communication templates.

**Ethiopian Document Processing**:

- **PDF Generation with Amharic Support**: Creation of Land Ownership Certificate in Amharic with QR codes.
- **Ethiopian QR Code Standard**: Implementation of Ethiopia's national QR code specification for document verification.

### *1.4.3 Implementation Methodology*

The Digital Land Registry and Ownership Verification System's implementation was guided by the following tools and technologies to ensure efficient development, debugging, and maintenance:

1. Programming Language:

- TypeScript: Chosen for its strong type definitions, enhancing code quality and reducing runtime errors.
- JavaScript: Used alongside TypeScript for rapid prototyping and integration with Ethiopian payment gateway APIs.

2. Database Management System (DBMS):

- PostgreSQL: Selected for its robustness and efficiency in handling complex queries and large datasets of Ethiopian land records, citizen information, and transaction histories.
- Blockchain: as an immutable ledger for ownership verification. This ensures that land titles and transaction histories are tamper-proof, transparent, and cryptographically secured against fraud.

3. Application Framework:

- Next.js: Employed for full-stack web application development, ensuring a responsive and scalable Citizen Portal with server-side rendering capabilities.
- React Native: Utilized for cross-platform mobile application development, providing

accessibility for citizens in both urban and rural areas of Ethiopia.

4. Styling Framework:
   - Tailwind CSS: Provided a utility-first CSS framework for consistent and efficient styling across the platform, ensuring accessibility for users with varying digital literacy levels.
5. Object Relational Mapping Tool:
   - Prisma: Implemented for ORM, enabling efficient database operations and seamless integration between the backend and PostgreSQL database.
6. API Testing Tool:
   - Postman: Ensured robust testing and validation of API endpoints.
7. Version Control:
   - GitHub: Used for version control, allowing collaborative development and efficient code management.
8. Development Environment:
   - Visual Studio Code: Provided an efficient and customizable coding environment.

## 1.5 Feasiblity

The feasibility of the project is analyzed to ensure it is viable and practical to implement. This includes an assessment of its economic benefits, technical knowledge requirements, and the time available for successful completion.

**Economic Feasibility**

The project is economically feasible due to the following reasons:

- **Development Costs:** Moderate - uses standard web technologies.
- **Infrastructure:** Cloud hosting keeps upfront costs low.
- **Savings:** Reduces office queues, document handling, and fraud investigation costs.
- **Revenue Protection:** Digital payments reduce receipt fraud and improve revenue collection.

**Technical Feasibility**

The technical feasibility of the project is supported by the following:

- **Team Expertise:** The development team has proficiency in technologies such as TypeScript, Prisma, and PostgreSQL, which are core to the platform.
- **Tool Selection:** Modern frameworks like Next.js and Tailwind CSS provide the necessary tools to build a scalable, secure, and efficient system.
- **Resource Availability:** Adequate documentation and community support for the chosen technologies ensure that potential technical challenges can be resolved effectively.
- **Compatibility:** The selected tools and frameworks are compatible, enabling seamless integration and smooth operation of the system.

**Time Feasibility**

The project timeline has been carefully planned to ensure timely delivery:

- **Detailed Schedule:** Tasks have been broken down into manageable phases with clearly defined milestones.
- **Agile Development:** Adopting an iterative approach allows for flexibility, quick adjustments, and continuous progress tracking.
- **Resource Allocation:** Adequate resources, including skilled developers and testing tools, are allocated to ensure tasks are completed within the set timeframe.
- **Buffer Time:** Provisions for unforeseen delays have been incorporated into the schedule to minimize risks of overruns.

**Legal Feasibility**

- **Data Protection**: Complies with Ethiopia's Data Protection Proclamation
- **Digital Payments:** Follows National Bank of Ethiopia regulations
- **Land Transactions:** Supports existing Property Registration laws
- **Electronic Evidence:** Digital certificates with QR codes are legally acceptable

**Social Feasibility**

- **Demand:** Citizens want to avoid long queues and document loss
- **Trust:** Transparency in tracking builds confidence in the system

- **Accessibility:** Web and mobile access works for Ethiopia's growing smartphone users
- **Language**: Can support Amharic for wider accessibility

## 1.6 Beneficiaries and significance of the project

**Transformative Impact on Land Administration**

The Digital Land Registry and Ownership Verification System represents a pivotal advancement in Ethiopia's journey toward transparent, efficient, and inclusive land governance. This project directly addresses systemic inefficiencies that have long plagued Ethiopia's land administration system, offering tangible solutions with far-reaching implications.

**Key Significance**

**Citizen Empowerment & Access to Justice:**

- **Democratizes Land Information:** Shifts power from bureaucratic gatekeepers to citizens by providing direct access to their land records.
- **Reduces Vulnerability:** Minimizes citizens' exposure to fraudulent transactions and document manipulation through transparent tracking.
- **Enhances Property Rights:** Strengthens land tenure security, which is fundamental for economic development and social stability.

**Institutional Modernization**

- **Digital Transformation Catalyst**: Serves as a model for digitizing other government services in Ethiopia.
- **Process Standardization:** Establishes uniform procedures across different urban land administration offices.
- **Capacity Building:** Develops digital skills within government institutions that can be applied to other sectors.

**Economic Development Enabler**

- **Unlocks Capital:** Secure, verifiable land ownership enables property to be used as collateral for loans and investments.
- **Stimulates Real Estate Market:** Efficient transfers and transparent records encourage property

transactions

- **Attracts Investment**: Clear land titles reduce risks for domestic and foreign investors.

**Technological Advancement in Public Service**

- Innovation Benchmark: **Demonstrates how modern web technologies can solve long-standing governance challenges**
- **Digital Infrastructure:** Creates foundational systems that can be extended to other government-citizen interfaces
- **Data-Driven Governance:** Establishes basis for analytics and evidence-based policy making in land management

**Broader Impact**

**For Ethiopian Governance:**

- **Transparency Model:** Sets precedent for open, accountable government service delivery.
- **Anti-Corruption Tool**: Digital audit trails and automated workflows reduce opportunities for rent-seeking.
- **Trust Building**: Enhances citizen confidence in government institutions through reliable service delivery.

**For Urban Development:**

- **City Planning Support**: Accurate, up-to-date land data informs sustainable urban planning
- **Property Tax Base**: Improves identification and valuation of properties for municipal revenue
- **Infrastructure Planning**: Better land records support utilities, transportation, and public service planning

**For Social Equity:**

- **Equal Access:** Removes geographic and bureaucratic barriers to land services
- **Document Security:** Protects vulnerable populations from document loss or destruction
- **Dispute Reduction**: Clear, accessible records minimize land-related conflicts

**Contribution to the Legal Domain**

This project's contribution is transformative in bridging the gap between citizens and land administration, fostering transparency, and ensuring tenure security. It highlights how technology can make property transactions more approachable and efficient, ultimately promoting a more transparent and equitable urban land market.

## 1.7 Limitation of the project

This project aims to develop a digital portal that connects citizens with verified land records to streamline property ownership and transactions. The system is designed to address the transparency and security challenges citizens face when navigating the land market, as well as assist landowners in managing and listing their assets more effectively. Our solution provides an easy-to-use interface, helping users find and interact with verified property listings across a range of land administration services.The platform will provide:

- A searchable marketplace of land parcels based on their location, size, and price
- An ownership verification system powered by blockchain to facilitate trust between buyers and sellers.
- Secure communication tools for negotiations and inquiries regarding land listings.
- Payment and escrow processing for land deposits, registration fees, and legal transfers.
- Transaction history and title logs for users to assess the legitimacy and chain of custody of properties.

The system will leverage technologies such as Next.js, TypeScript, PostgreSQL, and Prisma, ensuring a responsive and scalable solution. The project will be delivered within the agreed-upon timeframe, with essential features being prioritized for the initial launch.

**Issues Encountered: Challenges and Constraints**

1. **Adoption by Citizens and Land Holders**: One of the primary challenges is encouraging adoption from both individual citizens and traditional landholders. Many users are accustomed to traditional, paper-based methods of land administration and informal brokerage, which may make transitioning to an online portal more difficult. This resistance to change could delay the adoption of the platform and hinder its effectiveness in solving the problem of land tenure insecurity.

2. **Internet Connectivity and Digital Literacy:** The platform relies heavily on internet access and digital interaction, which may be a barrier in regions with limited connectivity or among users with low digital literacy. Users in peri-urban or underserved areas may not be able to access the platform as intended, limiting its reach and inclusivity. This issue could result in disparities in the platform's effectiveness across different geographical locations..

3. **Limited Land Categories and Jurisdictions:** While the platform aims to cover a wide range of property types, it may not be able to accommodate all land classifications, particularly those

involving complex communal holdings or specialized rural lease types. Certain users with niche property needs may find the platform less useful, as not all administrative zones or land categories will be represented in the initial stages of development..

4. **Integration and Scaling:** Due to time constraints and limited resources, the platform's integration with official government land databases and various third-party services (e.g., bank escrow systems or national ID APIs, Electricity Provider ,water provider , tax authority, legal or regulatory body) may not be fully realized. The scaling of the system to handle a high volume of concurrent blockchain transactions and large-scale geospatial data may also pose challenges, requiring future upgrades and architectural adjustments.

**Activities Left Undone Due to Constraints**

- **Full Land Category Coverage:** The platform may not be able to accommodate all specialized land classifications initially. Some niche areas (e.g., communal rural holdings, industrial economic zones, or heritage sites) may be excluded from the service offering due to the complexity of their legal frameworks and the need for specialized administrative verification.
- **Limited Offline Access:** While the platform could theoretically function in offline mode for certain data-viewing features, full offline support and local synchronization were not implemented due to time and resource constraints. This means that users in areas with unstable internet access may experience disruptions when trying to verify titles or browse the marketplace in real-time.
- **Advanced Features for Landholders:** Some advanced features, such as automated property valuation models, AI-powered land use analysis, and integrated GIS mapping tools for boundary disputes, are planned for future versions but were not included in the initial launch due to time limitations.

## 1.8 Scope of the project

The platform will focus on connecting citizens with verified land records and registered property owners in specific urban jurisdictions. The core objective is to provide an efficient, secure, and user-friendly means for individuals to verify land titles and participate in the property marketplace based on their needs.

The system will feature:

- **Property and Ownership Matching:** A system that matches users (buyers) with land parcels based on location, zoning regulations, size, and other relevant criteria.
- **Title and Transaction Management Tools:** Tools for managing the land acquisition process,

including tracking title verification status, documentation, and communication between buyers and sellers.

- **Secure Payment and Escrow:** A secure payment gateway for handling deposits, registration fees, and property payments, ensuring transparency and efficiency in financial transactions via blockchain-backed audit trails.
- **Land Administration Knowledge Base:** A repository of land laws, urban planning FAQs, and general property transfer information to assist users in understanding land-related terms and processes. This will serve as a resource to support users without replacing formal legal consultation.
- **Regulatory and Jurisdictional Compliance:** The platform will respect and comply with the land lease laws and administrative regulations of the respective regions, ensuring the platform's legality in each urban center it operates in.

**Business Processes/Subsystems to be Implemented**

**User Registration and Profile Management**

- **Service Provided:** Citizens and landowners will create and manage personal profiles, which will include relevant information such as verified identity credentials for sellers and property interests for citizens.
- **Features:** User authentication (KYC integration), profile editing, and secure storage of personal and ownership credentials.

**Search and Marketplace Engine**

- **Service Provided**: A search engine that matches citizens with land parcels based on predefined criteria such as location, land grade, size, and price.
- **Features:** Advanced geospatial search filters, property recommendation algorithms, and a matching system for buyers and sellers.

**Land Title & Transaction Management System**

- **Service Provided:** Landowners and citizens will have access to a management system for tracking the status of land verification and transfer processes.
- **Features:** Real-time title status tracking (via Blockchain), document management for deeds/leases, site visit scheduling, and transaction milestone management.

**Secure Payment and Escrow Subsystem**

- **Service Provided:** A payment gateway for secure and transparent transactions between buyers, sellers, and administrative bodies.
- **Features**: Digital invoice generation, secure payment processing (mobile money/bank integration), transaction history logs, and automated escrow for funds release upon title transfer..

**Land Administration Knowledge Base**

- **Service Provided**: A repository of articles, FAQs, and general property law information to help users understand land tenure, lease terms, and legal processes.
- **Features**: Searchable knowledge base, categorized land administration content, and a user-friendly glossary of legal land terms.

**Communication and Negotiation Tools**

- **Service Provided**: Secure communication tools to allow citizens and landowners to negotiate and discuss property details in a protected environment.
- **Features**: Encrypted messaging system, secure document sharing (e.g., site plans), and meeting coordination for physical inspections.

**Jurisdictional and Regulatory Compliance System**

- **Service Provided**: Ensures that the platform's services and land listings adhere to the specific urban land policies and lease regulations of each administrative zone.
- **Features**: Regional settings (e.g., Addis Ababa vs. regional city rules), legal compliance checks for land use, and terms of service tailored to Ethiopian land law.

**Boundary of the System**

The platform is primarily intended for:

- Citizens seeking to register, verify and acquire land based on specific investment or housing needs.
- Landowners and Developers offering verified property for sale or lease through a transparent digital channel.

**What the System Will Not Do**

- **Provide Legal Adjudication**: The platform will not resolve land disputes or provide professional legal rulings. The role of adjudicating land claims remains solely with the relevant judicial and administrative authorities.

- **Replace Official Land Bureaus:** The platform is a digital interface to streamline access and verification but does not replace the formal legal role of government land offices in issuing final titles
- **Operate Outside Specific Urban Boundaries:** The platform's services will be confined to specific urban jurisdictions where digital records are available, and all services remain subject to local land lease laws and national regulations.

## 1.9 Organization of the project

This project is organized into five main chapters, each addressing a critical aspect of the platform's development and implementation. Below is a comprehensive summary of the work covered in each chapter:

**Chapter One: Introduction**

This chapter introduces the project, outlining its purpose, objectives, and significance. It provides an overview of the problem the platform aims to solve—improving access to land services through digital transformation. Additionally, this chapter explains the project's scope, limitations, methodology, and feasibility, setting the foundation for the subsequent development phases.

**Chapter Two: System Features**

This chapter details the functional and non-functional requirements of the system. It includes use case modeling, business rule documentation, user interface prototypes, and various UML diagrams (state chart, activity, sequence, and class diagrams) that illustrate the system's behavior and structure. This chapter translates user needs into clear system specifications..

**Chapter Three: System Design**

This chapter covers the architectural design of the platform, providing a detailed description of the system's components and their interactions. It includes the system architecture, database design and user interface layout.

The design choices are explained, considering factors such as scalability, security, and user experience. This chapter also includes data flow diagrams, entity-relationship diagrams, and other technical documentation that illustrate how the platform is structured.

# Chapter Two: System features

## 2.1 The Existing System

Currently, the process of verifying land ownership and engaging in property transactions operates through traditional methods, primarily involving physical presence and direct interaction with government bureaus. Individuals seeking to verify titles or purchase land typically navigate the system by traveling from one administrative office to another or relying on word-of-mouth recommendations from informal brokers and acquaintances. This method of managing land assets often lacks transparency, as potential buyers and citizens do not have access to sufficient information about a property's legal status, historical transactions, or ownership validity before visiting offices in person.

**How the Existing System Functions**

The existing system is predominantly manual, with some semi-automated efforts, such as using social media platforms like Telegram , Facebook or in person for property promotion by some informal brokers and landowners. Below is a detailed step-by-step breakdown of how the system works:

**Searching for Land and Property:**

- **Manual Process**:A potential buyer usually begins the search for land by asking for recommendations from friends, family, or local "Delalas" (informal brokers). This informal process is often time-consuming and carries a high risk, as it may not lead to finding land with verified legal titles or clear ownership history.
- **Semi-Automated Process**: Some landowners and brokers try to expand their reach by advertising land parcels through platforms like Telegram, social media, or local listing websites. While this allows for a wider audience, it still depends on the individual's initiative to verify the legitimacy of these listings through physical office visits
- **Physical office visits.** Even when a lead is found online, the citizen must personally travel to the local Land Development and Administration Bureau to cross-reference the digital claim against the physical Registry Book, as there is no public-facing digital API for instant verification.

**Landowner Engagement and Verification:**

Once a citizen finds a property through recommendations or social media, they typically contact the seller or broker directly to arrange an in-person site visit and a trip to the local land bureau.

- **Manual Negotiations**: The buyer and seller usually engage in negotiations that involve in-person meetings, discussions of price, payment terms, and the division of transfer fees. The process is often lengthy and inefficient, as it relies heavily on direct face-to-face interaction and manual verification of paper documents.

- **Lack of Transparency**: Due to the reliance on word-of-mouth or informal brokers, buyers often have limited knowledge about the land's encumbrance status, tax history, or existing disputes before meeting the seller. This leaves citizens with little information to assess the security of their investment effectively.

**Information Gathering and Document Exchange:**

- **Manual Process**: Citizens and administrators must exchange relevant physical documents, such as site plans, title deeds, and ID copies, during in-person meetings at land bureaus. There is no centralized digital portal where this information can be easily accessed or updated by the public.
- **Inadequate Record-Keeping**: While land bureaus may keep records in physical files or simple standalone databases, these records are not integrated into a public-facing system that could help citizens make informed, data-driven decisions.

**Challenges for Landowners and Sellers:**

- **Promoting Assets**: While some sellers use semi-automated systems like Telegram channels to promote their property, this approach still relies on manual effort to reach credible buyers. There is no comprehensive platform to connect sellers with verified buyers based on specific needs, such as location, land grade, or budget.
- **Time-Consuming Adminstration**: Sellers spend significant time on administrative tasks like title verification, document collection from various offices, and multiple in-person consultations, which could be optimized with a more efficient system.

**Communication and Consultation:**

- The buyer and seller meet in person to discuss legal matters and boundary details, with no digital record or secure communication system to manage interactions. This lack of digital infrastructure means that important details, payment agreements, and document copies may be lost or miscommunicated.
- **In-Person Meetings:** For most land services, initial site visits, document signing at the notary, and even basic status inquiries are all conducted in person. This requires all parties to spend time traveling, which is inefficient and costly, especially in congested urban areas like Addis Ababa.

**Payment and Billing:**

- **Manual Payment System**: Payments for land and administrative fees are typically handled in cash or via manual bank transfers that require physical proof of deposit. Receipts and invoices are generated manually, and citizens must often visit the land bureau or bank multiple times to confirm financial clearance.
- **Lack of Integration**: There is no automated system for tracking land payments or generating secure digital receipts, which can result in delays, fraud through fake receipts, or errors in the final title transfer process.

**Drawbacks of the Existing System**

1. **Inefficiency**: The existing process involves significant amounts of time, money, and effort from the citizen, requiring them to physically visit various land bureaus, coordinate with informal brokers, and rely on word-of-mouth recommendations. This makes the property acquisition process slow, bureaucratic, and often frustrating.

2. **Lack of Transparency**: Citizens often do not have enough information about a land parcel's legal status, historical ownership, or encumbrances before visiting an office in person. This lack of pre-verified data leads to high uncertainty and a significant risk of falling victim to land-related fraud

3. **Limited Access** Individuals in remote areas, members of the diaspora, or those with limited mobility struggle to access land administration services due to the heavy dependence on physical presence and in-person meetings at local urban land offices

4. **Manual Processes**: The existing system is primarily manual, requiring administrators to manage land records using traditional methods such as paper files and physical ledgers. This results in high susceptibility to human error, physical record damage, and a lack of scalability for a rapidly growing urban land market.
5. **Fragmented Communication**: The lack of an integrated communication system between citizens and land administrators means that crucial details regarding property boundaries, legal requirements, or application status might be lost or miscommunicated during in-person meetings. This reliance on verbal exchanges and unrecorded phone calls leads to frequent misunderstandings, misplaced documentation, and missed opportunities for secure investment.

6. **Difficulty in Reaching the Right Clients**: Landowners and verified developers face challenges in attracting genuine potential buyers, especially when they rely on informal platforms like Telegram that do not provide targeted matching or a secure way to connect with citizens based on specific land needs. Without a dedicated portal, legitimate sellers are often overshadowed

by unverified listings, making it difficult to establish the trust necessary for high-value transactions.

## 2.2 Proposed System

To address the issues highlighted in the statement of the problem, the proposed system introduces a comprehensive digital platform designed to modernize and enhance the urban land administration and marketplace process. This platform aims to provide an efficient, accessible, and secure solution that overcomes the barriers currently faced by citizens and property owners in Ethiopia

**Key Solutions to the Identified Problems:**

1. **Streamlined Land Discovery and Verification:** The platform will feature a searchable database of land parcels, allowing citizens to easily find and verify property details online. This empowers users to check land authenticity and boundaries without the need for physical office visits.

2. **Transparent Property Profiles:** Each land listing will include verified details such as geospatial data, parcel boundaries, and historical ownership records. By validating details against official registry databases, the system ensures transparency and trustworthiness in the land selection process.

3. **Real-Time Tracking System:** The platform will offer a built-in tracking tool that allows citizens to monitor their application status in real-time. This eliminates the need to visit offices repeatedly, showing exactly which officer is handling the request and providing estimated completion times.

4. **Secure Digital Payment System:** A secure payment gateway (integrating TeleBirr, CBE Birr, and EthSwitch) will be integrated to handle financial transactions. This ensures automatic receipt generation and reduces the risks associated with cash-based transactions and fraudulent receipts.

5. **Professional and Secure Digital Certificates**: The platform supports the issuance of digital land certificates equipped with QR codes and digital signatures. This ensures that titles are downloadable anytime, cannot be lost, and are easily verifiable by relevant authorities.

6. **Equal Opportunity for Marketplace Participation:** The system will promote a fair marketplace by allowing all verified landowners to list properties based on merit and data rather than informal connections. This creates a more balanced and equitable urban land market for all citizens.

7. **Standardized Security and Immutability:** The platform will implement a centralized digital repository that maintains an immutable chain of ownership. This ensures that land records remain accurate, consistent, and protected from unauthorized alterations or conflicts.

8. **Comprehensive Legal Knowledge Base**: To assist clients in understanding their legal situations, the platform will provide access to a library of legal resources, including articles, FAQs, and case studies. This feature will educate users and reduce reliance on guesswork or uninformed decisions.

9. **Automated Document and Case Management:** Citizens and land officers will have access to tools for digital document uploading, application processing, and inter-agency coordination (such as automatic tax clearance). This streamlines the workflow and reduces reliance on manual, paper-based processes.

**Benefits of the Proposed System:**

- **Accessibility**: The platform ensures that citizens can access land services and verify ownership from anywhere, breaking down geographical and bureaucratic barriers.
- **Transparency**: Citizens gain access to reliable, real-time information about land parcels and application progress, fostering trust in the administration.
- **Efficiency**: Automated tools and digital payments reduce the time and effort required for site visits, document handling, and inter-agency clearances.
- **Security**: By utilizing digital signatures and immutable records, the system creates a high-integrity environment where land titles are secure and fraud is minimized.

## 2.3 Requirement Analysis

### *2.3.1 Functional requirement*

**User & Access Management Requirements**

**FREQ-1: Priority: High**

- The system shall allow Ethiopian citizens to register and create personal accounts by providing valid identification (Kebele ID), mobile number, and email address. The registration process shall include identity verification through government databases.

**FREQ-2: Priority: High**

- The system shall provide secure login mechanisms for registered users, including password authentication, SMS-based one-time passwords (OTP), and optional biometric verification for mobile applications.

**FREQ-3: Priority: High**

- The system shall implement role-based access control (RBAC) with distinct permission levels for Citizens, Land Officers, Marketplace Moderators, System Administrators, and Government Auditors.

**Land Ownership Verification Requirements**

**FREQ-4: Priority: High**

- The system shall enable citizens to submit official land ownership verification requests through digital channels, eliminating the need for physical office visits.

**FREQ-5: Priority: High**

- The system shall provide land officers with a dedicated interface to review, prioritize, and process verification requests with access to all relevant documentation.

**FREQ-6: Priority: High**

- The system shall allow land officers to officially approve or reject verification requests, with mandatory comments for rejections and digital signature capabilities for approvals.

**FREQ-7: Priority: High**

- The system shall provide citizens with real-time tracking of verification request status through a user dashboard with visual progress indicators and estimated completion times.

**FREQ-8: Priority: High**

- The system shall provide citizens with real-time tracking of verification request status through a user dashboard with visual progress indicators and estimated completion times.

**Digital Land Registry Requirements**

**FREQ-9: Priority: High**

- The system shall maintain a secure, centralized digital repository of all verified land ownership records, including geospatial data, parcel boundaries, and owner information.

**FREQ-10: Priority: Medium**

- The system shall preserve complete historical ownership records for audit purposes, maintaining an immutable chain of ownership transfers for each land parcel.

**Land Marketplace Requirements**

**FREQ-11: Priority: High**

- The system shall allow verified land owners to create, publish, and manage property listings for sale, including detailed descriptions, photographs, pricing, and terms.

**FREQ-12: Priority: Medium**

- The system shall provide users with comprehensive search and filtering capabilities to browse available land listings by location, price, size, zoning, and other relevant criteria.

**FREQ-13: Priority: High**

- The system shall facilitate secure communication between buyers and sellers, allowing buyers to submit formal purchase requests with terms and conditions.

**Transaction & Ownership Transfer Requirements**

**FREQ-14: Priority: High**

- The system shall support end-to-end secure land purchase transactions with integrated payment processing through Ethiopian financial systems (TeleBirr, CBE Birr, EthSwitch).

**FREQ-15: Priority: High**

- The system shall automatically update official land ownership records upon successful transaction completion, ensuring real-time synchronization of the digital registry.

**Administration & Monitoring Requirements**

**FREQ-16: Priority: High**

- The system shall provide administrators with comprehensive tools to manage user accounts, moderate marketplace listings, update land records, and configure system
- parameters.

**FREQ-17: Priority: Low**

- The system shall generate analytical reports and dashboards for administrative oversight, including transaction volumes, user statistics, revenue reports, and system performance metrics.

**Notifications & Security Requirements**

**FREQ-18: Priority: Medium**

- The system shall automatically notify users via multiple channels (SMS, email, in-app notifications) about important status changes, required actions, and transaction updates.

**FREQ-19: Priority: High**

- The system shall implement enterprise-grade security measures including data encryption, secure API communications, regular security audits, and vulnerability assessments.

**FREQ-20: Priority: Medium**

- The system shall maintain comprehensive audit logs documenting all critical system actions, user activities, data modifications, and security events for compliance and forensic analysis.

### *2.3.2 System Usecase*

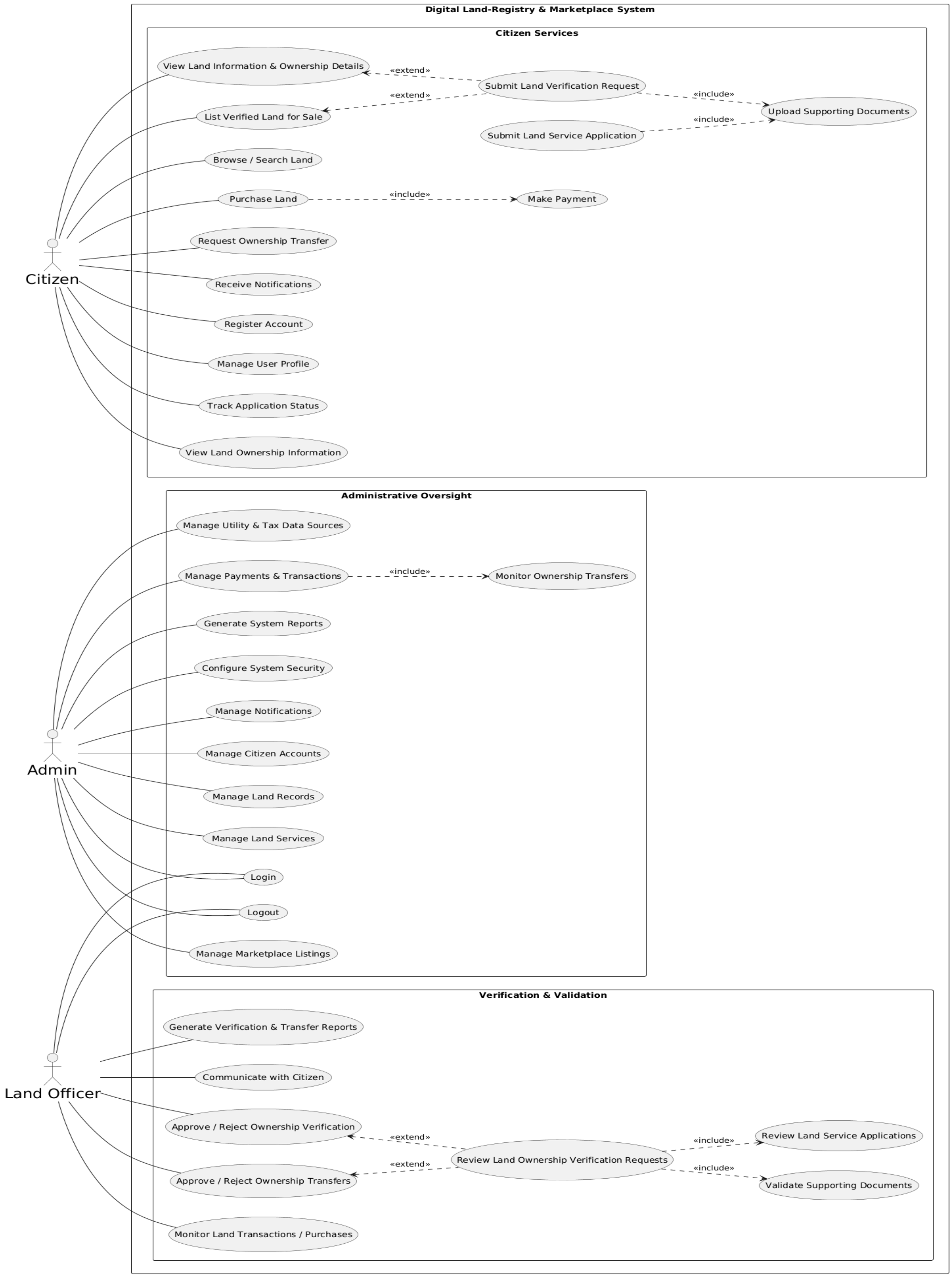

Digital Land-Registry & Marketplace System
Citizen Services
View Land Information & Ownership Details
«extend»
Submit Land Verification Request
List Verified Land for Sale
«extend»
«include»
Upload Supporting Documents
Submit Land Service Application
«include»
Browse / Search Land
Purchase Land
«include»
Make Payment
Request Ownership Transfer
Receive Notifications
Register Account
Manage User Profile
Track Application Status
View Land Ownership Information
Citizen
Administrative Oversight
Manage Utility & Tax Data Sources
Manage Payments & Transactions
«include»
Monitor Ownership Transfers
Generate System Reports
Configure System Security
Manage Notifications
Manage Citizen Accounts
Manage Land Records
Manage Land Services
Login
Logout
Manage Marketplace Listings
Admin
Verification & Validation
Generate Verification & Transfer Reports
Communicate with Citizen
Approve / Reject Ownership Verification
«extend»
Review Land Ownership Verification Requests
«include»
Review Land Service Applications
Approve / Reject Ownership Transfers
«extend»
«include»
Validate Supporting Documents
Monitor Land Transactions / Purchases
Land Officer

Figure 1 System Usecase

## Use Case documentation

Table 1 Description of register use case

| **ID** | **UC01.1** |
|---|---|
| *Name* | Register Account |
| *Actor* | Citizen |
| **Use Case Description** | Allows new users to create an account to access land management services |
| **Flow of Events** | 1. Access registration page<br>2. Fill personal details<br>3. Create username/password<br>4. Submit form<br>5. System validates data<br>6. Account created<br>7. Verification sent |
| **Alternative Flow** | A1: Duplicate account → Show error<br>A2: Invalid data → Show validation errors<br>A3: System error → Display error message |
| **Post Condition** | Account created, verification pending, user can log in |

Table 2 Description of Login Use Case

| *Id* | **UC01.2** |
|---|---|
| *Name* | Login |
| *Actors* | Citizen |
| *Pre condition* | 1. User has a registered account.<br>2. User's account is **not** locked, suspended, or deactivated. |
| **Flow of Events** | 1. Enter credentials<br>2. Submit login<br>3. System authenticates<br>4. Redirect to dashboard |
| **Alternative Flow** | A1: Invalid credentials → Show error<br>A2: Account locked → Show lock message<br>A3: Inactive account → Prompt activation |
| **Post Condition** | User logged in, session created, dashboard displayed |
| **Use Case Description** | Authenticates registered users to access system |

**Table 3 Description of Submit Verification Request**

| | |
|---|---|
| Id | **UC01.3** |
| Name | Submit Verification Request |
| Actors | Citizen |
| Use Case Description | Submits land ownership for official verification |
| Pre condition | 1. User is logged in.<br>2. User has necessary documents (deed, ID, survey map) ready for upload. |
| Flow Of Events | 1. Access verification form<br>2. Enter land details<br>3. Upload documents<br>4. Submit request<br>5. Receive confirmation |
| **Alternative Flow** | A1: Invalid documents → Reject upload<br>A2: Incomplete form → Show missing fields<br>A3: System error → Save draft |
| **Post Condition** | Request submitted, status pending, documents stored |

**Table 4 Description of List Verified Land for Sale usecase**

| | |
|---|---|
| *Id* | **UC01.4** |
| *Name* | List Verified Land for Sale |
| *Actors* | Citizen |
| *Use Case Description* | Lists a verified land parcel on the public marketplace for potential buyers to view and purchase. |
| *Pre condition* | 1. User is logged in.<br>2. User owns at least one land parcel with "Verified" status.<br>3. Land is not already listed or under another transaction. |
| *Flow Of Events* | 1. Go to "My Lands" dashboard.<br>2. Select verified land parcel.<br>3. Choose "Sell This Land".<br>4. Enter sale details (price, description, contact). |

| | |
|---|---|
| | 5. Upload land photos.<br>6. Set listing as active.<br>7. Confirm and publish. |
| Alternative Flow | A1: Land Not Verified<br>- System checks land status.<br>- Shows error "Land must be verified first".<br>- Redirects to verification request page.<br>A2: Draft Saved<br>- User saves as draft before publishing.<br>- Listing stored as "Draft".<br>- Can be edited and published later. |
| **Post Condition** | 1. Land listing is active on marketplace.<br>2. Listing status is "For Sale".<br>3. Public can view listing details.<br>4. User can manage the listing. |

**Table 5 Description of Browse or Search Land usecase**

| | |
|---|---|
| *Id* | **UC01.5** |
| *Name* | Browse or Search Land |
| *Actors* | Citizen |
| *Use Case Description* | Searches, filters, and views land listings available for sale in the marketplace. |
| *Pre condition* | 1. Marketplace module is active. (No login required for basic browse). |
| *Flow Of Events* | 1. Access marketplace homepage.<br>2. Use search bar or filters (location, price, area).<br>3. View list of results.<br>4. Click on a listing to see details.<br>5. View land photos and map location.<br>6. Check ownership verification badge. |
| *Alternative Flow* | A1: No Results Found<br>- System shows "No listings match your criteria".<br>- Suggests broadening filters.<br>- Shows popular listings instead.<br>A2: Advanced Search<br>- User clicks "Advanced Search".<br>- More specific filters appear (land type, zoning).<br>- User refines search. |
| *Post Condition* | 1. List of relevant land parcels is displayed.<br>2. User has information to make a purchase decision.<br>3. Search criteria may be saved for logged-in users. |

**Table 6 Description of Browse or Search Land usecase**

| | |
|---|---|
| *Id* | **UC01.6** |

| | |
|---|---|
| *Name* | Browse or Search Land |
| *Actors* | Citizen |
| *Use Case Description* | Searches, filters, and views land listings available for sale in the marketplace. |
| *Pre condition* | 1. Marketplace module is active. (No login required for basic browse). |
| *Flow Of Events* | 1. Access marketplace homepage.<br>2. Use search bar or filters (location, price, area).<br>3. View list of results.<br>4. Click on a listing to see details.<br>5. View land photos and map location.<br>6. Check ownership verification badge. |
| *Alternative Flow* | - System shows "No listings match your criteria".<br>- Suggests broadening filters.<br>- Shows popular listings instead.<br>A2: Advanced Search<br>- User clicks "Advanced Search".<br>- More specific filters appear (land type, zoning).<br>- User refines search. |
| *Post Condition* | 1. List of relevant land parcels is displayed.<br>2. User has information to make a purchase decision.<br>3. Search criteria may be saved for logged-in users. |

**Table 7 Description of Purchase Land usecase**

| | |
|---|---|
| *Id* | **UC01.7** |
| *Name* | Purchase Land |
| *Actors* | Citizen |
| *Use Case Description* | Initiates and completes the process to buy a specific land parcel listed on the marketplace. |
| *Pre condition* | 1. User is logged in.<br>2. Selected land is verified and status is "For Sale".<br>3. Land is not under contract with another buyer.<br>4. User has a valid payment method on file. |
| *Flow Of Events* | 1. From land details page, click "Buy Now".<br>2. Review sale agreement terms.<br>3. Enter payment details or select saved method.<br>4. Confirm purchase.<br>5. System processes payment.<br>6. <<include>> Make Payment (UC-107).<br>7. Receive purchase confirmation.<br>8. Ownership transfer |
| *Alternative Flow* | A1: Concurrent Purchase<br>- System detects another user buying same land. |

| | |
|---|---|
| | - Shows error "Land no longer available".<br>- Redirects user back to marketplace.<br>A2: Purchase Cancelled<br>- User cancels before confirmation.<br>- Redirects to land details page.<br>- Transaction is not created. |
| *Post Condition* | 1. Purchase transaction is recorded.<br>2. Land status changes to "Sold" or "Under Contract".<br>3. Request Ownership Transfer (UC-109) is auto-created.<br>4. Seller and buyer receive notifications. |

**Table 8 Description of Make Payment usecase**

| *Id* | **UC01.8** |
|---|---|
| *Name* | Make Payment |
| *Actors* | Citizen |
| *Use Case Description* | Processes the financial transaction for a purchase or service fee. **(**Often included in other use cases) |
| *Pre condition* | 1. A pending transaction exists (purchase, fee).<br>2. User payment method is valid.<br>3. Payment gateway connection is available. |
| *Flow Of Events* | 1. System redirects to secure payment page.<br>2. User selects payment method (card, bank transfer, digital wallet).<br>3. Enters payment details.<br>4. Confirms the payment amount.<br>5. System processes via gateway.<br>6. Receives authorization response. |
| *Alternative Flow* | A1: Payment Declined<br>- Gateway returns decline.<br>- System shows "Payment failed".<br>- Asks user to try another method.<br>A2: Payment Gateway Error<br>- Connection to gateway fails.<br>- System shows "Temporary error".<br>- Saves cart/transaction |
| *Post Condition* | 1. Payment is marked as "Successful" in the system.<br>2. Transaction record is updated.<br>3. Receipt is generated and sent to user.<br>4. The main use case (e.g., purchase) proceeds |

**Table 9 Description of Manage User Profile usecase**

| *Id* | **UC01.9** |
|---|---|
| *Name* | Manage User Profile |
| *Actors* | Citizen |

| | |
|---|---|
| *Use Case Description* | Allows the user to view and update their personal information, contact details, and account settings. |
| *Pre condition* | 1. User is logged in. |
| *Flow Of Events* | 1. Navigate to "My Profile".<br>2. View current information.<br>3. Click "Edit".<br>4. Update fields (phone, address, password).<br>5. Save changes.<br>6. System validates and updates. |
| *Alternative Flow* | A1: Invalid Data<br>- System validates new data (e.g., email format).<br>- Shows error on invalid field.<br>- User corrects and retries.<br>A2: Cancel Edit<br>- User cancels before saving.<br>- All changes are discarded.<br>- Returns to profile view. |
| *Post Condition* | 1. User profile is updated in the database.<br>2. User may receive a confirmation email for critical changes (email/password).<br>3. Changes are reflected immediately in the session. |

**Table 10 Description of Request Ownership Transfer usecase**

| *Id* | **UC01.10** |
|---|---|
| *Name* | Request Ownership Transfer |
| *Actors* | Citizen |
| *Use Case Description* | Initiates the official process to transfer legal ownership of a land parcel from one party to another (e.g., after sale, gift, inheritance). |
| *Pre condition* | 1. User is logged in.<br>2. User is the current legal owner or has legal authority (buyer, heir, agent).<br>3. Land parcel exists and is verified in the system.<br>4. Required legal documents are available. |
| *Flow Of Events* | 1. Navigate to "Ownership Transfer".<br>2. Select the land parcel to transfer.<br>3. Enter details of the new owner (transferee).<br>4. Select transfer reason (Sale, Gift, Inheritance).<br>5. Upload supporting legal documents.<br>6. Pay applicable transfer fees (<<include>> Make Payment).<br>7. Submit to Land Officer for review. |
| *Alternative Flow* | A1: Missing Documents<br>- System validates required docs per transfer type.<br>- Prevents submission, lists missing items.<br>- User uploads before proceeding. |

| | |
|---|---|
| | A2: Fee Payment Failed<br>- Payment for transfer fees fails.<br>- Transfer request is saved as "Pending Payment".<br>- User can retry payment from dashboard. |
| *Post Condition* | 1. Transfer request is created with "Pending Review" status.<br>2. Land Officer is notified.<br>3. Land status may change to "Transfer Pending".<br>4. Request is trackable by both parties. |

**Table 11 Description of Submit Land Service Application usecase**

| | |
|---|---|
| *Id* | **UC01.11** |
| *Name* | Submit Land Service Application |
| *Actors* | Citizen |
| *Use Case Description* | Applies for a government land-related administrative service (e.g., land subdivision, mutation, zoning change). |
| *Pre condition* | 1. User is logged in.<br>2. User owns or has legal rights to the concerned land.<br>3. Land is verified.<br>4. User knows the specific service required. |
| *Flow Of Events* | 1. Go to "Land Services" section.<br>2. Select the type of service needed.<br>3. Select the land parcel.<br>4. Fill the application form.<br>5. Upload required documents.<br>6. Pay service fee (<<include>> Make Payment).<br>7. Submit application. |
| *Alternative Flow* | A1: Service Not Available for Land<br>- System validates land eligibility for the service.<br>- Shows error "This service not applicable".<br>- User selects a different service or land.<br>A2: Save as Draft<br>- User saves incomplete application.<br>- Application saved in "Drafts".<br>- Can be completed later. |
| *Post Condition* | 1. Service application is submitted.<br>2. Status is set to "Received".<br>3. Assigned to a Land Officer queue.<br>4. User can track its status |

**Table 12 Description of Track Application Status usecase**

| | |
|---|---|
| *Id* | **UC01.12** |
| *Name* | Track Application Status |
| *Actors* | Citizen |

| | |
|---|---|
| *Use Case Description* | Allows the user to monitor the current status and history of their submitted applications (verification, service, transfer). |
| *Pre condition* | 1. User is logged in.<br>2. User has at least one submitted application |
| *Flow Of Events* | 1. Navigate to "My Applications" dashboard.<br>2. View list of all applications with current status.<br>3. Click on a specific application.<br>4. View detailed status timeline and history. |
| | 5. See officer comments or notes if any. |
| *Alternative Flow* | A1: No Applications<br>- Dashboard shows empty state message.<br>- Provides links to submit new requests.<br>A2: Filter Applications<br>- User filters list by type (Verification, Service, Transfer) or status.<br>- System shows filtered results. |
| *Post Condition* | 1. User is informed of the progress.<br>2. User can see if any action is required on their part. |

**Table 13 Description of resolve dispute use case**

| | |
|---|---|
| *Id* | **UC01.13** |
| *Name* | Upload Supporting Document |
| *Actors* | Citizen |
| *Use Case Description* | Adds required documents to an existing application that is in draft state or where the officer has requested more information. (Often included in other use cases) |
| *Pre condition* | 1. An application exists and is in "Draft", "Submitted", or "Info Required" status.<br>2. User has the document file ready. |
| *Flow Of Events* | 1. From application details page, click "Upload Documents".<br>2. Select document type from dropdown.<br>3. Choose file from device.<br>4. Upload file.<br>5. System validates file type and size.<br>6. File is attached to the application. |

| | |
|---|---|
| *Alternative Flow* | A1: Invalid File<br>- System rejects file (wrong type, too large, corrupted).<br>- Shows error with requirements.<br>- User selects a different file.<br>A2: Replace Document<br>- User deletes an old uploaded file.<br>- Uploads a new version in its place. |
| *Post Condition* | 1. Document is stored and linked to the application.<br>2. Application status may update (e.g., from "Draft" to "Submitted").<br>3. Land Officer may be notified of new uploads. |

**Table 14 Description of access law guide use case**

| | |
|---|---|
| *Id* | **UC01.14** |
| *Name* | View Land Ownership Information |
| *Actors* | Citizen |
| *Use Case Description* | Accesses the official record, proof, and details of ownership for land parcels registered under their name. |
| *Pre condition* | 1. User is logged in.<br>2. User is the registered owner of at least one land parcel. |
| *Flow Of Events* | 1. Navigate to "My Lands".<br>2. System lists all land parcels owned by the user.<br>3. Select a specific land parcel.<br>4. View ownership certificate (digital), registration date, history.<br>5. Option to download a summary PDF. |
| *Alternative Flow* | A1: No Land Owned<br>- "My Lands" section shows empty state.<br>- Provides link to "Submit Verification Request".<br>A2: Co-Ownership<br>- Land has multiple owners.<br>- System shows all co-owner names and shares. |
| *Post Condition* | 1. User views verified ownership details.<br>2. User can use this information as proof. |

**Table 15 Description of Provide feedback**

| | |
|---|---|
| *Id* | **UC01.15** |
| *Name* | Receive Notification |
| *Actors* | Citizen |
| *Use Case Description* | The system alerts the user about important updates, status changes, or messages related to their account and transactions. (A system-triggered use case) |
| *Pre condition* | 1. User has an active account.<br>2. A system event (status change, new message, deadline) occurs that requires user awareness. |

| | |
|---|---|
| *Flow Of Events* | 1. System Event Triggered (e.g., application approved).<br>2. System generates notification.<br>3. Notification is queued for delivery (in-app, email, SMS).<br>4. User receives the notification in their preferred channel.<br>5. Notification appears in "Notification Center". |
| *Alternative Flow* | A1: Notification Channel Failed<br>- Primary channel (e.g., SMS) fails.<br>- System retries or uses fallback channel (e.g., email).<br>A2: User Disables Notifications<br>- User has opted out.<br>- Notification is only logged, not actively sent. |
| *Post Condition* | 1. User is informed of the update.<br>2. Notification is marked as "Delivered".<br>3. Unread notification count increases. |

**Table 16 Description of receive payment use case**

| | |
|---|---|
| *Id* | **UC01.16** |
| *Name* | View Land Information & Details |
| *Actors* | Citizen |
| *Use Case Description* | Views comprehensive technical and legal details of a specific land parcel, including map location, boundaries, and attributes. |
| *Pre condition* | 1. For personal land: User is logged in and owns the land.<br>2. For marketplace listings: No login required. |
| *Flow Of Events* | 1. From search results or "My Lands", select a land parcel.<br>2. View detailed page with tabs: Overview, Map, Documents, History.<br>3. View interactive map with plot boundaries.<br>4. Check land attributes (area, soil type, zoning). |
| | 5. See public remarks or notes. |
| *Alternative Flow* | A1: Map Unavailable<br>- GIS service is down.<br>- System shows static image or "Map temporarily unavailable".<br>A2: Limited Access<br>- User tries to view private land of another.<br>- System shows "Access Denied" or limited public info. |

| | |
|---|---|
| *Post Condition* | 1. User gains complete information about the land parcel.<br>2. User can make informed decisions (e.g., to purchase). |

Table 17 Description of Logout usecase

| | |
|---|---|
| *Id* | **UC01.17** |
| *Name* | Logout |
| *Actors* | Citizen |
| *Use Case Description* | Securely terminates the current user session to protect account access |
| *Pre condition* | 1. User is currently logged in. |
| *Flow Of Events* | 1. User clicks "Logout" button or avatar menu.<br>2. System confirms intent (optional).<br>3. System invalidates the session token.<br>4. Clears session data from browser.<br>5. Redirects user to login page or homepage. |
| *Alternative Flow* | A1: Auto-Logout<br>- User session expires due to inactivity.<br>- System shows "Session expired" message.<br>- Redirects to login page.<br>A2: Cancel Logout<br>- User cancels the logout confirmation.<br>- Returns to the previous page, session remains active. |
| *Post Condition* | 1. User session is securely terminated.<br>2. Server-side session is destroyed.<br>3. User must log in again to access protected features. |

## Land Adminstritive

Table 18 Description of Review Verification Requests usecase

| | |
|---|---|
| *Id* | **UC01.18** |
| *Name* | Review Verification Requests |
| *Actors* | Land Officer |
| *Use Case Description* | Examines newly submitted land ownership verification requests from citizens. |
| *Pre condition* | 1. Officer is logged into the system. |
| | 2. There is at least one request in the "Pending" queue assigned to them or the general pool. |

| | |
|---|---|
| *Flow Of Events* | 1. Navigate to "Verification Queue".<br>2. System displays list of pending requests, sorted by date.<br>3. Officer selects a request to review.<br>4. Views citizen details, land details, and uploaded documents.<br>5. <<include>> Validate Supporting Documents (UC-202).<br>6. Makes a preliminary assessment. |
| *Alternative Flow* | A1: Assign to Another Officer<br>- Officer is not specialized in the request's region/type.<br>- Reassigns request to a different officer's queue.<br>- Request status remains "Pending". |
| *Post Condition* | 1. Officer is familiar with the request details.<br>2. Request is ready for validation and decision. |

**Table 19 Description of Validate Supporting Documents usecase**

| | |
|---|---|
| *Id* | **UC01.19** |
| *Name* | Validate Supporting Documents |
| *Actors* | Land Officer |
| *Use Case Description* | Systematically checks the authenticity and completeness of documents submitted with an application. **(Often included)** |
| *Pre condition* | 1. Officer is reviewing a specific application (verification, transfer, service).<br>2. Application has attached documents. |
| *Flow Of Events* | 1. Officer opens the document viewer.<br>2. Checks for required documents per application type.<br>3. Verifies document authenticity (signatures, stamps, dates).<br>4. Cross-references with internal or external databases if available.<br>5. Marks each document as "Accepted" or "Rejected" |
| *Alternative Flow* | A1: Documents Insufficient/Invalid<br>- Key documents are missing or fraudulent.<br>- Officer marks application as "Requires More Info" or "Reject".<br>- System notifies the citizen.<br>A2: External Check Needed<br>- Officer needs to consult physical archives or another department.<br>- Puts application "On Hold" and adds an internal note. |
| *Post Condition* | 1. All documents are marked as validated. |
| | 2. A validation note is added to the application record.<br>3. Application can proceed to final decision. |

**Table 20 Description of Approve/Reject Verification usecase**

| | |
|---|---|
| *Id* | **UC01.20** |
| *Name* | Approve/Reject Verification |
| *Actors* | Land Officer |
| *Use Case Description* | Makes the final decision to approve or reject a land ownership verification request after review. |
| *Pre condition* | 1. Officer has reviewed the request and validated all documents (UC-201, UC-202).<br>2. Officer has the required authority to make the decision. |
| *Flow Of Events* | 1. On the request review page, officer selects "Approve" or "Reject".<br>2. If rejecting, enters a mandatory reason for the citizen.<br>3. System prompts for final confirmation.<br>4. Officer confirms the decision.<br>5. If Approved: System updates land status to "Verified" and triggers notifications.<br>6. If Rejected: System updates request status to "Rejected". |
| *Alternative Flow* | A1: Conditional Approval<br>- Approval requires minor corrections.<br>- Officer selects "Approve with Conditions".<br>- System sets status to "Approved (Pending Corrections)" and notifies citizen.<br>A2: Decision Saved as Draft<br>- Officer saves decision without submitting. |
| *Post Condition* | 1. Verification request status is finalized (Approved/Rejected).<br>2. Land ownership status is updated in the central record.<br>3. Citizen receives a formal notification (UC-114).<br>4. Case is moved to the "Completed" log. |

**Table 21 Description of Review & Process Transfer Requests usecase**

| | |
|---|---|
| *Id* | **UC01.21** |
| *Name* | Review & Process Transfer Requests |
| *Actors* | Land Officer |
| *Use Case Description* | Reviews legal requests for ownership transfer and updates the official land registry upon approval. |
| *Pre condition* | 1. Officer is logged in.<br>2. A transfer request with status **"Pending Review"** exists in their queue. |
| *Flow Of Events* | 1. Access "Transfer Requests" queue. |

| | |
|---|---|
| | 2. Open a specific request.<br>3. Review transfer details (parties, land, reason, documents).<br>4. Validate legal documents (sale deed, NOC, inheritance proof).<br>5. Approve or reject the transfer.<br>6. If Approved: <<include>> Update Land Ownership Records (UC-205).<br>7. If Rejected: Notify citizen with reason. |
| *Alternative Flow* | A1: Require Additional Stamp Duty/Fees<br>- System calculates outstanding dues.<br>- Officer puts request "On Hold" pending payment.<br>- Notifies citizen to pay.<br>A2: Legal Discrepancy Found<br>- Officer finds a legal impediment (e.g., court injunction).<br>- Rejects request and advises citizen to resolve legally. |
| *Post Condition* | 1. Transfer request is resolved.<br>2. If approved, land ownership is legally transferred in the system.<br>3. All parties and the administrator are notified.<br>4. Audit log is updated. |

**Table 22 Description of Update Land Ownership Records usecase**

| | |
|---|---|
| *Id* | **UC01.22** |
| *Name* | Update Land Ownership Records |
| *Actors* | Land Officer |
| *Use Case Description* | Officially modifies the land registry to reflect a change in ownership. **(Critical system function)** |
| *Pre condition* | 1. A transfer or mutation request has been officially approved.<br>2. Officer has the administrative privilege to edit the land registry. |
| *Flow Of Events* | 1. System auto-populates a registry update form from the approved request.<br>2. Officer verifies the new owner's details and effective date.<br>3. Officer digitally signs the registry update.<br>4. System executes the update: changes owner ID, updates history timeline.<br>5. System generates a new digital ownership certificate for the new owner. |
| *Alternative Flow* | A1: System Update Conflict<br>- Another process is modifying the same record.<br>- System shows error, asks officer to retry.<br>A2: Manual Override Required<br>- System auto-update fails.<br>- Officer accesses the backend registry for manual correction following a strict procedure. |
| *Post Condition* | 1. Land ownership is legally changed in the |

| | |
|---|---|
| | master database.<br>2. Old owner's rights are revoked, new owner's rights are established.<br>3. A new ownership record version is created.<br>4. Previous owner's "My Lands" list is updated. |

**Table 23 Description of Review Service Applications usecase**

| | |
|---|---|
| *Id* | **UC01.23** |
| *Name* | Review Service Applications |
| *Actors* | Land Officer |
| *Use Case Description* | Processes applications for land-related government services (subdivision, mutation, etc.). |
| *Pre condition* | 1. Officer is logged in.<br>2. Service applications exist in the "Services" queue. |
| *Flow Of Events* | 1. Navigate to "Service Applications".<br>2. Filter by service type.<br>3. Review application form and attached documents.<br>4. Conduct field checks or internal consultations if needed.<br>5. Make a decision: Approve, Reject, or Request More Info.<br>6. Process any applicable fees.<br>7. If approved, trigger the relevant service workflow (e.g., generate new plot numbers for subdivision) |
| *Alternative Flow* | A1: Site Visit Required<br>- Officer cannot decide based on documents alone.<br>- Marks application as "Site Visit Required" and schedules it.<br>A2: Inter-Department Coordination<br>- Application requires input from another department (e.g., zoning).<br>- Forwards application internally with comments. |
| *Post Condition* | 1. Service application is resolved.<br>2. If approved, the requested service is initiated and its outcome is recorded.<br>3. Citizen is notified of the decision.<br>4. All actions are logged for the audit trail. |

## Adminstritive

**Table 24 Description of Manage Users & Roles usecase**

| | |
|---|---|
| *Id* | **UC01.24** |

| | |
|---|---|
| *Name* | Manage Users & Roles |
| *Actors* | Administrator |
| *Use Case Description* | Creates, modifies, deactivates user accounts and assigns system roles & permissions. |
| *Pre condition* | 1. Administrator is logged in with super-user privileges |
| *Flow Of Events* | 1. Navigate to "User Management" dashboard.<br>2. View list of all users with filters.<br>3. To Add: Click "Add User", fill details, assign role (Citizen, Officer, Admin), set temporary password.<br>4. To Edit: Select user, modify details/role/status.<br>5. To Deactivate: Toggle user status to "Inactive".<br>6. Save all changes. |
| *Alternative Flow* | A1: Duplicate User<br>- System detects duplicate email/ID.<br>- Shows warning, prevents creation.<br>A2: Last Admin Deactivation<br>- System prevents deactivating the only active admin account.<br>- Shows error "At least one active admin required". |
| *Post Condition* | 1. User database is updated.<br>2. Role-based permissions are enforced.<br>3. Affected users receive relevant notifications (e.g., account created, role changed). |

**Table 25 Description of Monitor Marketplace & Payments usecase**

| | |
|---|---|
| *Id* | **UC01.25** |
| *Name* | Monitor Marketplace & Payments |
| *Actors* | Administrator |
| *Use Case Description* | Oversees all marketplace listings and financial transactions for compliance, fraud detection, and dispute resolution. |
| *Pre condition* | 1. Administrator is logged in.<br>2. Marketplace module is active. |
| *Flow Of Events* | 1. Navigate to "Marketplace Monitor".<br>2. View dashboards: active listings, recent transactions, revenue.<br>3. Filter and inspect suspicious listings (e.g., unrealistic prices).<br>4. Review transaction logs and payment gateway reports.<br>5. Resolve disputes flagged by users or the system. |
| *Alternative Flow* | A1: Flag Suspicious Activity<br>- System or admin flags a transaction.<br>- Admin investigates, can suspend the transaction and freeze involved accounts temporarily.<br>A2: Generate Financial Report<br>- Admin runs a report for a period. |

| | |
|---|---|
| | - System generates a summary of commissions/fees |
| *Post Condition* | 1. Marketplace operates in a secure and trustworthy manner.<br>2. Financial discrepancies are identified and acted upon.<br>3. Disputes are logged and resolved. |

**Table 26 Description of Generate Reports & Audit Logs usecase**

| | |
|---|---|
| *Id* | **UC01.26** |
| *Name* | Generate Reports & Audit Logs |
| *Actors* | Administrator |
| *Use Case Description* | Creates operational, financial, and audit reports for analysis, compliance, and decision-making. |
| *Pre condition* | 1. Administrator is logged in.<br>2. Relevant data exists in the system for the chosen report type. |
| *Flow Of Events* | 1. Go to "Reports & Analytics".<br>2. Select report type (User Activity, Transaction Summary, Verification Stats, System Audit).<br>3. Set parameters (date range, user role, region).<br>4. Click "Generate".<br>5. View report on screen, export to PDF/Excel, or schedule automatic delivery |
| *Alternative Flow* | A1: Complex Report<br>- Report requires long processing time.<br>- System processes in background and notifies admin via email when ready.<br>A2: Data Privacy Filter<br>- Report contains sensitive personal data.<br>- System automatically masks/makes data anonymous in the export based on rules. |
| *Post Condition* | 1. A formal report is generated and stored.<br>2. Insights are gained for system improvement and compliance.<br>3. Audit trail is maintained for regulatory purposes. |

**Table 27 Description of Configure System & Security usecase**

| | |
|---|---|
| *Id* | **UC01.27** |
| *Name* | Configure System & Security |
| *Actors* | Administrator |
| *Use Case Description* | Sets up and modifies global system settings, security policies, and notification templates. |
| *Pre condition* | 1. Administrator is logged in with configuration privileges. |
| *Flow Of Events* | 1. Navigate to "System Configuration".<br>2. Adjust modules: enable/disable features like marketplace or verification.<br>3. Set security policies: password rules, login attempt limits, session timeout.<br>4. Configure notification templates for emails/SMS.<br>5. Integrate with external services (payment gateway, SMS API).<br>6. Test and save all configurations. |
| *Alternative* | A1: Invalid Configuration |

| | |
|---|---|
| *Flow* | - Entered settings cause a conflict or error.<br>- System validates and prevents saving, highlighting the issue.<br>A2: Rollback Configuration<br>- New settings cause system issues.<br>- Admin uses "Restore Previous Settings" from a backup. |
| *Post Condition* | 1. System behavior and rules are updated globally.<br>2. Security posture is defined.<br>3. All users experience the new configuration. |

### *2.3.3 Business rule Documentation*

**[BR-1]:** Account Creation Rule

- **Identifier:** BR-1
- **Rule Name:** Valid Citizen Registration
- **Description:** Citizens must provide a valid Kebele ID, active mobile number, and email address to create an account.
- **Enforcement:** System validation against government databases during registration.
- **Scope**: Citizen Registration.

**[BR-2]:** Unique Account Rule

- **Identifier:** BR-2
- **Rule Name:** One Account per Citizen
- **Description:** Each citizen is allowed only one registered account per Kebele ID.
- **Enforcement:** System prevents duplicate Kebele ID registrations.
- **Scope:** User Management

**[BR-3]:** Identity Verification Rule

- **Identifier:** BR-3
- **Rule Name:** Mandatory Identity Verification
- **Description:** Citizens must verify their identity through SMS OTP or biometric verification before accessing sensitive land data.
- **Enforcement**: Required step before dashboard access.
- **Scope:** Authentication & Security

[BR-4]: Password Security Rule

- **Identifier:** BR-4
- **Name:** Strong Password Requirement
- **Description:** Passwords must be at least 8 characters long and include uppercase, lowercase, numbers, and special characters.

- **Enforcement:** Frontend and backend validation during password creation/change.
- **Scope:** Security

[BR-5]: Account Lockout Rule

- **Identifier:** BR-5
- **Rule Name**: Failed Login Attempt Lockout
- **Description**: After 5 consecutive failed login attempts, the account is locked for 30 minutes.
- **Enforcement**: Automatic lockout; unlock via SMS OTP or admin.
- **Scope**: Security

[BR-6]: Document Upload Rule

- **Identifier:** BR-6
- **Rule Name**: Supported Document Formats
- **Description**: Citizens can only upload documents in PDF, JPEG, or PNG formats, with a maximum size of 10MB per file.
- **Enforcement:** File validation during upload.
- **Scope:** Document Management

**[BR-7]:** Land Verification Request Rule

- **Identifier:** BR-7
- **Rule Name**: Verified Land Ownership Required
- **Description:** Only citizens with verified land ownership in the registry can submit verification requests.
- **Enforcement**: System checks land registry before allowing request submission.
- **Scope**: Land Verification

[BR-8]: Payment Method Rule

- **Identifier:** BR-8
- **Name:** Rule Approved Payment Channels
- **Description:** All payments must be made through integrated Ethiopian payment systems: Tele Birr, CBE Birr, or Eth Switch.
- **Enforcement**: Only approved payment gateways are available.
- **Scope**: Payment Processing

[BR-9] : Receipt Generation Rule

- **Identifier:** BR-9
- **Rule Name:** Automatic Digital Receipt
- **Description:** Upon successful payment, the system must generate and store a digital receipt with a unique transaction ID.
- **Enforcement:** Automatic post-payment process.
- **Scope:** Financial Transactions

[BR-10] : Marketplace Listing Rule

- **Identifier:** BR-10
- **Rule Name:** Verified Owners Only for Listings
- **Description:** Only citizens with verified land ownership can list properties for sale on the marketplace.
- **Enforcement:** Ownership verification check before listing creation.
- **Scope**: Land Marketplace

[BR-11] : Listing Moderation Rule

- **Identifier:** BR-11
- **Rule Name:** Mandatory Listing Approval
- **Description:** All new marketplace listings must be reviewed and approved by a land officer before being published.
- **Enforcement**: Listing status set to "Pending" until approved.
- **Scope**: Marketplace Management

[BR-12]፡ Communication Rule

- **Identifier:** BR-12
- **Rule Name:** Secure In-System Messaging
- **Description:** All buyer-seller communications must occur within the platform's secure messaging system.
- **Enforcement**: No external contact details shared until transaction approval.
- **Scope:** Marketplace Communication

[BR-13]፡ Ownership Transfer Rule

- **Identifier:** BR-13
- **Name:** Lawyer Professionalism
- **Rule Name:** Full Payment Before Transfer
- **Description:** Land ownership can only be transferred after full payment is confirmed and cleared.
- **Enforcement:** System checks payment status before initiating transfer.
- **Scope:** Transaction Processing

[BR-14]፡ Data Privacy Rule

- **Identifier:** BR-14
- **Rule Name:** Confidentiality of Land Records
- **Description**: Land records and citizen data must be kept confidential and accessible only to authorized users.
- **Enforcement**: Role-based access control (RBAC) and encryption.
- **Scope:** Data Security

[BR-15]፡ Officer Verification Rule

- **Identifier:** BR-15
- **Rule Name**: Two-Officer Verification for High-Value Transactions

- **Description**: Transactions above 1,000,000 ETB require approval from two different land officers.
- **Enforcement:** System enforces dual approval workflow.
- **Scope:** Transaction Security

[BR-16]፡ Notification Rule

- **Identifier:** BR-16
- **Rule Name:** Mandatory Status Updates
- **Description:** Citizens must receive SMS/email notifications at each major stage of their application (submitted, under review, approved, rejected).
- **Enforcement**: Automated notifications triggered by status changes.
- **Scope:** User Communication

[BR-17]፡ Audit Log Rule

- **Identifier ፡** BR-17
- **Rule Name:** Immutable Audit Trail
- **Description**: Land records and citizen data must be kept confidential and accessible only to authorized users.
- **Enforcement**: Role-based access control (RBAC) and encryption.
- **Scope:** Data Security

[BR-18]፡ System Backup Rule

- **Identifier:** BR-18
- **Rule Name:** Daily Automated Backups
- **Description:** The system must perform automated daily backups of all land records and transaction data.
- **Enforcement:** Scheduled backup jobs with verification.
- **Scope:** Data Integrity

[BR-19]፡ Account Deactivation Rule

- **Identifier:** BR-19 Identifier.
- **Rule Name:** Inactive Account Deactivation
- **Description:** Accounts inactive for more than 12 months will be deactivated (but not deleted).
- **Enforcement:** Automated monthly review process.
- **Scope:** User Management

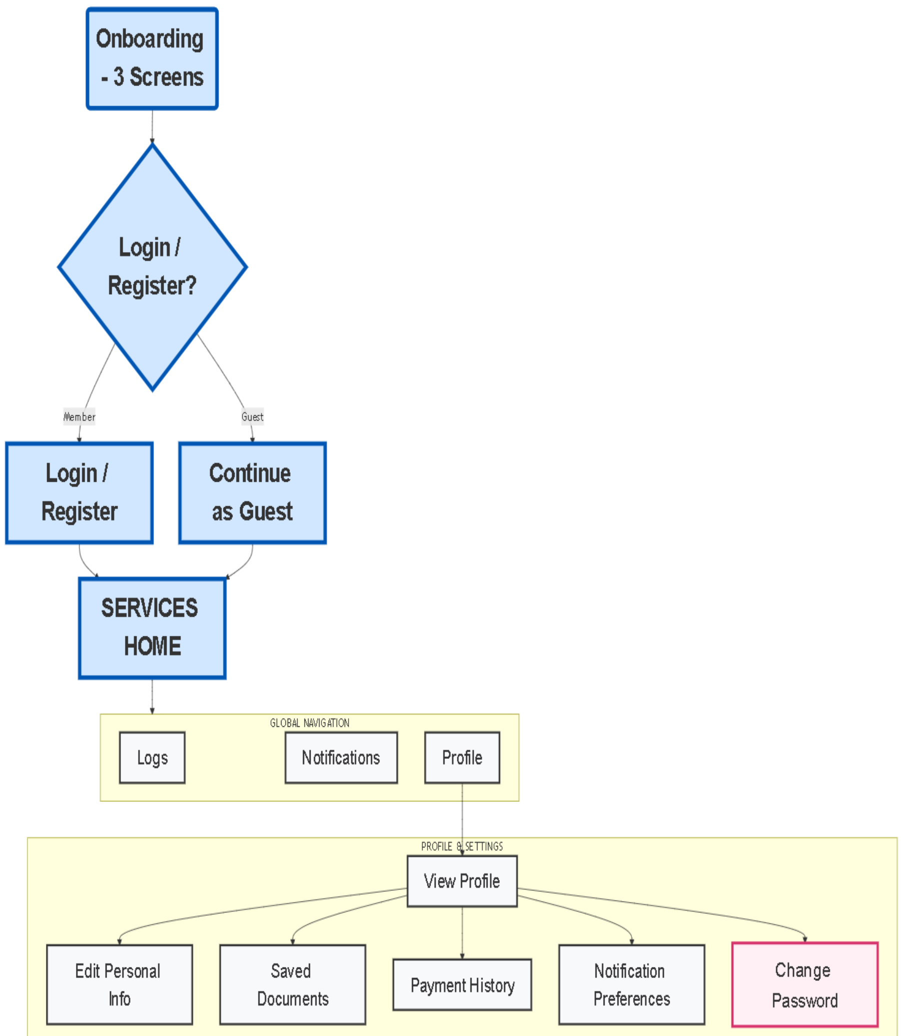
[BR-20]፡ Data Privacy Rule

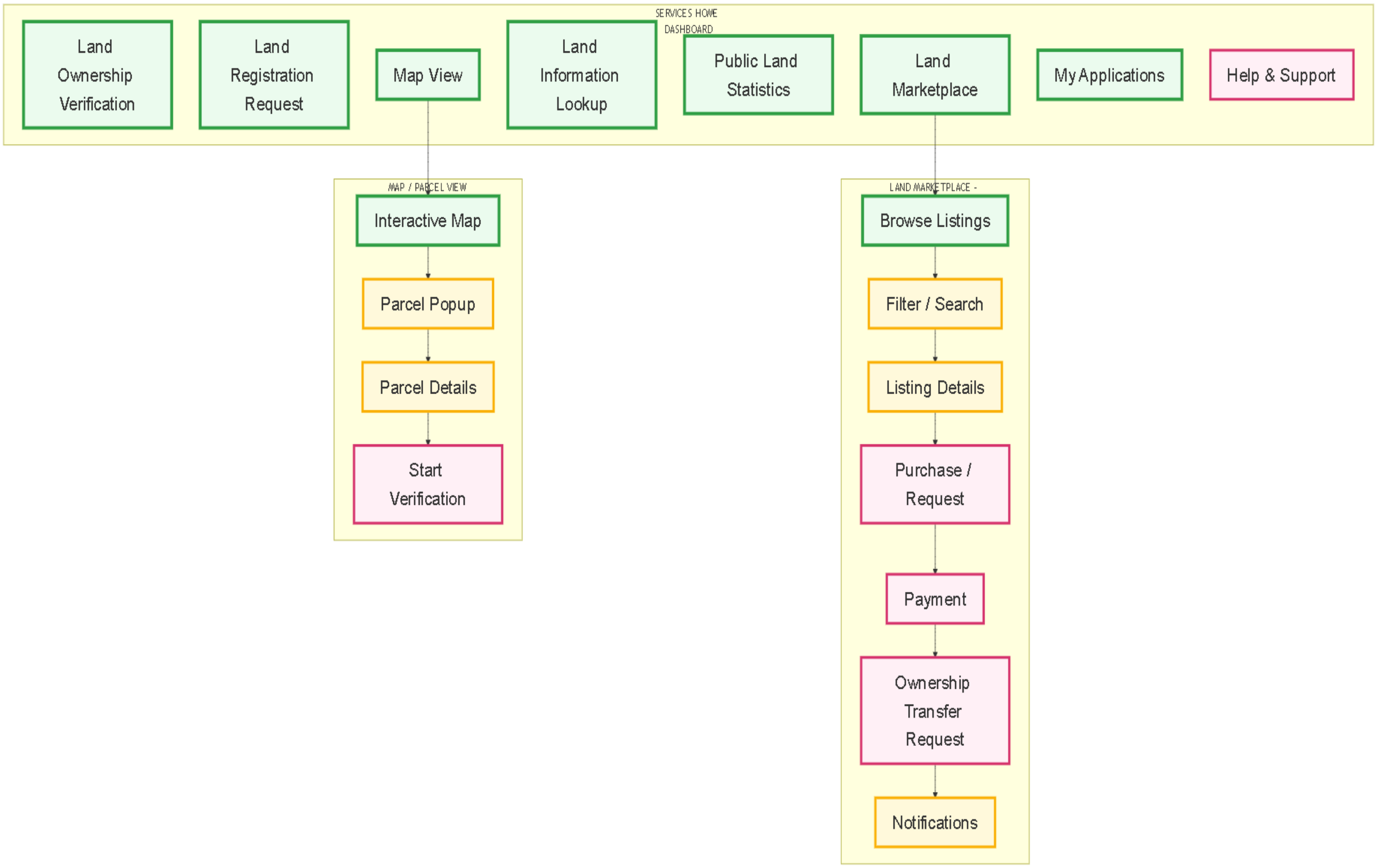
- **Identifier:** BR-20
- **Rule Name:** System as Tool, Not Legal Authority
- **Description:** The system facilitates land transactions but does not replace legal land ownership

verification by authorized government bodies.

- **Enforcement:** Displayed disclaimer on dashboard and during transactions.
- **Scope:** Legal Compliance

### *2.3.4 User Interface Prototype*




SERVICES HOME DASHBOARD
Land Ownership Verification
Land Registration Request
Map View
Land Information Lookup
Public Land Statistics
Land Marketplace
My Applications
Help & Support
MAP / PARCEL VIEW
Interactive Map
Parcel Popup
Parcel Details
Start Verification
LAND MARKETPLACE -
Browse Listings
Filter / Search
Listing Details
Purchase / Request
Payment
Ownership Transfer Request
Notifications

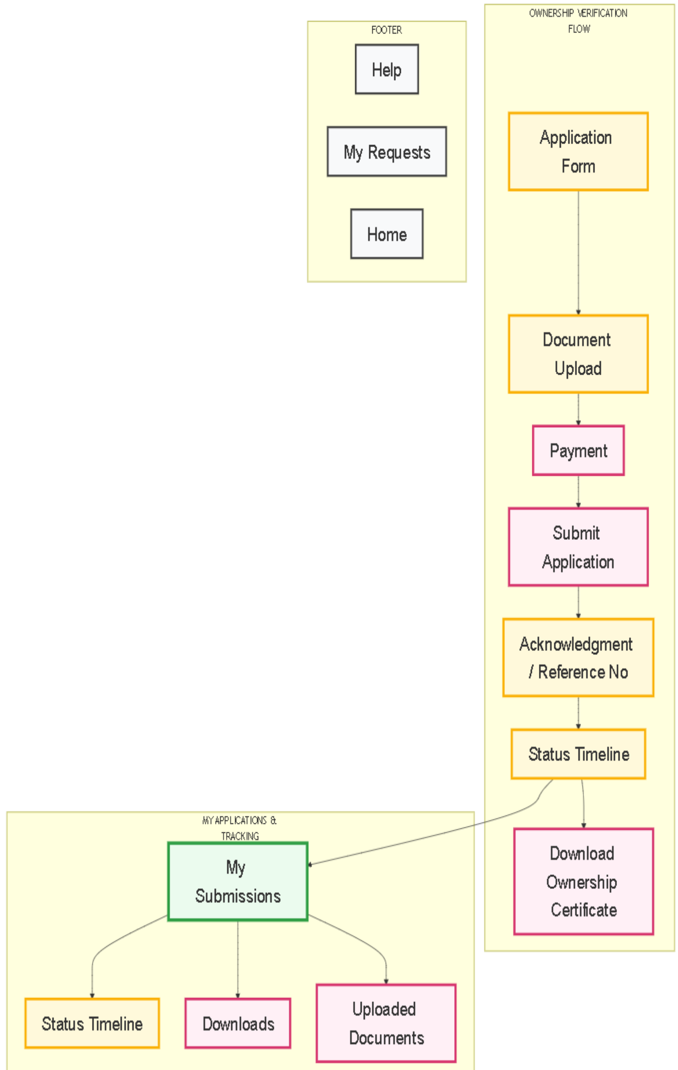



Figure 2 Client UI Prototype

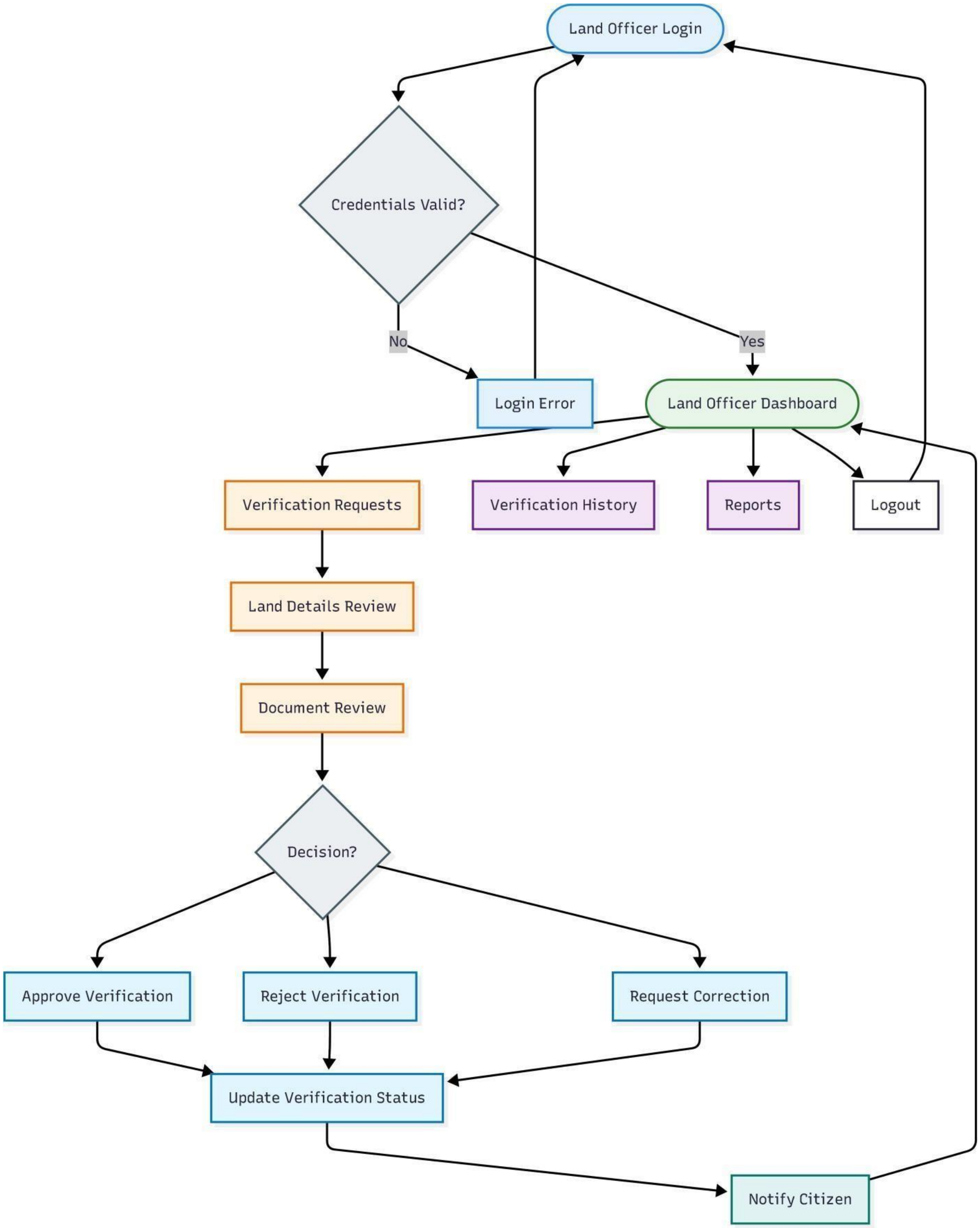
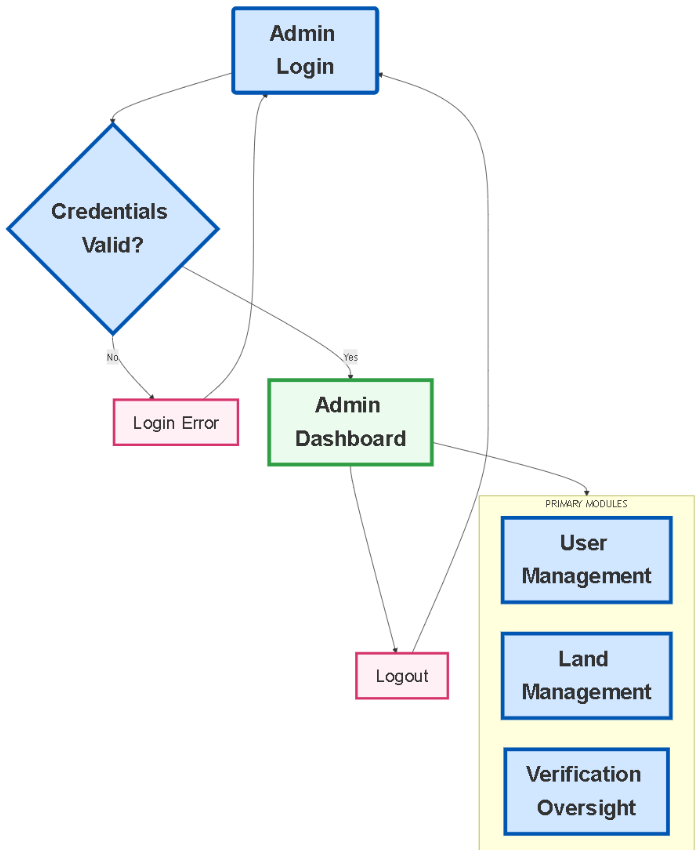
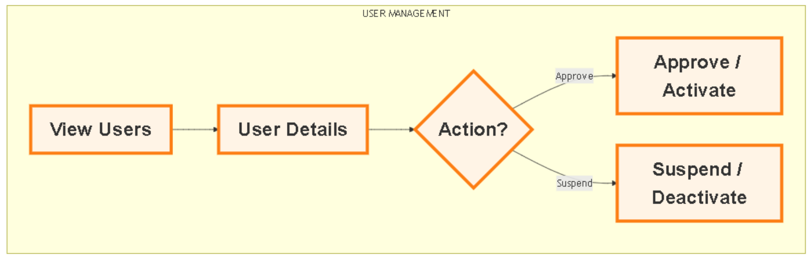



**Figure 3 Land Officer UI Prototype**


Admin Login
Credentials Valid?
No
Yes
Login Error
Admin Dashboard
PRIMARY MODULES
User Management
Land Management
Verification Oversight
Logout


LAND MANAGEMENT
View Lands
Land Details
Update Records


USER MANAGEMENT
View Users
User Details
Action?
Approve
Suspend
Approve / Activate
Suspend / Deactivate

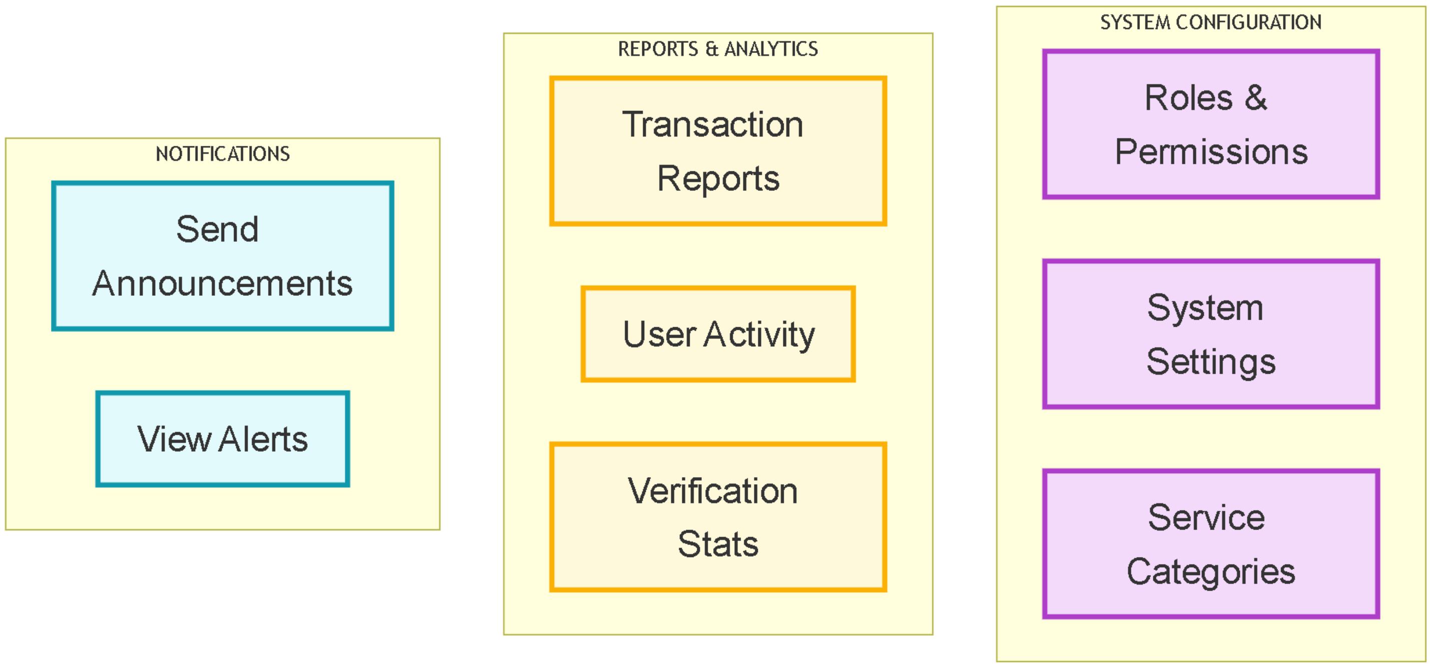



Figure 4 Admin UI prototype

### *2.3.5 State Chart Diagram*

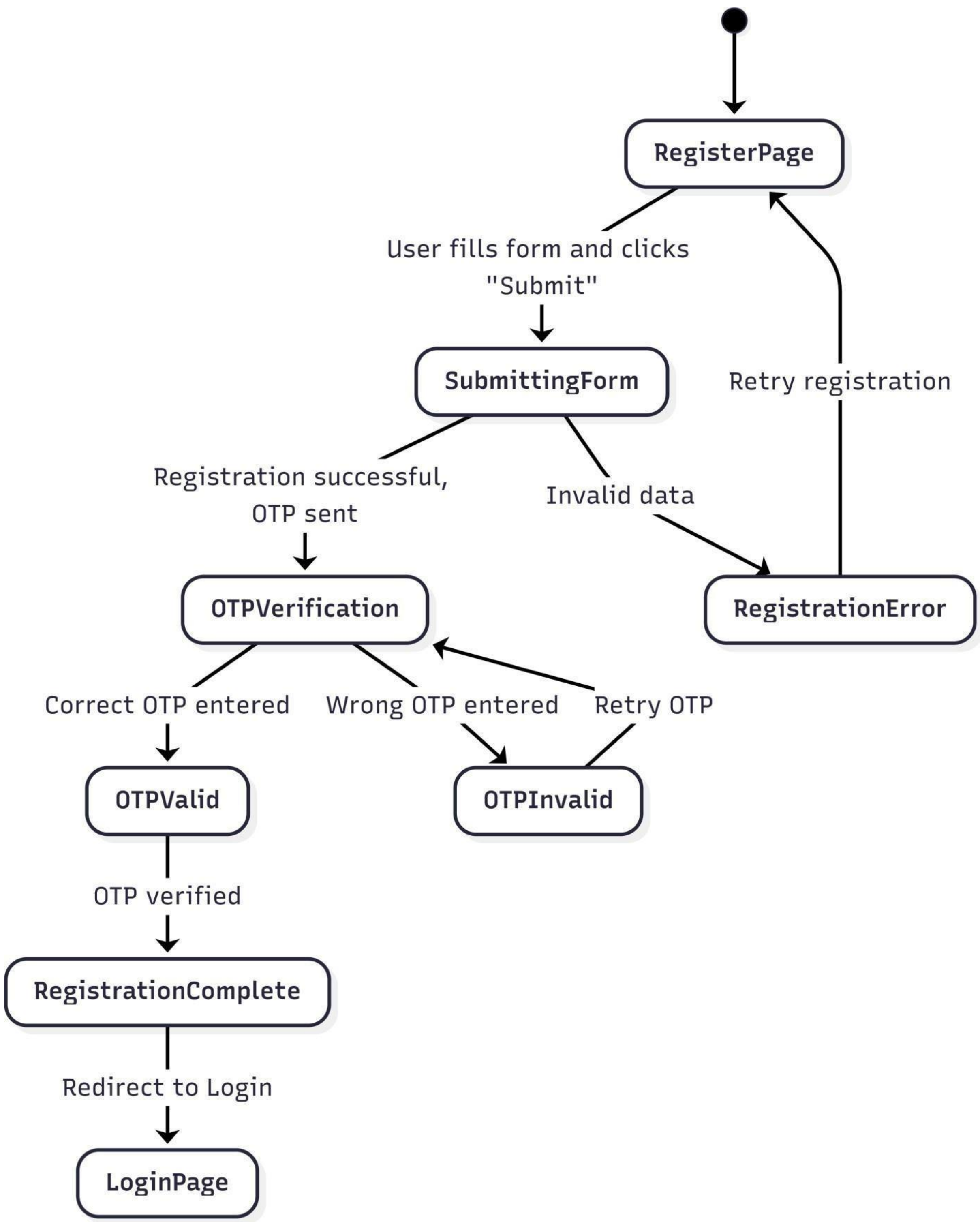



**Figure 5 Register State Diagram**

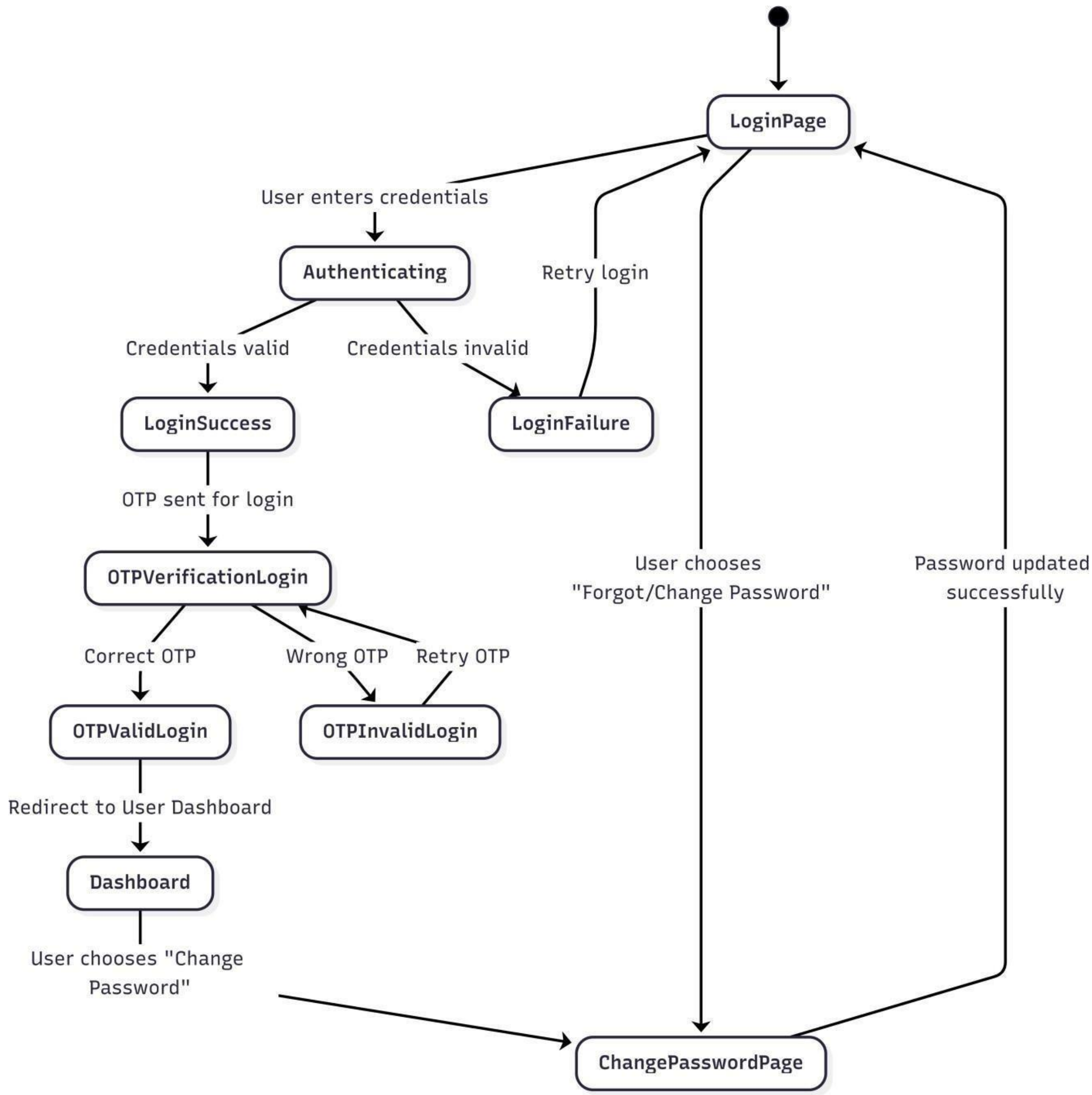



**Figure 6 Login State Diagram**

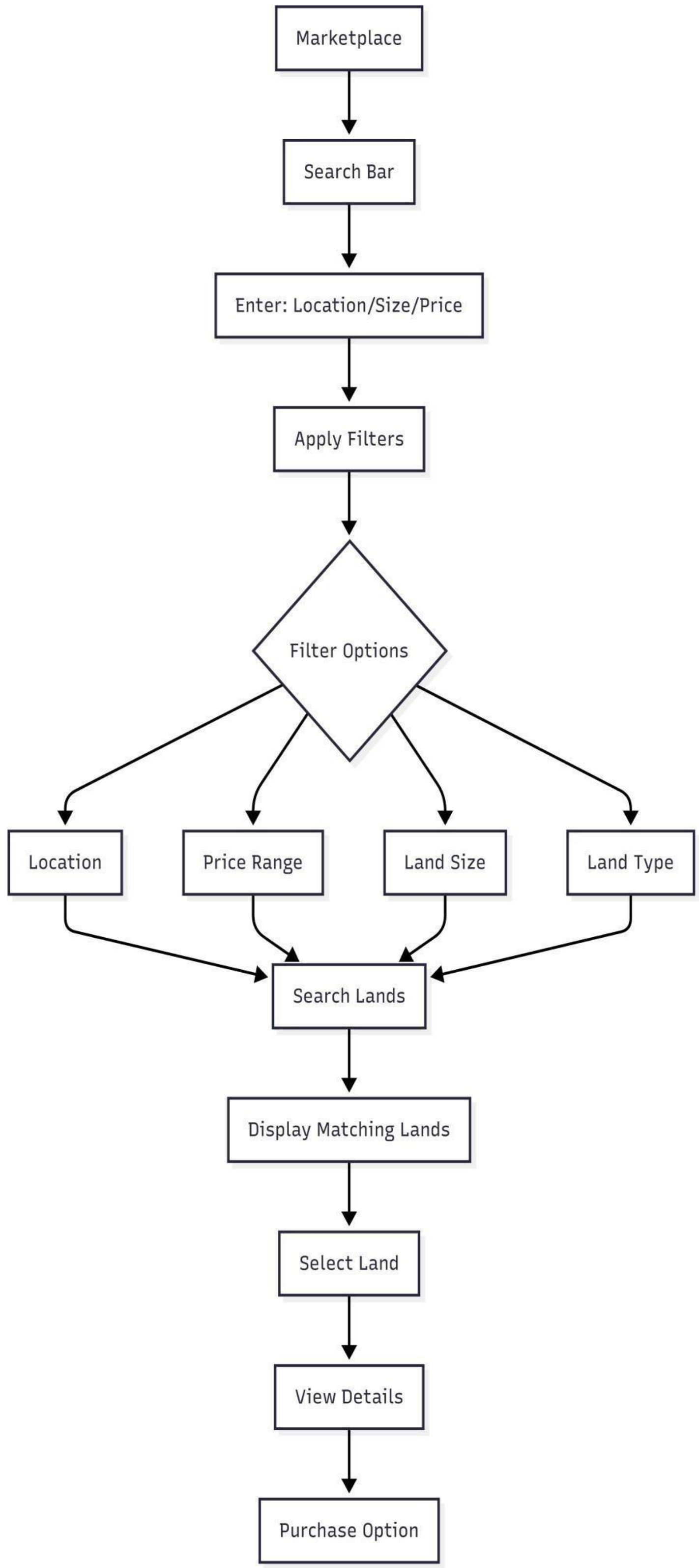



**Figure 7 Search State Diagram**

Figure 8 Land Verification state Diagram

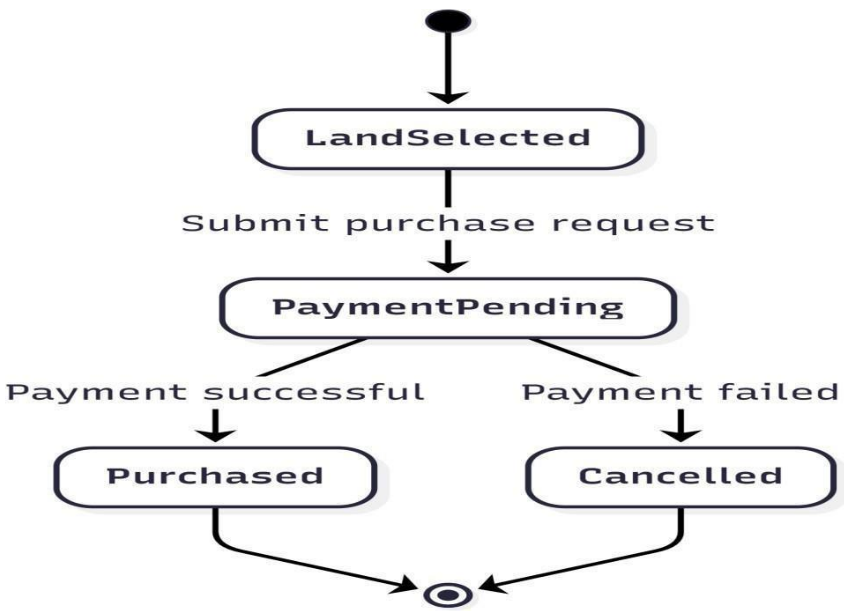


Figure 9 Land Purchase State Diagram


**Figure 10 Land Tansfer State diagram**

### *2.3.6 Activity Diagram*

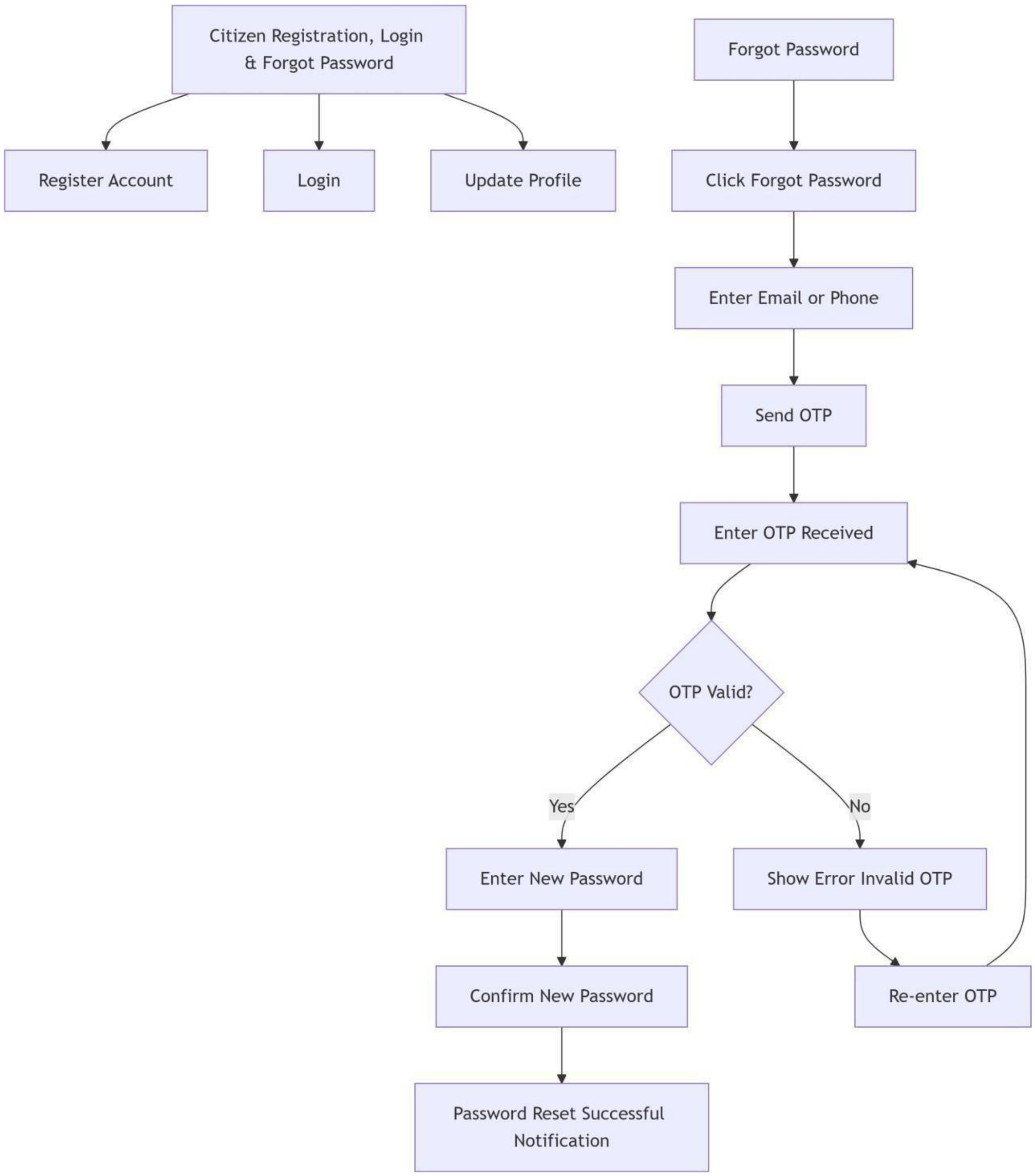



**Figure 11 Citizen registration, login, forget password**

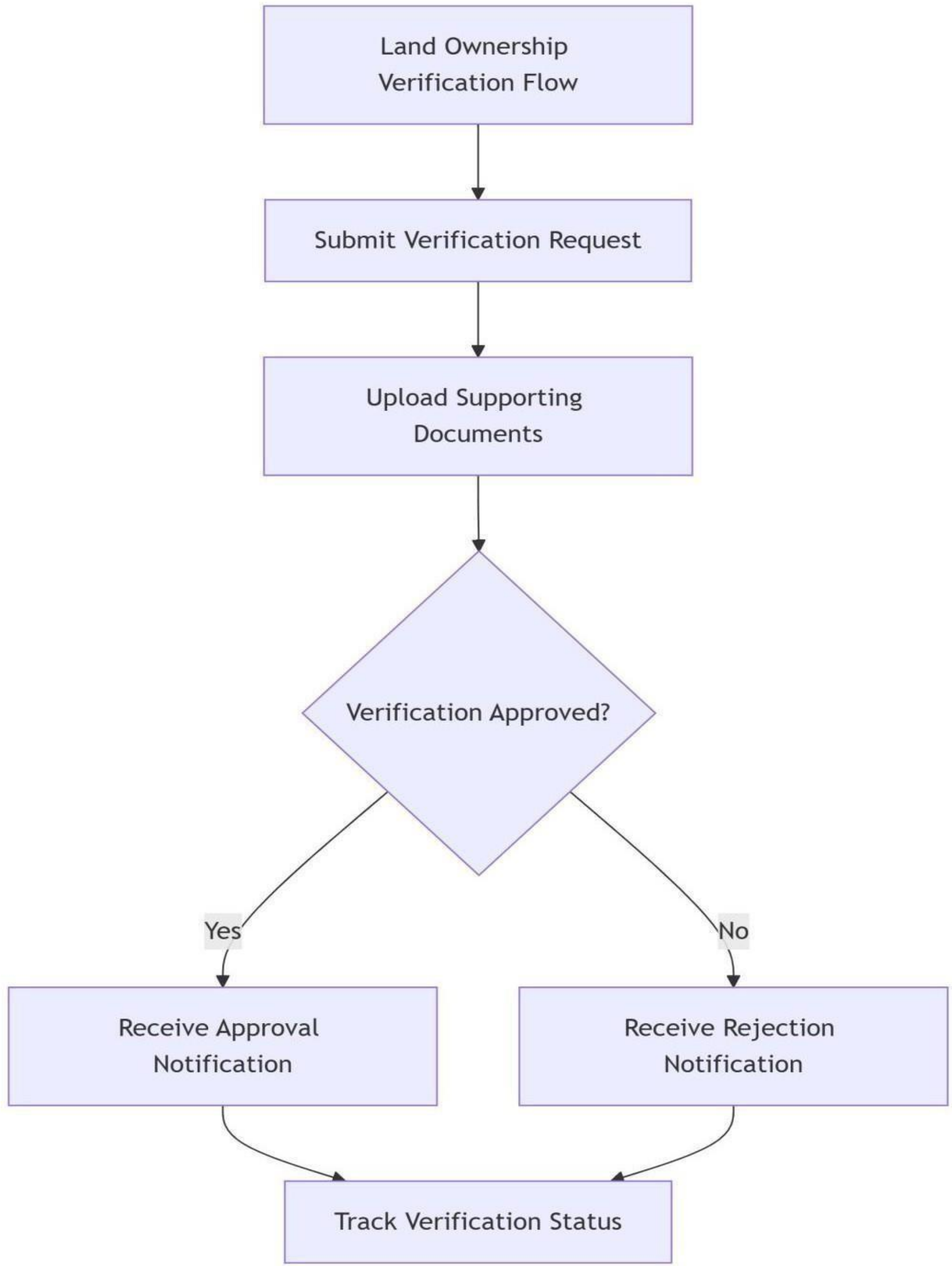



Figure 12 Land ownership verification flow

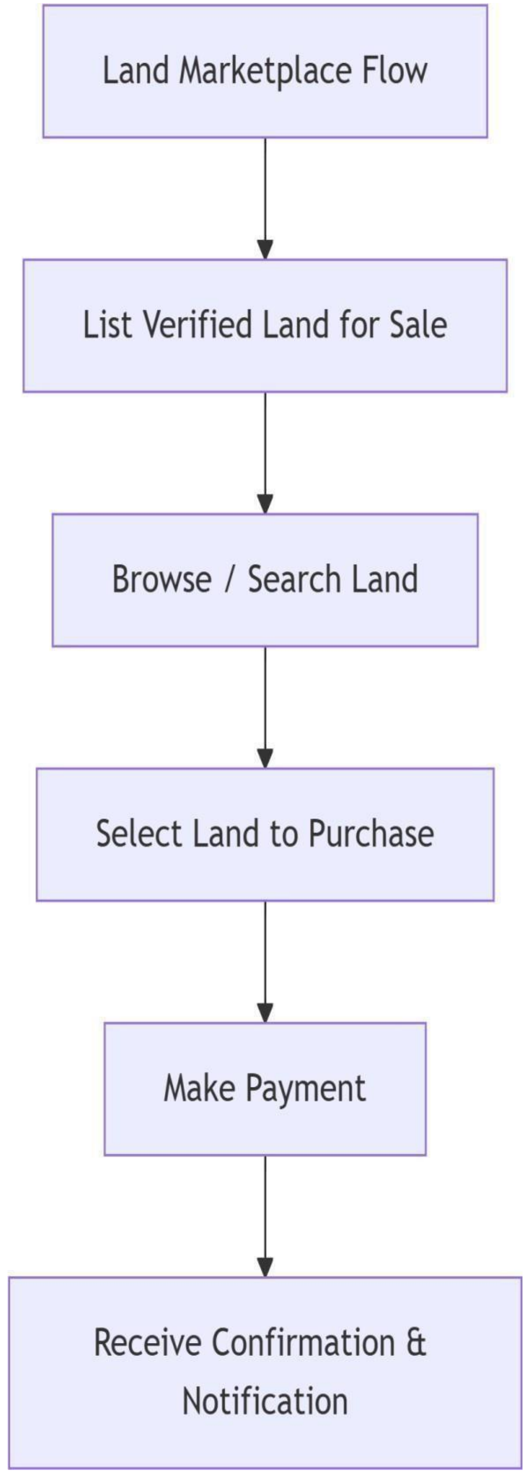



**Figure 13 Land Marketplace Flow**

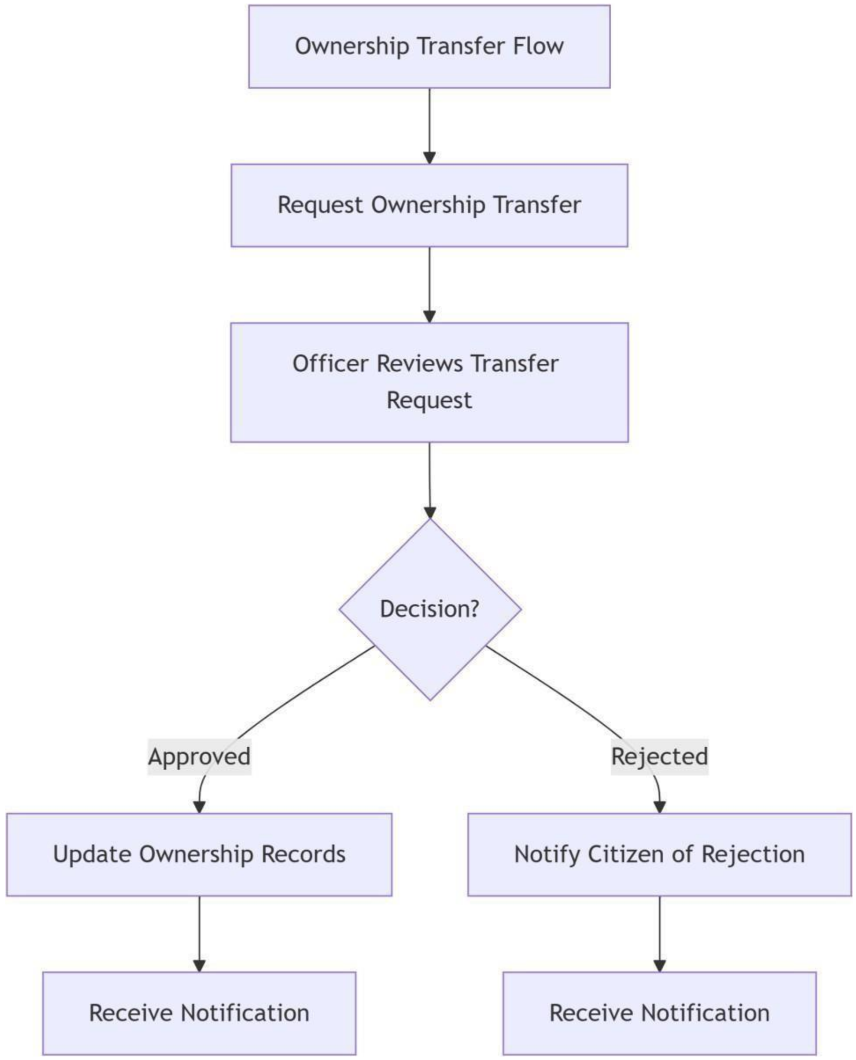



**Figure 14 ownership transfer flow**

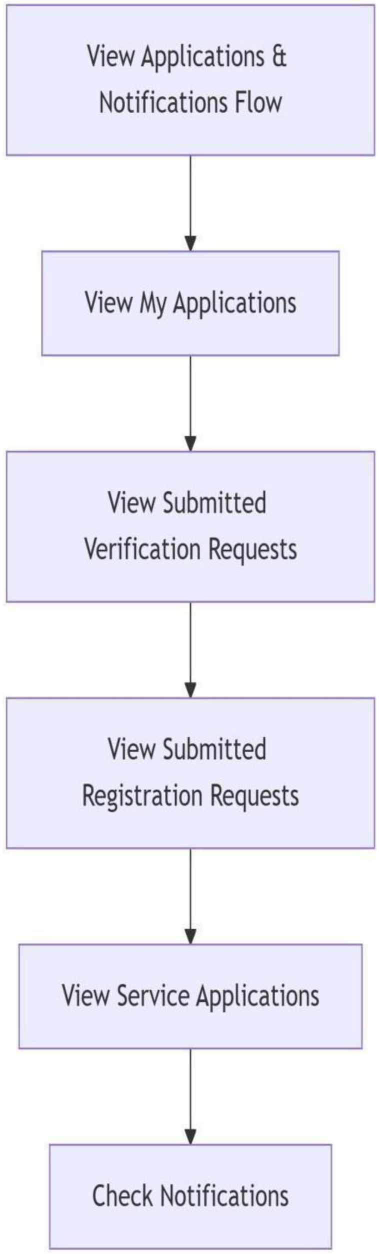



**Figure 15 View application and notification flow**

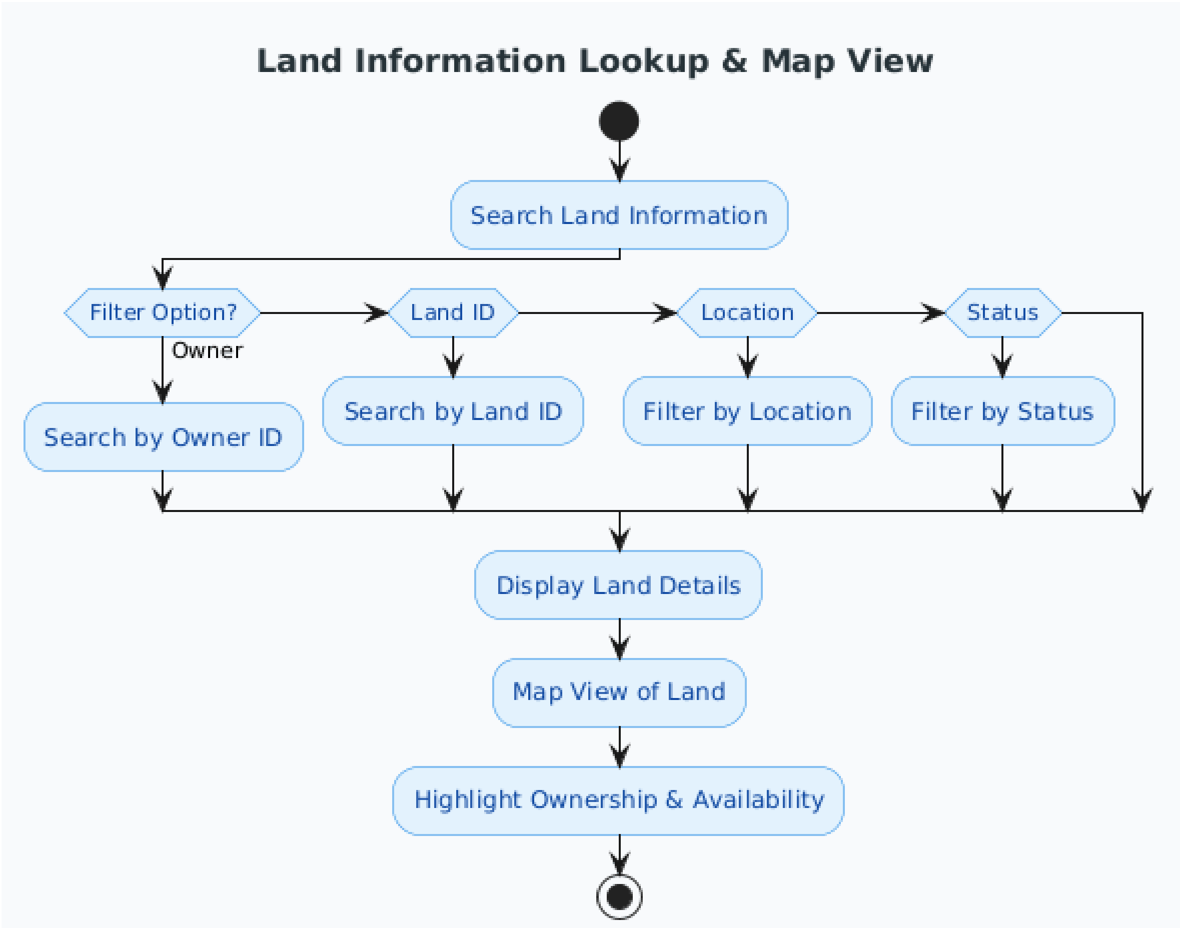



Figure 16 Land information lookup and map view

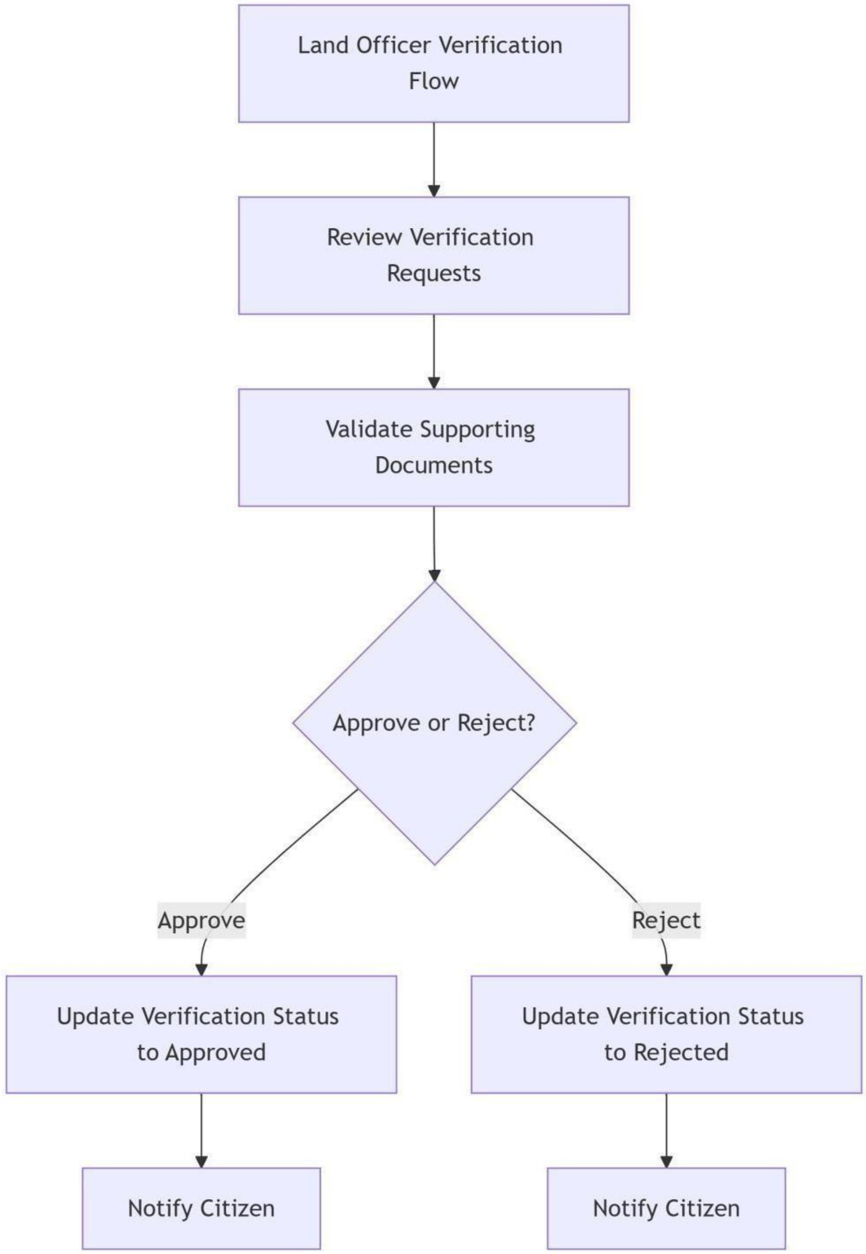



Figure 17 Land officer verification flow

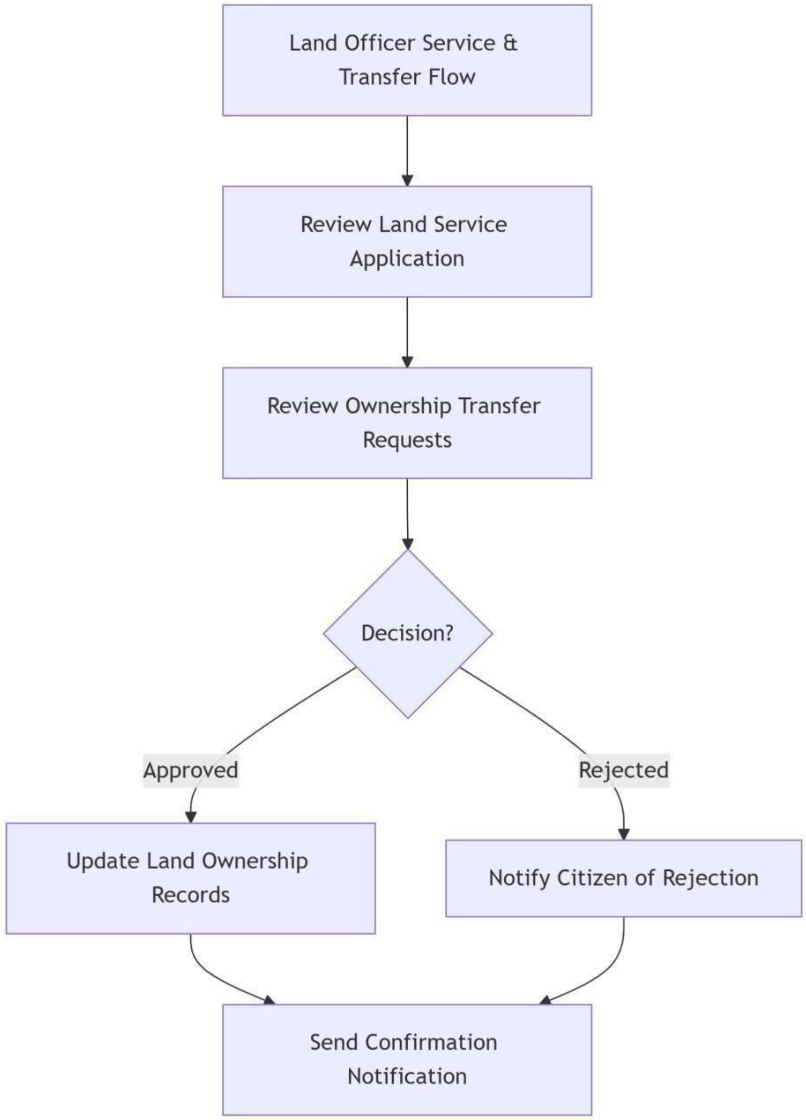



Figure 18 Land officer service and transfer flow

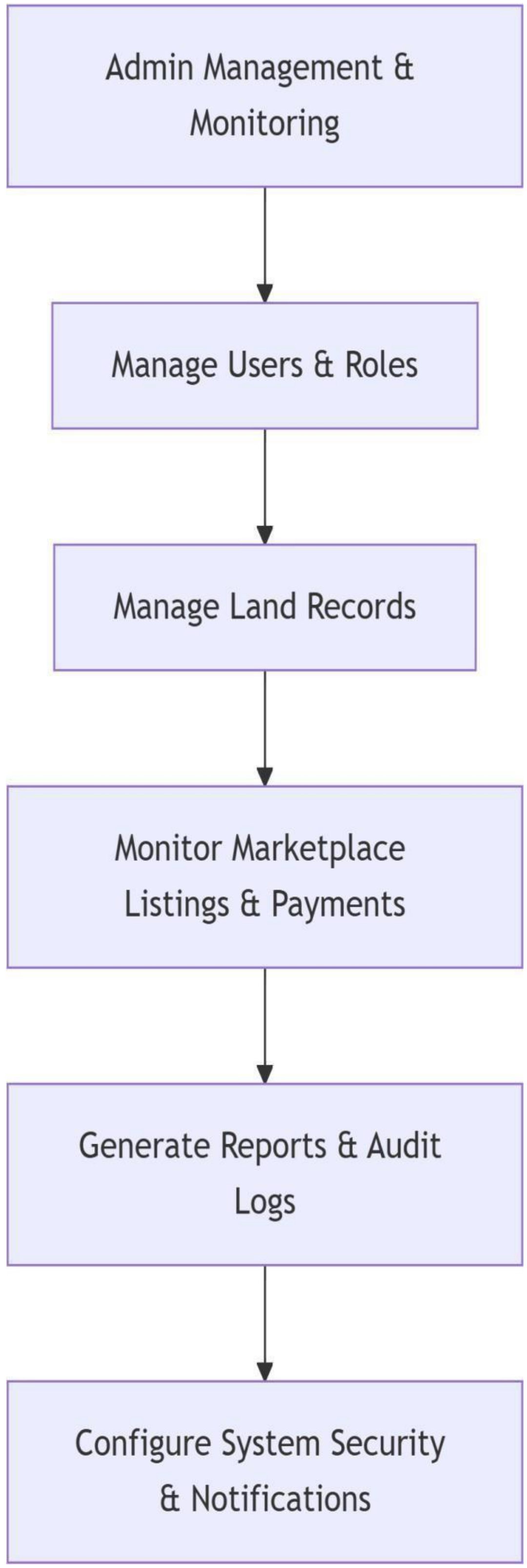



Figure 19 Admin management and monitoring

### *2.3.7 Sequence Diagram*

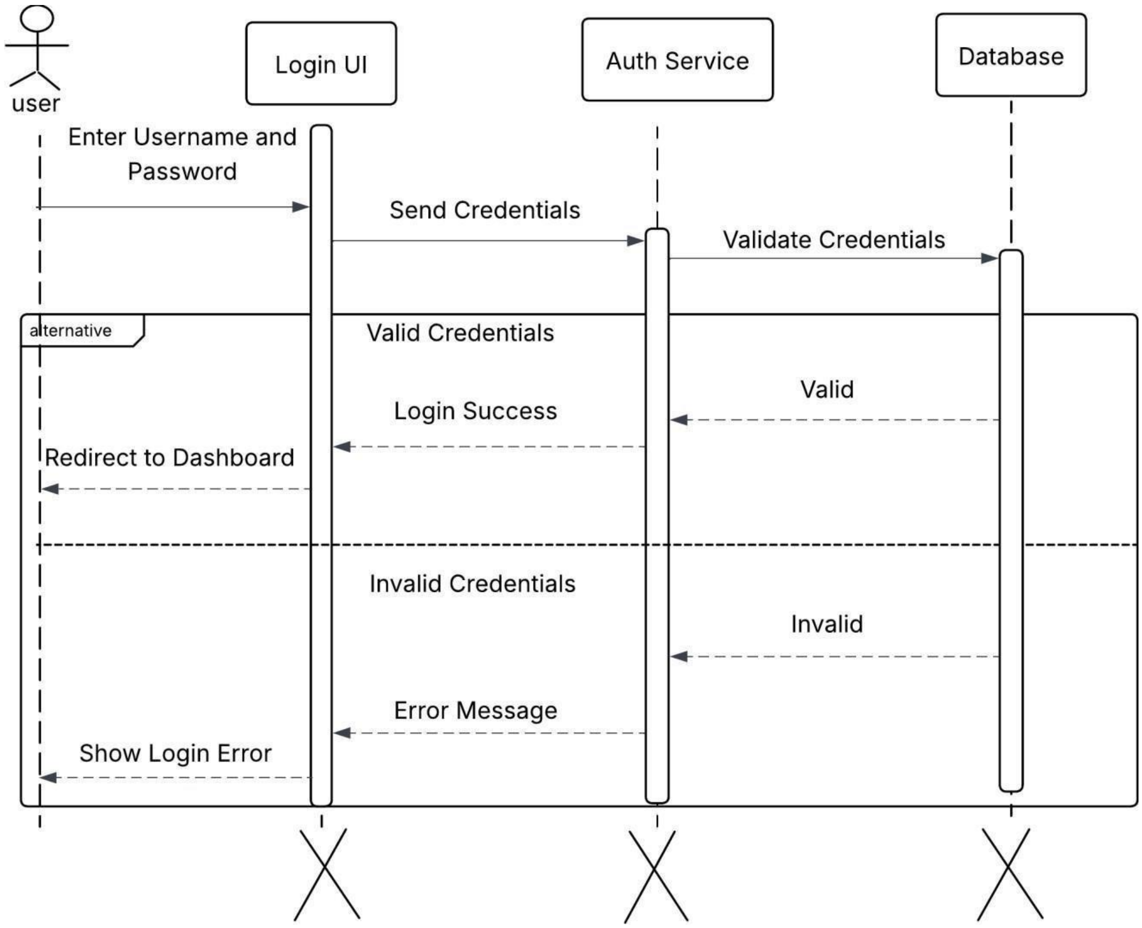



Figure 20 Login Sequence Diagram

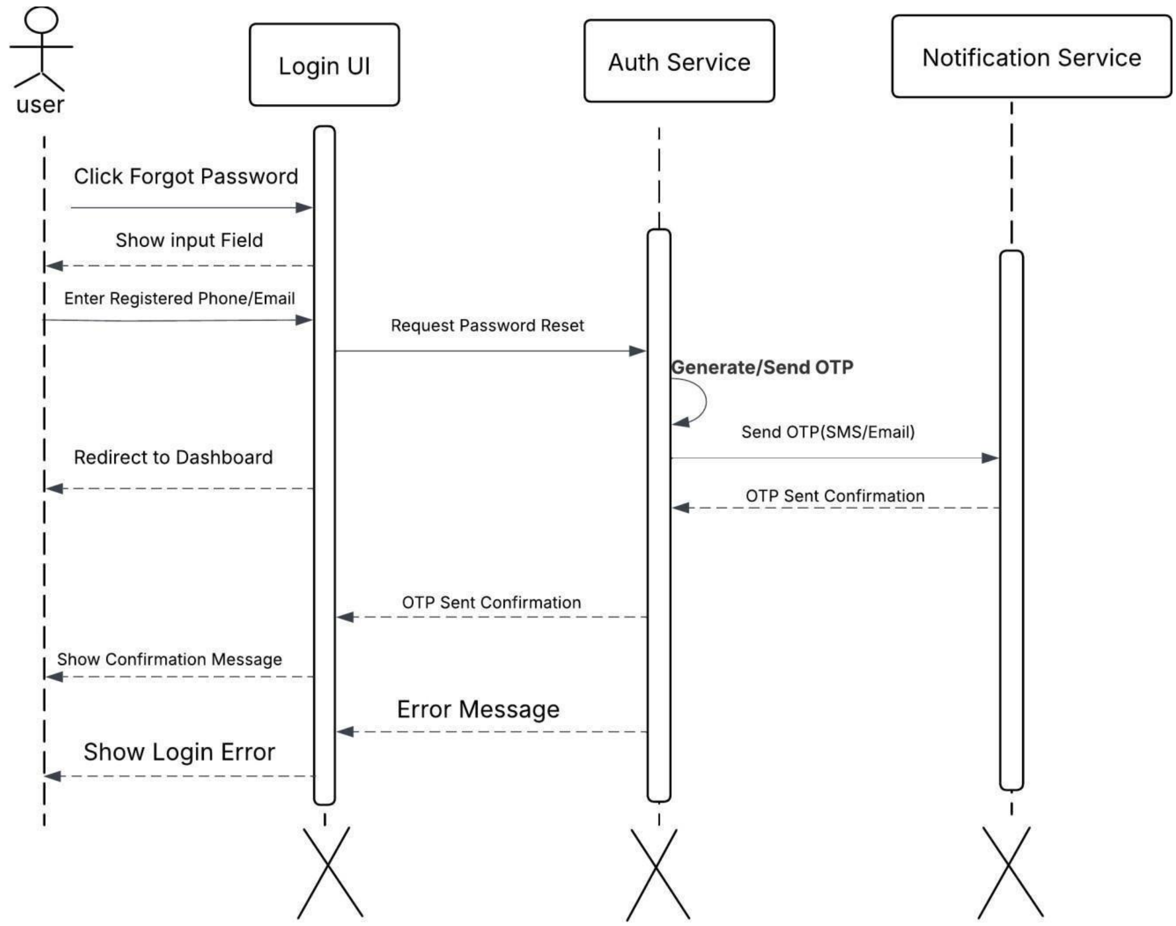



Figure 21 Forget Password Sequence

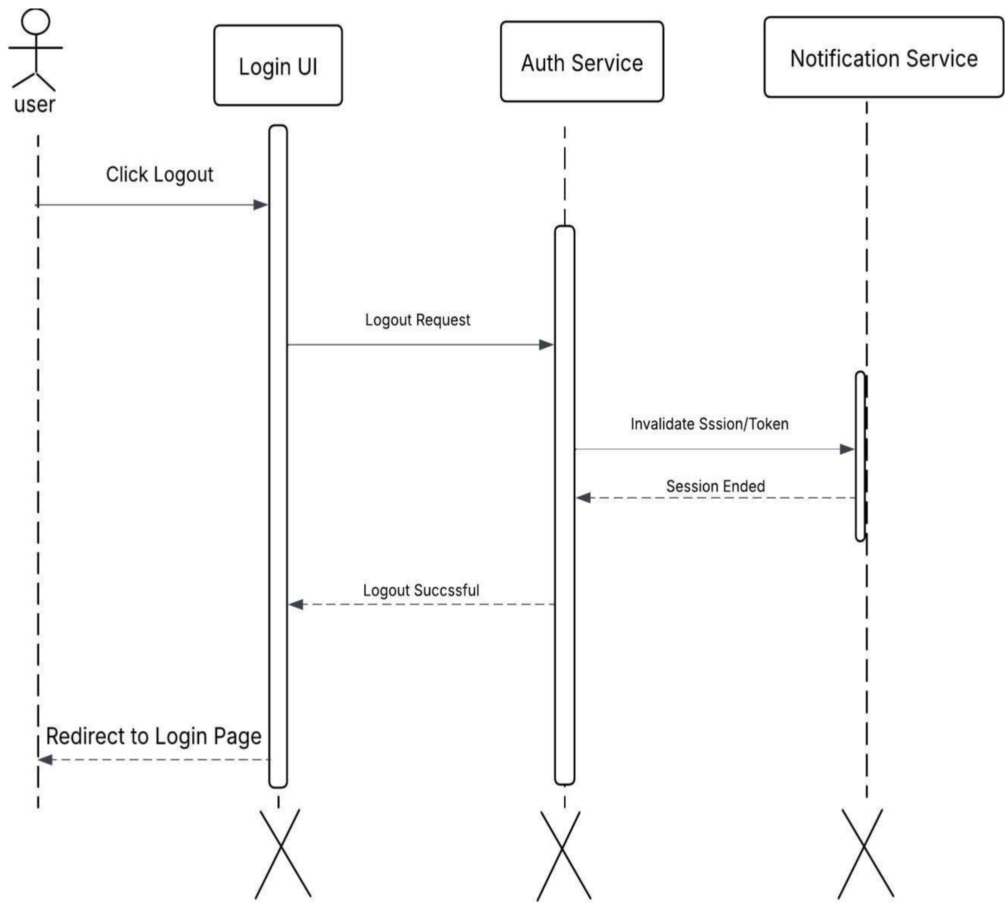



**Figure 22 Logout Sequence**

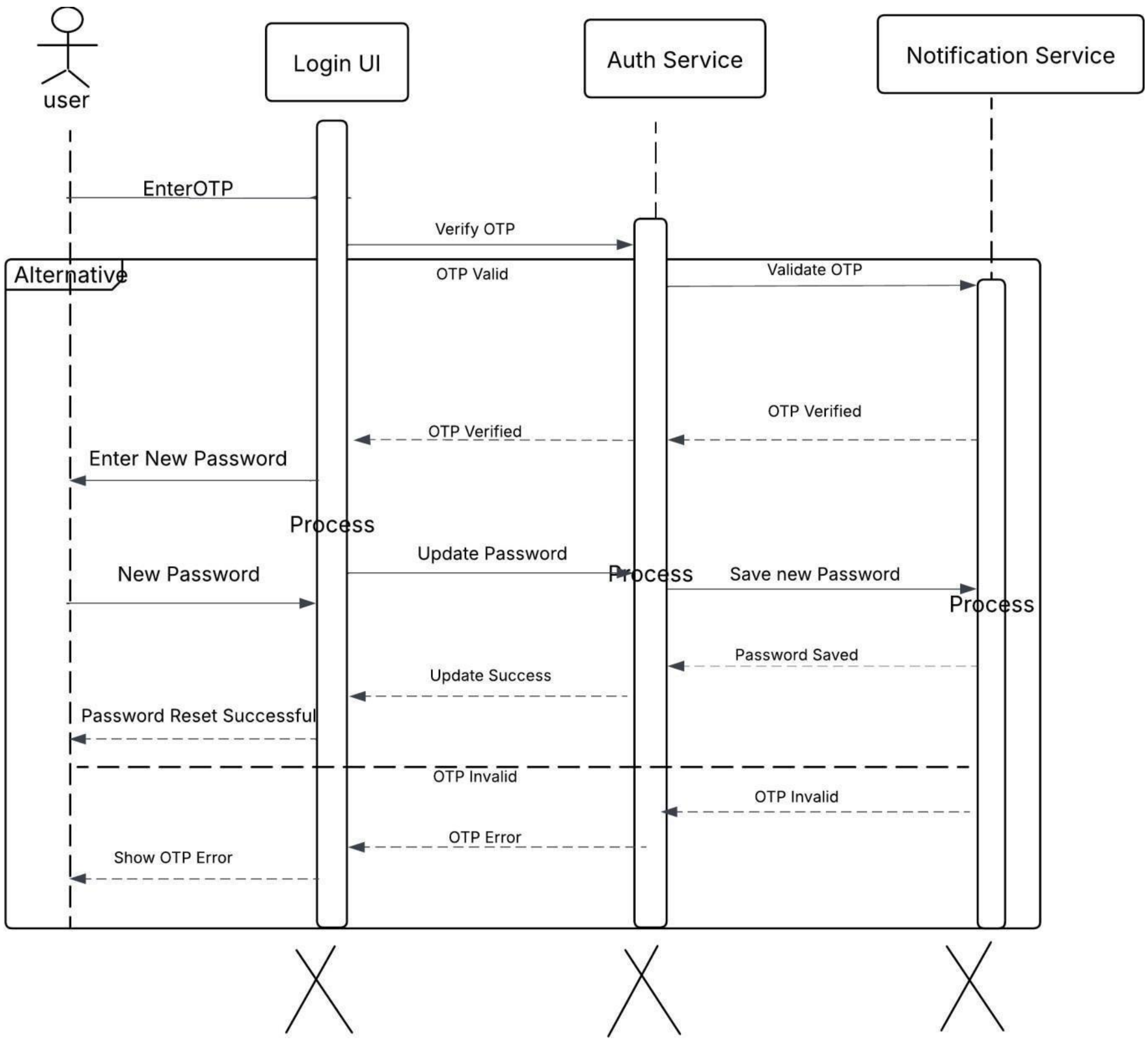



**Figure 23 OTP Verification Sequence**

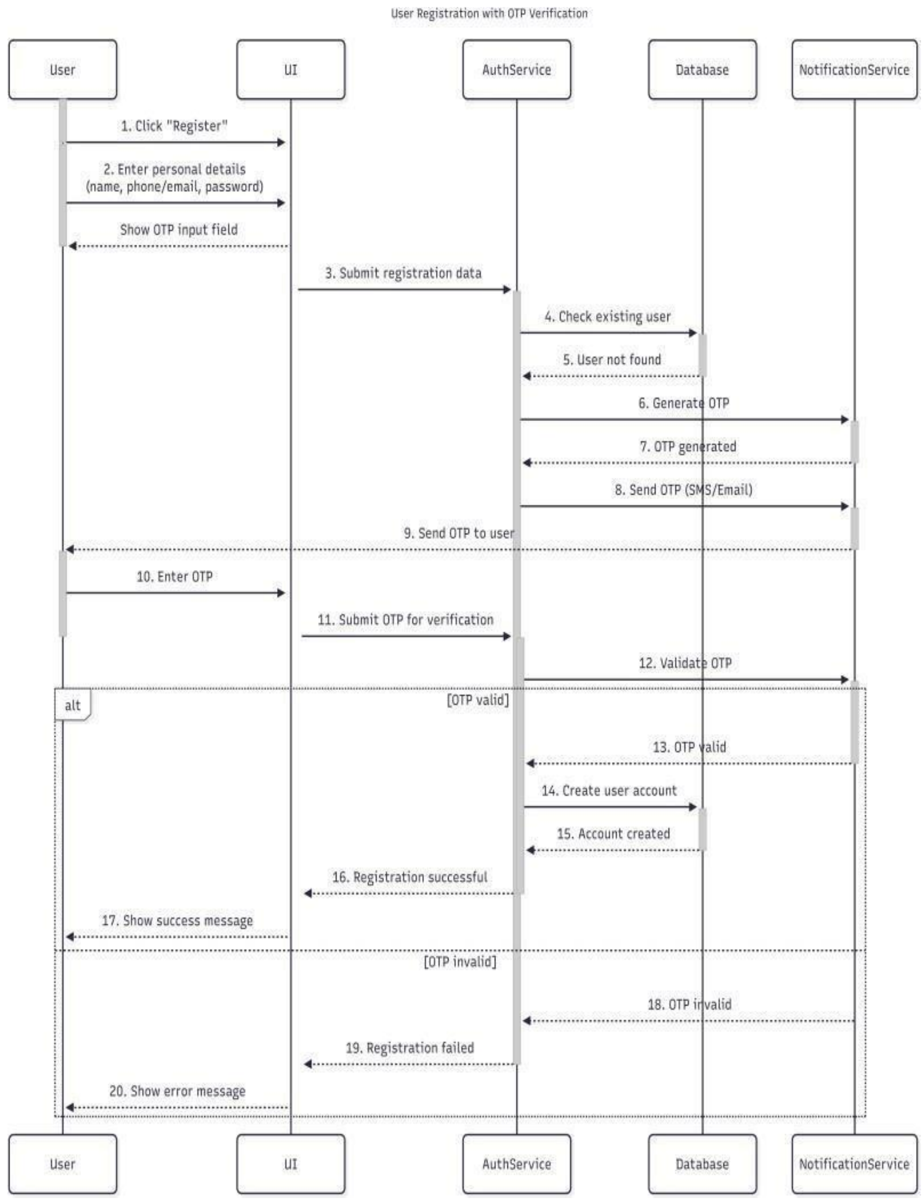



**Figure 24 User Registration Sequence**

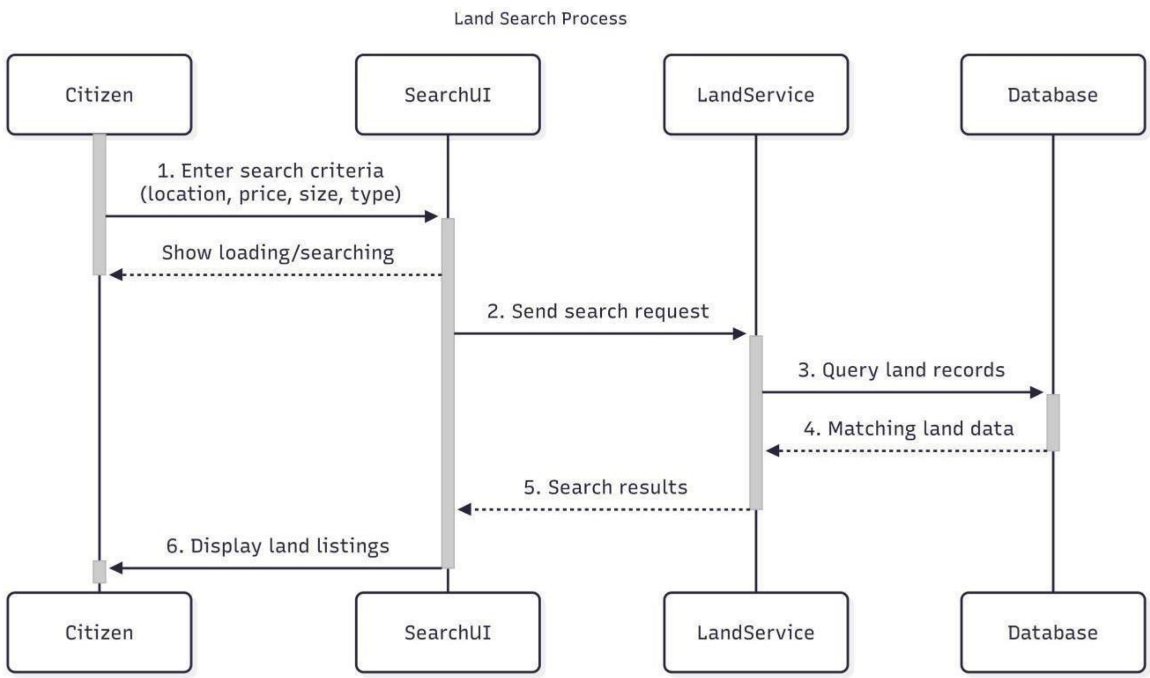



Figure 25 Land Search Sequence

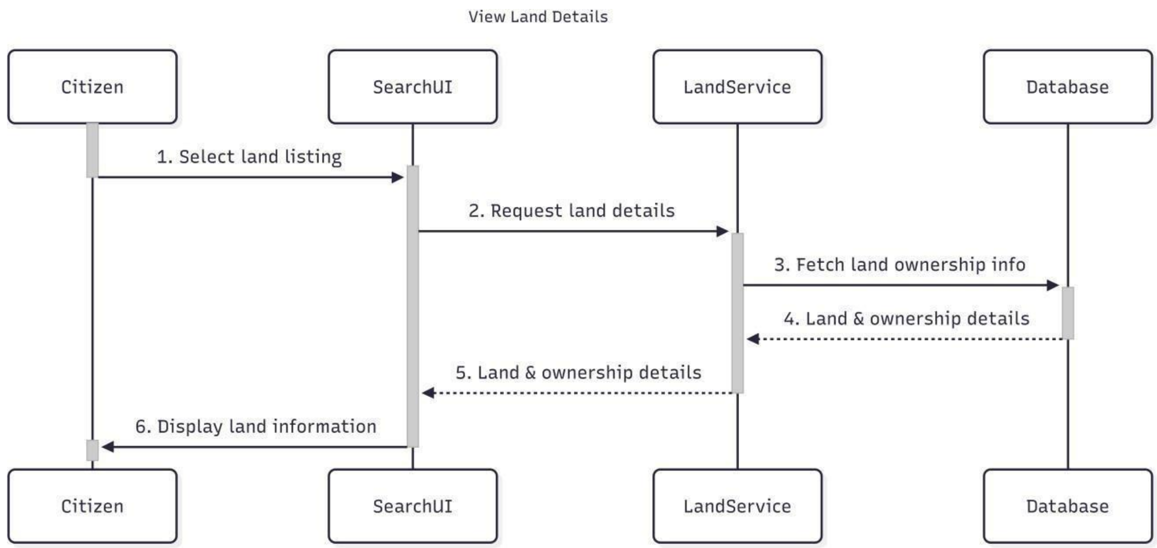



Figure 26 View Land Detail Sequence

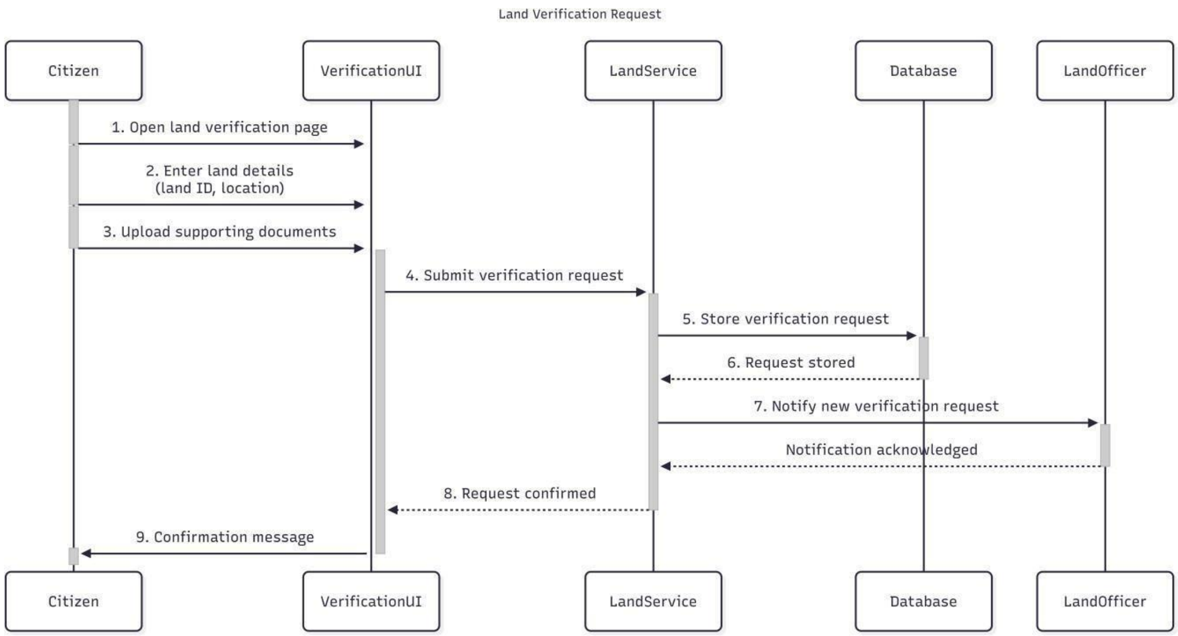



**Figure 27 Submit land Verification Request Sequence**

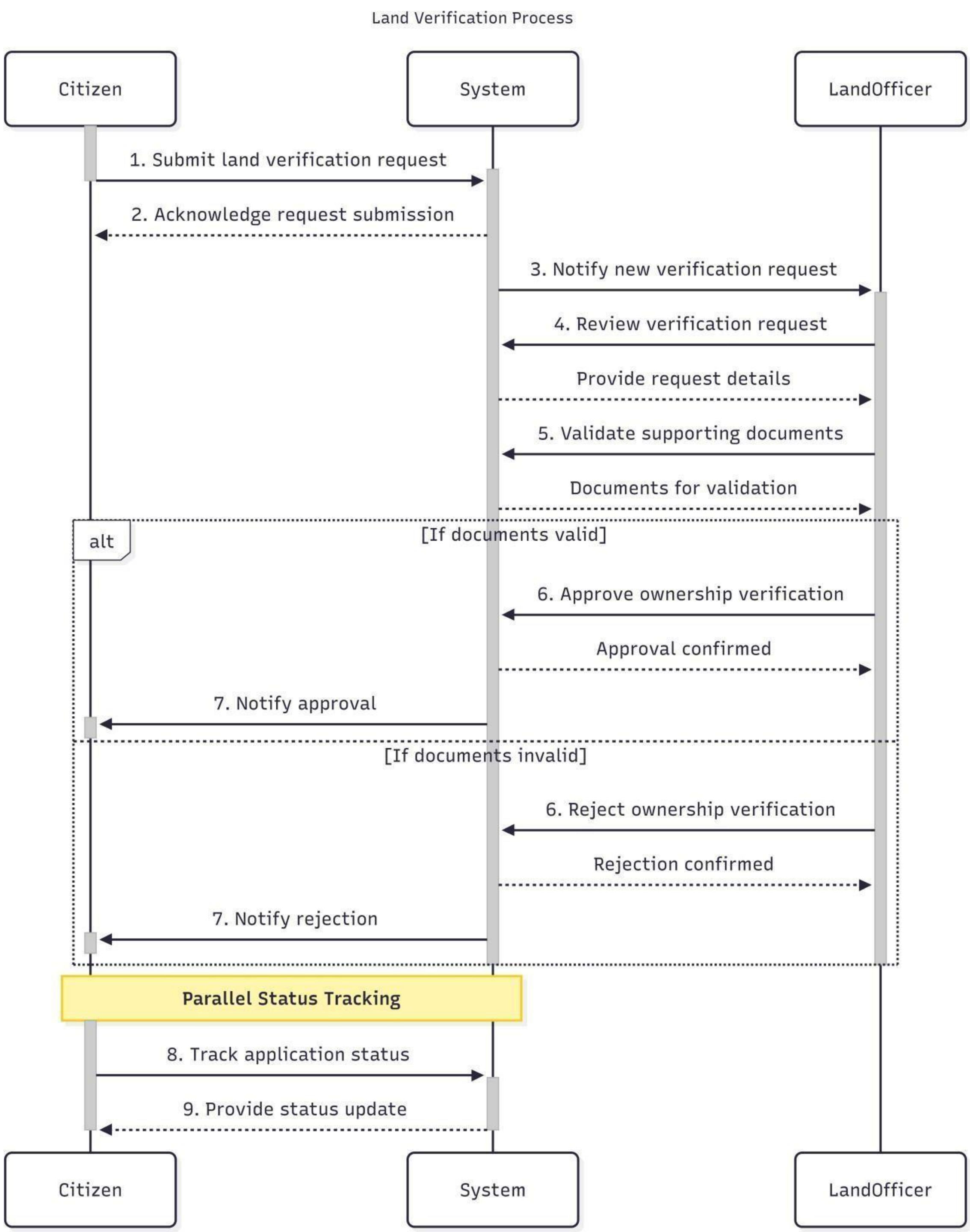



Figure 28 Land Verification (Land Officer)

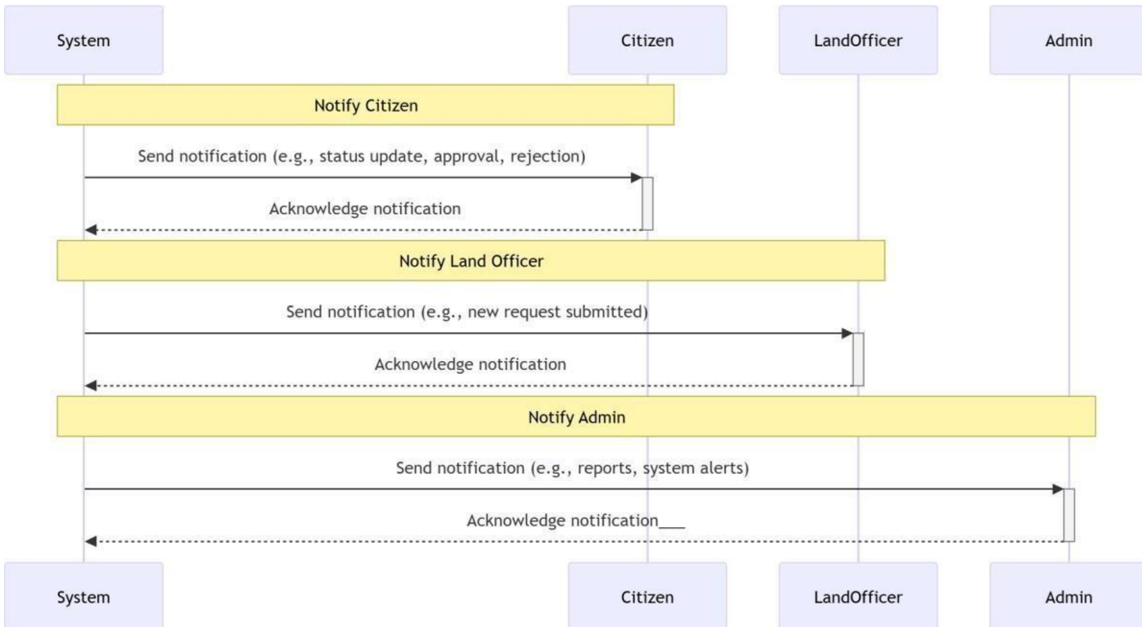



**Figure 29 Notification Sequence**

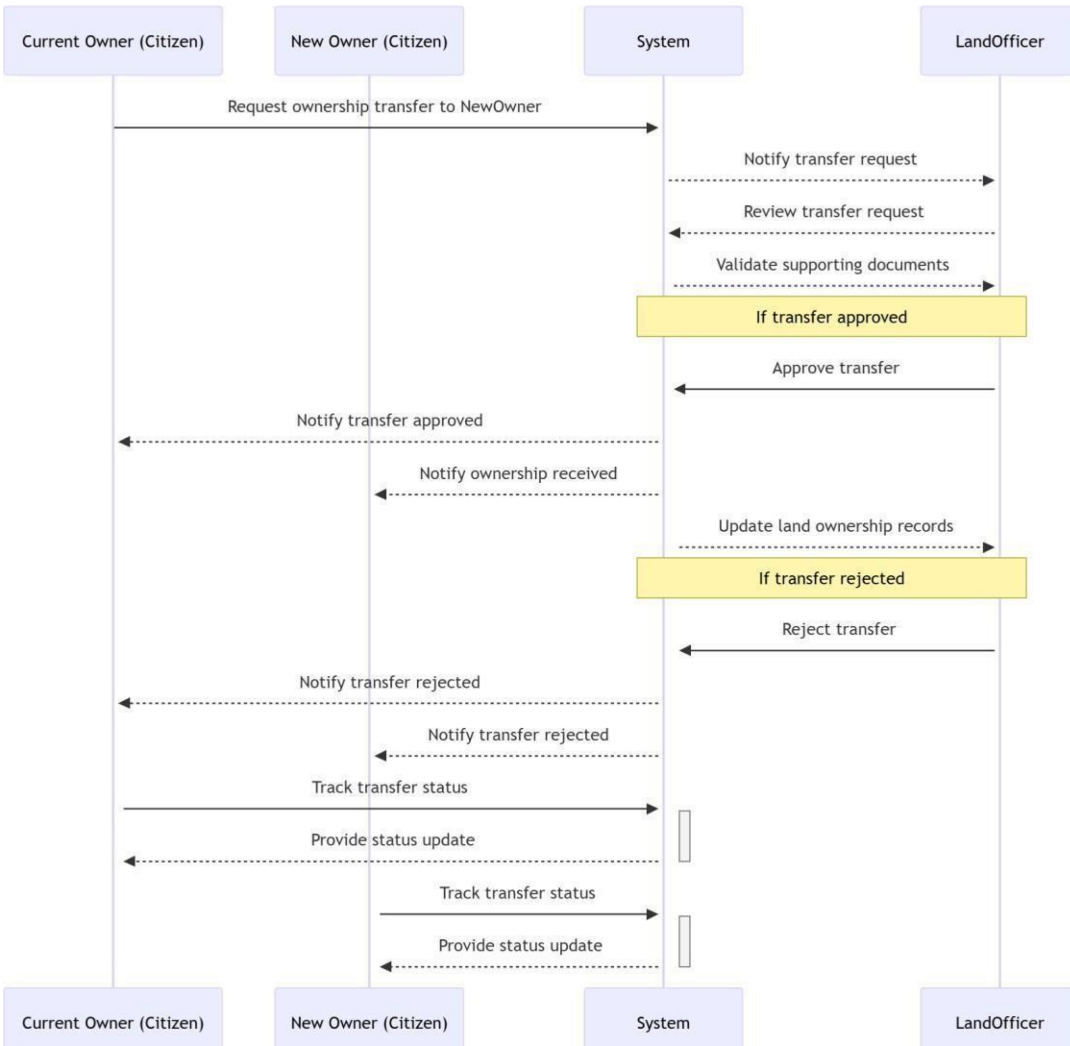



**Figure 30 Ownership Transfer Sequence**

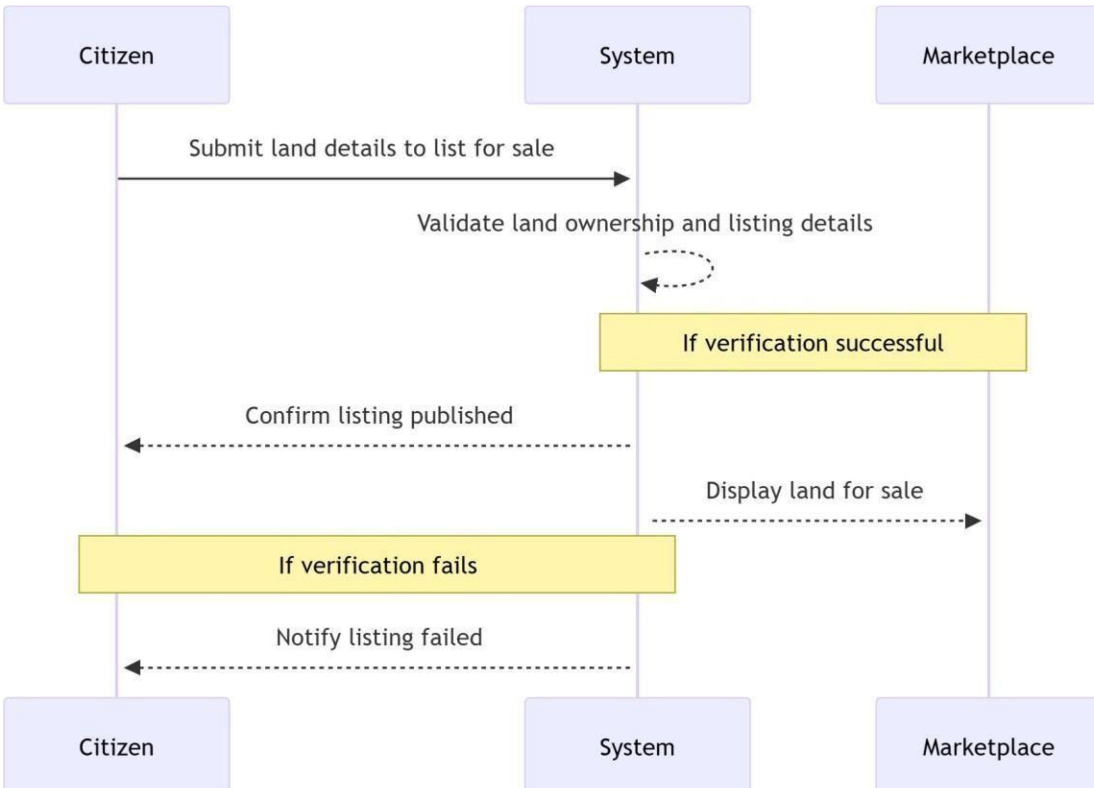



Figure 31 List Land For Sale Sequence

### *2.3.8 Analysis Class Model*

**Classes:**

**Land Owner:**

- **Attributes:** owner ID, name, email, phone, nationalID, address.
- **Methods:** updateProfile(), viewLandRecords(), initiateTransfer(), raiseDispute().

**Land Officer(admin/staff):**

- **Attributes:** officerID, name, department, role, officeLocation.
- **Methods:** verifyOwnership(), updateLandRegistry(), resolveDispute(), manageUsers().

**Land Record:**

- **Attributes:** recordID, parcelNumber, areaSize, locationCoordinates, currentOwnerID, status.
- **Methods:** validateData(), getHistory().

**Document:**

- **Attributes:** documentID, type, uploadDate, ownerID, parcelID.

**Transfer Request:**

- **Attributes:** transferID, sellerID, buyerID, parcelID, status, timestamp.
- **Methods**: submitRequest(), approveTransfer(), rejectTransfer().

**Message:**

- **Attributes:** notificationID, recipientID, message, dateStatus.
- **Methods:** initiate(), send().

**Dispute:**

- **Attributes:** disputeID, parcelID, complainantID, description, resolvedStatus.

**Account:**

- **Attributes:** accountID, username, password, role..
- **Methods:** login(), resetPassword() .

**Payment:**

- **Attributes:** paymentID, amount, transactionDate, payerID, parcelID.
- **Methods:** verifyTransaction().

**FAQ:**

- **Attributes:** guideID, topic, description, category.
- **Methods:** searchGuide().

**Sub-Class Notifications**

**TransferNotification:**

- **Methods:** notifyTransferStatus()

**DisputeNotification:**

- **Methods:** notifyDisputeUpdate()

**VerificationNotification:**

- **Methods:** notifyVerificationResult()

**PaymentNotification:**

- **Methods:** notifyPaymentSuccess()

**Relationships**

**LandOwner:**

- Has many Documents, LandRecords, Notifications, and Disputes.
- Can communicate with LandOfficers regarding verification.
- Can initiate TransferRequests for specific LandRecords.
- Can rate/provide feedback on the registration process.

**LandOfficer:**

- Manages many LandOwners, LandRecords, and Disputes.
- Can update the Registry and verify Documents.
- Can communicate with LandOwners to request additional data.

**System/Admin:**

- Oversees all Accounts, Logs, and System Notifications.
- Can add/remove staff (LandOfficers) and manage system-wide FAQs.

**Relationships**

**Land Owner(Applicant):**

- Has many Documents, land records, notifications, transfer requests, and disputes.
- Can communicate/chat with Land Officer and Admin regarding land verification.
- Can give Feedback to Land Officer on service delivery.
- Can rate the efficiency of the Land Officer.
- Can ask Questions/Inquiries to Land Officer and Admin about land laws and registry status.

**Land Officer (Registry Staff):**

- Has many Documents, reviews, messages, notifications, and land disputes to resolve.
- Can communicate/chat with Land Owner regarding verification and document corrections.
- Can communicate with Admin for system overrides or reporting.
- Can file internal reports and ask Admin for guidance on complex land cases.

**Admin:**

- Has many Land Owners, Land Officers, system messages, and audit notifications.

- Can add/remove Land Officer and Land Owner accounts.
- Has full oversight of the Land Registry database and dispute resolutions.

### *2.3.9 Logic Model*

## Sign Up

**Begin**
Get: Name, Email, Phone Number, Password, User Type (Owner or Officer), National ID
Load Sign Up Page
Enter Name, Email, Phone Number, Password, and National ID Details
Click the Sign Up button

If (Form is filled)
If (Valid input values)
If (Email/Phone Number does not exist in the database) Hash the password
Save details to the database based on user type
Send verification link or OTP
Return: Success message Else
Return: Email/Phone Number already exists
ENDIF
Else
Return: Invalid input values ENDIF
Else
Return: Empty submission, please fill valid input
ENDIF
**End**

## Sign In

**Begin**
Get: Login Email and Phone Number Get: Login Password
Load Login Page
Enter Email and Phone Number Enter Password
Click the Login button

If (Form is filled)
If (Valid input values)
If (Email/Phone Number and Password match records in the database) Generate session/authentication token
Redirect to the appropriate page based on user role Else
Return: Login Failed ENDIF
Else
Return: Invalid Input Value ENDIF
Else

Return: Empty submission, please fill valid input ENDIF
**End**

## Search Land Record

**Begin**
Get: Search Filters (e.g., Parcel Number, Owner Name, Location )
Load Search Page
Enter filters and Click Search
If (Filters are provided) If (Valid filters)
Query the land registry database with filters Sort and paginate results
Display a list of matching land records with their Ownership Status Else
Return: Invalid filters, please refine your search ENDIF
Else
Return: No filters provided, displaying all Records ENDIF
**End**

**Owner-OfficerCommunication Begin**

Land Offiicer's Profile
Click "Message" or "Request"
**For Messaging:**
Get: Message
Content Click
Send

If (Message Content is
provided) Save message
to the database
Notify the recipient of the new
message Display: Message sent
successfully Else
Return: Empty message, please type
content ENDIF

**Forgot Password**

**Begin**
Get: Email/Phone
Number Load Forgot
Password Page Enter
Email/Phone Number
Click Reset Password

If (Email/Phone Number is provided)
If (Email/Phone Number exists in the
database) Generate reset token/OTP
Send token/OTP to the
user Load Reset
Password Page

sql

Get: New Password
Enter New Password and Confirm Password
Click Submit

If (New Password matches requirements)
Update the database with the new hashed password Invalidate token/OTP
Return: Password reset successfully Else
Return: Invalid Password, please meet security requirements ENDIF

Else
Return: Email/Phone Number does not
exist ENDIF
Else
Return: Empty submission, please provide valid input
ENDIF
**End**

## 2.4 Non functional requirements

**User Interface and Human Factors**

The platform will have a user-friendly interface that is easy to navigate, even for users with limited technical skills .We will use clear and intuitive design elements, such as maps, search bars, and parcel filters, to help users find land records and connect with land officers efficiently. Documentation Maintaining comprehensive documentation throughout the entire development process was a crucial aspect of our project .This ensured transparency, facilitated collaboration, and enabled future maintenance and updates.

**Documentation**

Maintaining comprehensive documentation throughout the entire development process was a crucial aspect of our project. This ensured transparency, facilitated collaboration, and enabled future maintenance and updates.

**Documentation Tools**

**Google Docs:** We used Google Docs for collaborative editing and version control of key documents.

**Git hub**: We used Git hub for code version control and documentation. Every code commit was linked to relevant requirements and documentation within Click Up, ensuring traceability and transparency throughout the development lifecycle.

**Mermaid chart , Plant UML and Lucid app,:** We used this tools for preparing diagrams.

**Performance Characteristics**

1. **Enhanced Speed and Responsiveness :**The platform will leverage advancements in technology to deliver even faster and more responsive user experience. Users will be able to access land titles and communicate with registry staff in real-time, ensuring an instantaneous and seamless interaction.

2. **Advanced Reliability and Availability:** The platform will employ cutting-

edge infrastructure and hosting solutions to ensure unparalleled reliability and availability. Users can rely on the platform to be accessible 24/7, with minimal to no downtime, providing Uninterrupted access to ownership verification whenever they need it.

3. **Scalability for Growing Demand:** With the projected increasein users and land transfers, the platform will be designed to handle significant growth in traffic and user base. It will seamlessly scale its resources to accommodate a large number of users and title searches Simultaneously, ensuring uninterrupted service as demand Expands.

4. **Enhanced Security and Privacy Measures:** The platform will employ advance dsecurity measures to safeguard land records and maintain owner-registry confidentiality. This includes implementing state-of-the-art encryption techniques, robust authentication protocols, and proactive threat measures to protect against common cyberse.

**Error Handling and Extreme Conditions**

Effective error handling is the cornerstone of a robust and reliable platform. It ensures that when things go wrong, the platform gracefully recovers, minimizing disruption and providing users with a clear understanding of the issue.

Error Handling Strategies:

- Input validation: Implement strong input validation to prevent users from submitting invalid data, reducing the likelihood of errors.
- Exception handling: Use appropriate exception handling mechanisms to catch and handle unexpected errors gracefully, preventing them from crashing the platform.
- Fail gracefully: The platform will strive to remain functional even when errors occur, providing users with clear and helpful feed back instead of crashing or displaying cryptic error messages.
- Log errors: All errors will be logged with relevant information, including the time, type of error, user context, and any associated data, to facilitate troubleshooting and root cause analysis.
- Classify errors: Errors will be categorized based on their severity(critical, high, medium, low) and impact on user experience.
- Implement error messages: We’ll provide informative and actionable error

messages that explain the problem in simple language and suggest solutions or workarounds.

- Use error codes: Consistent error codes will be used to identify the specific type of error, allowing for easier debugging and troubleshooting.
- Monitor errors: We'll regularly monitor error logs to identify recurring issues, prioritize fixes, and improve the platform's overall stability.

**Quality Issues**

The system should be available 24 hours a day and 7 days a week. It is accessed by multiple users at a time and this will make sure that the operations and errors of one user will not affect the other or the performance of the entire system by using the error handling system.

**Security Issues**

As we develop our platform, it is crucial to consider potential security issues that could arise and implement solutions to address them. Here are some key areas we focused on

**Secure User Authentication:**

**Issue**: Weak passwords and compromised credentials are major security risks. Users often choose predictable passwords or reuse them across various platforms, making them vulnerable to hacking attempts.

Solutions**:**

- Enforce strong password requirements, including minimum length, complexity rules, and regular password changes..

**Vulnerability Management:**

**Issue**: Software vulnerabilities can provide attackers with opportunities to exploit the platform and gain access to user data.

**Solutions:**

- Keep all software up to date, including third-party libraries.
- Implement a secure coding practices program to minimize vulnerabilities introduced during development.

1. Secure Communication Channels:

- **Issue**: Unencrypted communication channels expose sensitive information to interception and eavesdropping.

**Solutions:**

- ○ Use HTTPS protocol for all communication channels to encrypt data in transit between users and the platform.
- ○ Implement TLS/SSL certificates for secure communication and to prevent man- in-the-middle attacks.

**Secure API Access:**

- **Issue**: APIs can provide access to sensitive data and functionality if not properly secured.

**Solutions:**

- ○ Implement API authentication and authorization mechanisms to control access and prevent unauthorized use.
- ○ Use rate limiting to prevent malicious use of APIs..
- ○ Monitor API activity logs and investigate any suspicious behavior

**Resource Management**

Efficient Resource Management Efficient resource management is integral to the optimal functioning of our platform encompassing CPU, memory, and database utilization. To ensure a responsive user experience, it's crucial to implement strategies that effectively allocate and monitor these resources.

**Constraints or Pseudo Requirements**

Constraints or Pseudo Requirements We have identified two primary constraints that may arise. Firstly, there is a potential challenge due to the poor internet connection in the country, making real-time parcel map loading on our system difficult. Secondly, there is no clearly defined law specifying whether digital land certificates are fully equivalent to physical deeds, considering the local administrative standards.

**System requirements**

***Hardware requirements***

- Processor: Minimum 1 GHz; Recommended 2GHz or more
- Hard Drive: Minimum 32 GB; Recommended 64 GB or more
- Memory (RAM): Minimum 2 GB; Recommended 4 GB or above

### *3.1.1 Software requirements*

**Supported Browsers**: People often ask what browser they should use. There is no single answer for this. Use whichever browser works best on your computer. However, we recommend downloading Firefox and chromium browsers.

Firefox, Chrome, Opera, Microsoft Edge

**Recommended Operating Systems Windows: 10 or above**

MAC: OS X v10.7 or higher

Linux: Ubuntu or other Linux distros that support node version 18

## 2.5 Key abstractions with CRC analysis

### *2.5.1 Conceptual modeling: Class diagram*

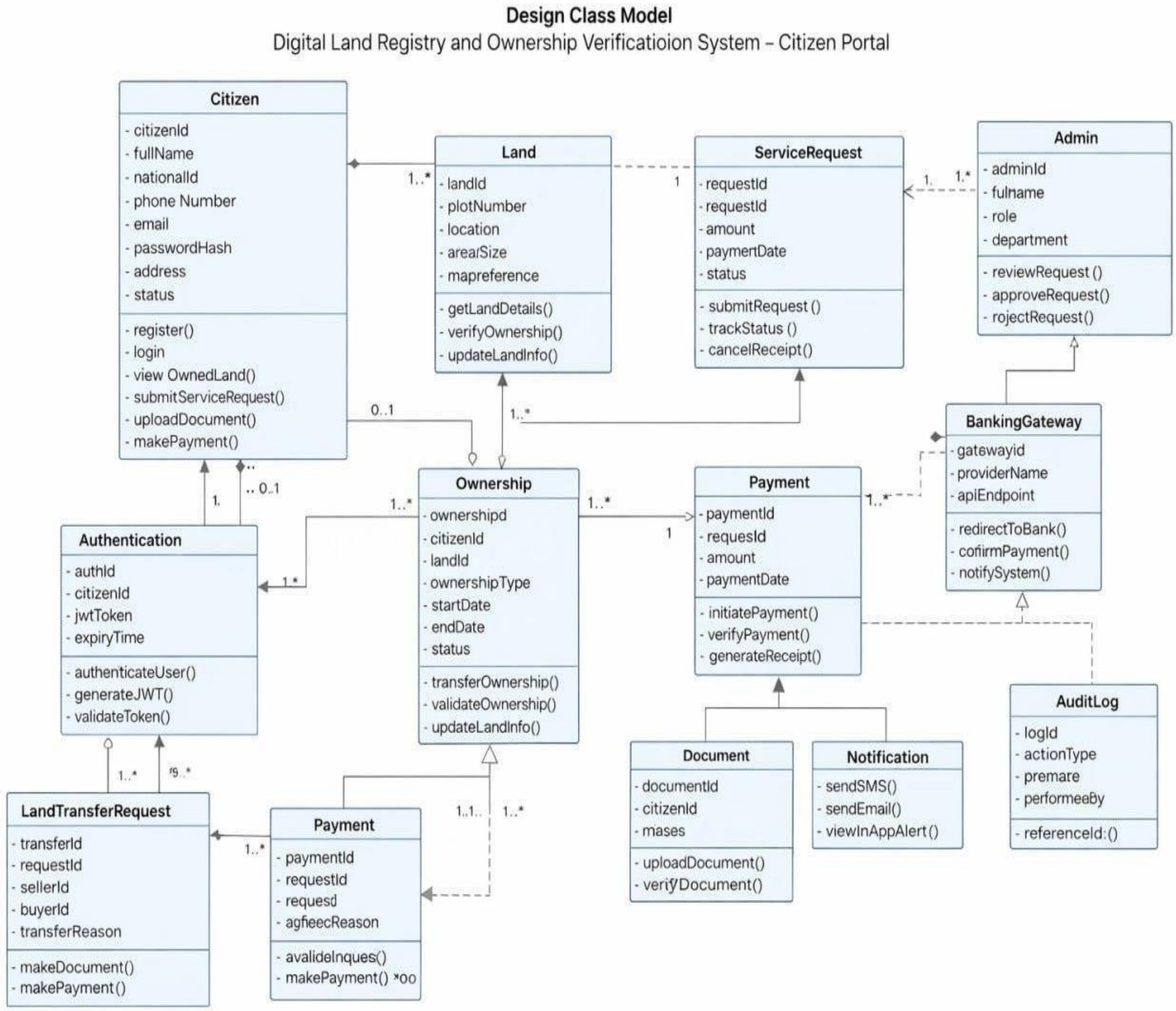



Figure 32 Class Diagram

### 2.5.2 *Identifying change cases*

**Change Case: Blockchain Integration**

**Change**: Utilizing blockchain technology for secure and transparent document storage, payment processing, and record-keeping.

**Likelihood**: Medium. Blockchain technology has the potential to revolutionize legal processes, but its widespread adoption may take time.

**Impact**: High: Enhanced security and transparency, improved trust between clients and land owners, reduced fraud.

Moderate: Complexity in implementing and maintaining blockchain-based systems.

Low: Potential for regulatory hurdles and challenges in integrating blockchain with existing legal frameworks.

**Change Case: Multilingual Support**

**Change**: Expanding language support to cater to a diverse client base.

**Likelihood**: Medium. Depending on the target market, demand for multilingual support may vary.

**Impact**: High: Increased accessibility for clients from diverse linguistic backgrounds, expanded market reach.

Moderate: Increased translation and localization costs.

Low: Potential for language-related errors and misunderstandings if not implemented with care.

*User Interface Prototyping*

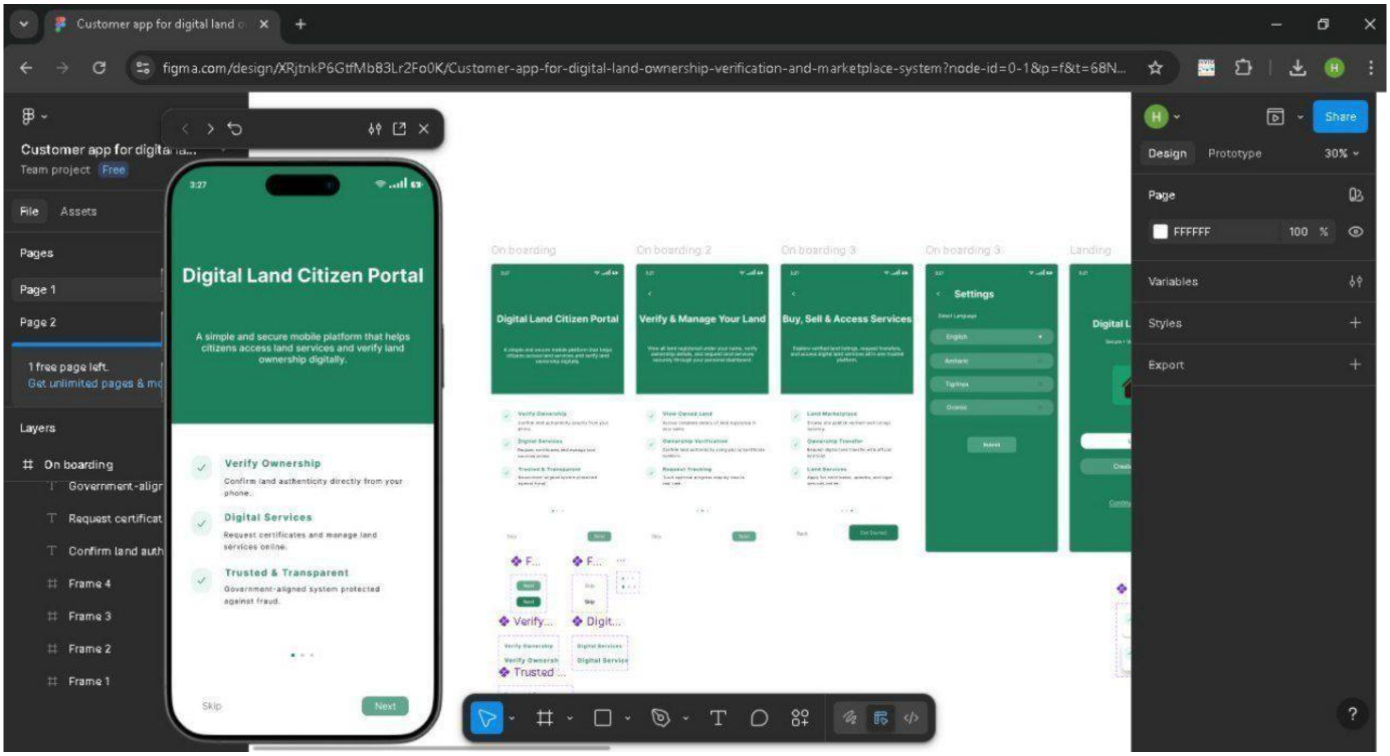


Figure 33 User Interface Prototype Onboarding 1

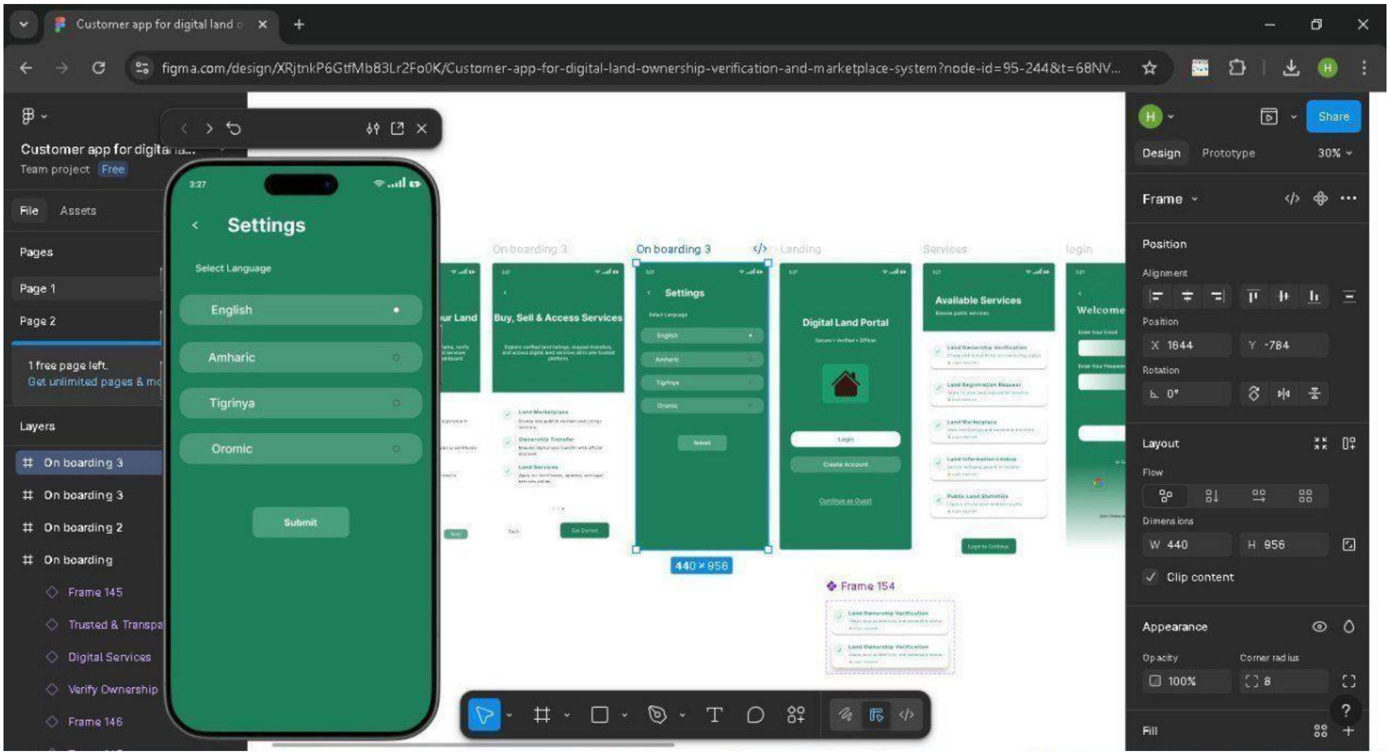



**Figure 34 User Interface Prototype Language Option**

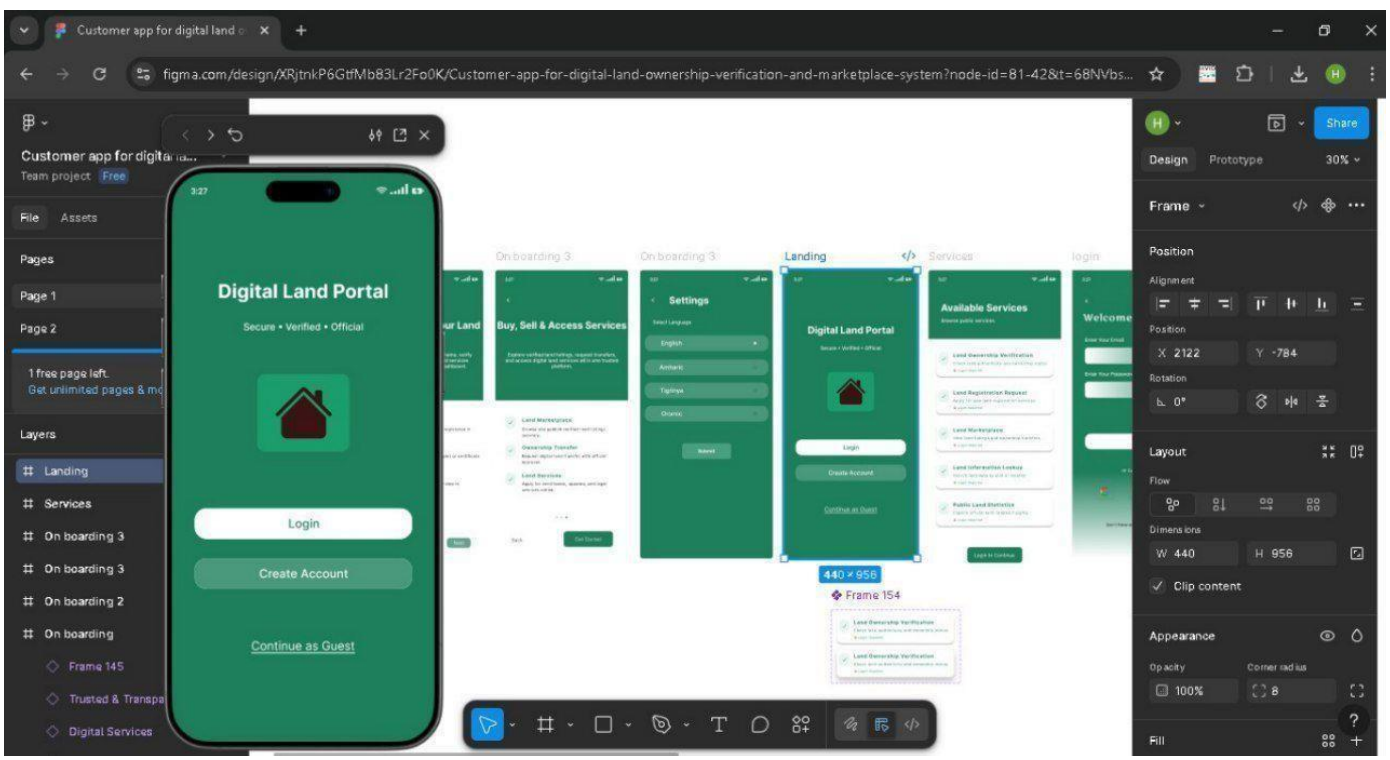



**Figure 35 User Interface Prototype Portal Home**

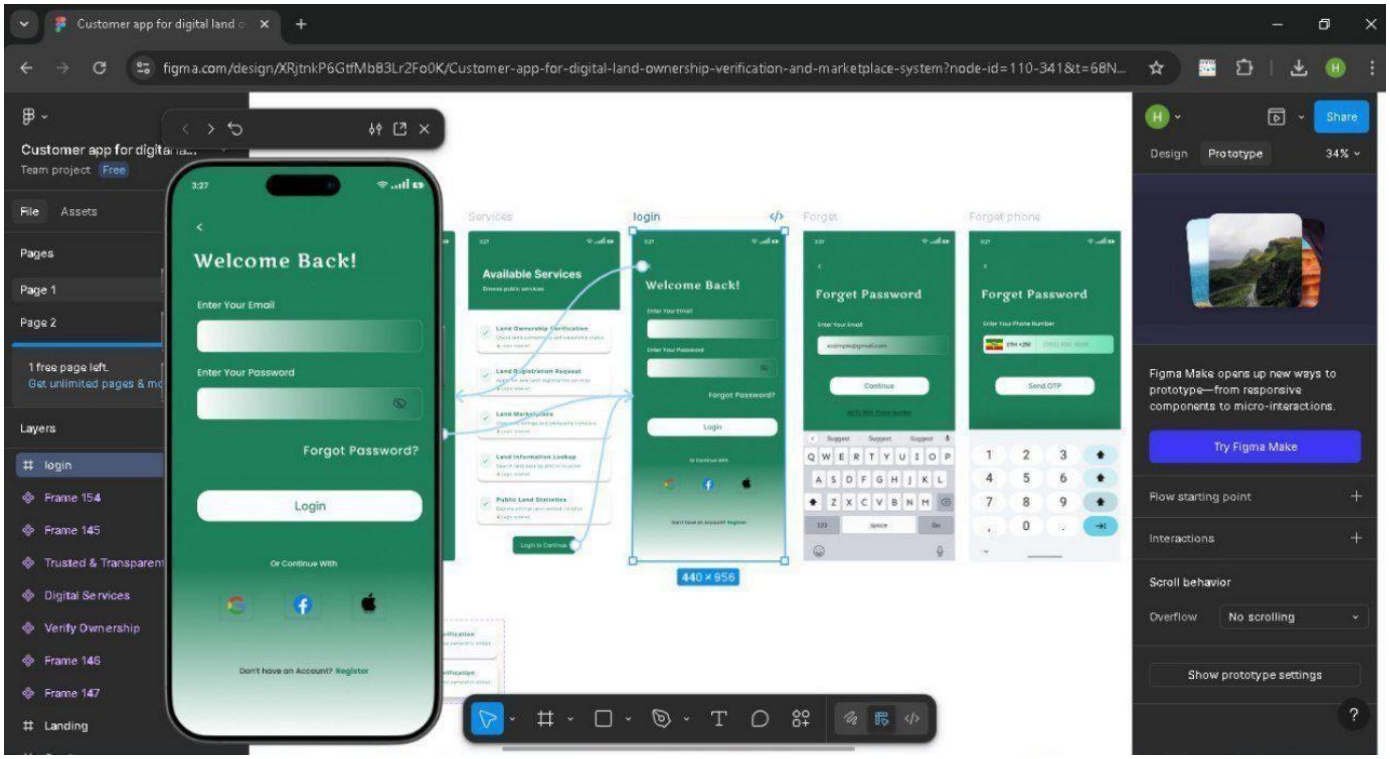


**Figure 36 User Interface Prototype Login**

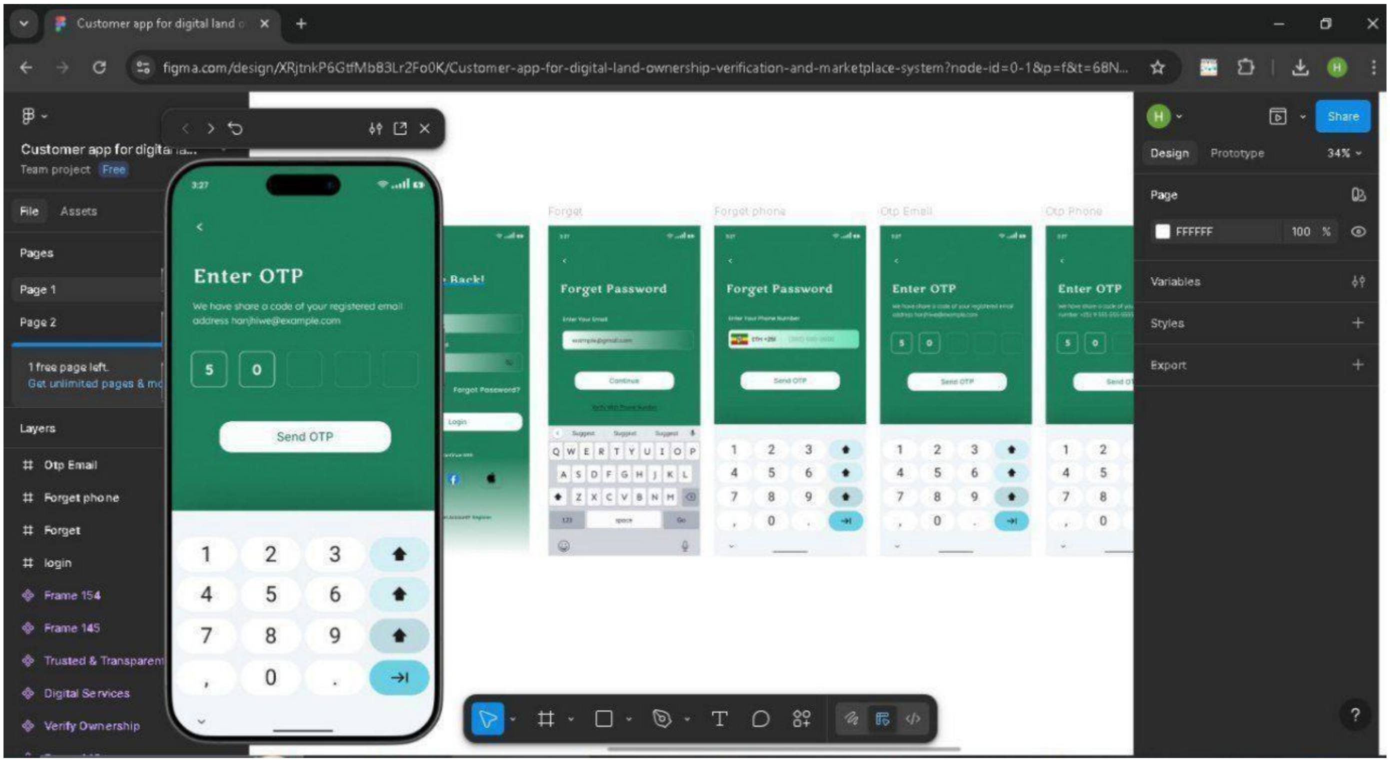



Figure 37 User Interface Prototype OTP Verfication

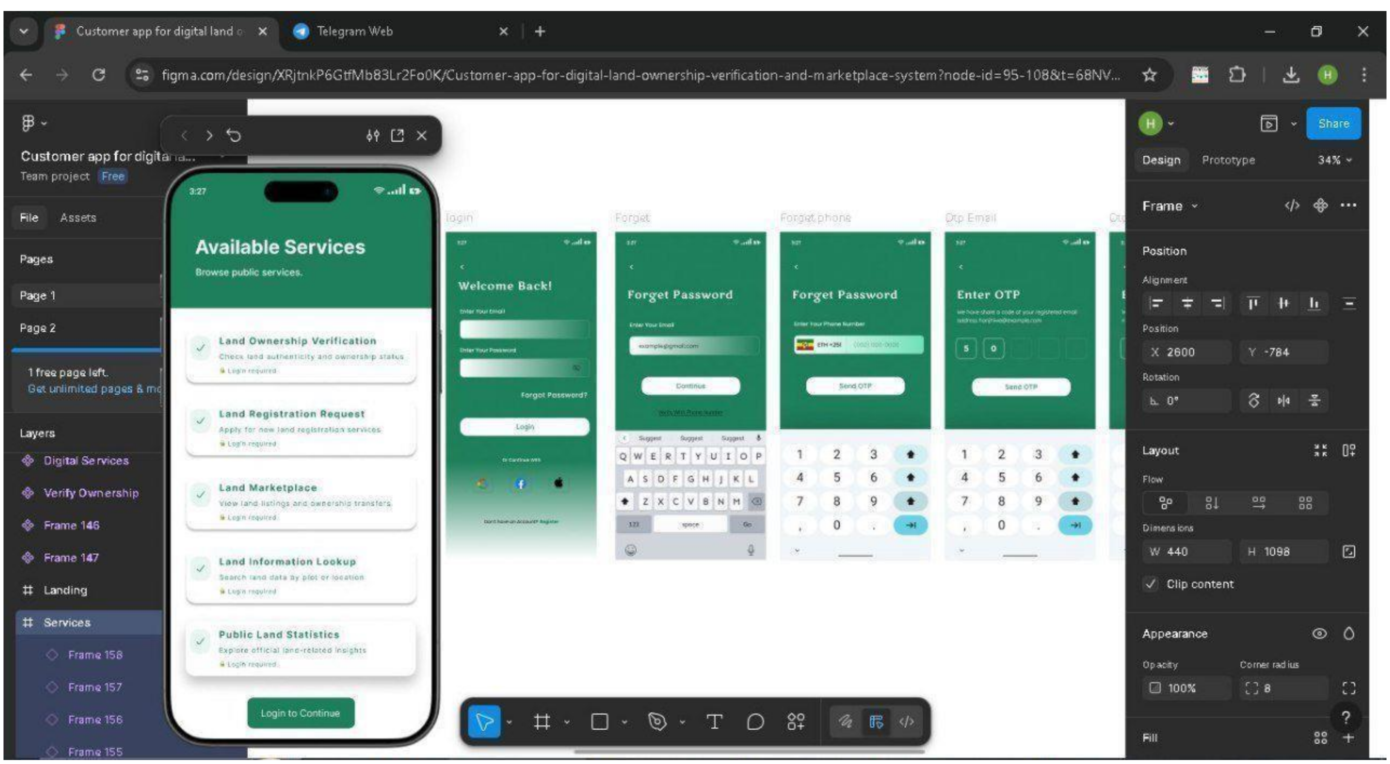



Figure 38 User Interface Prototype Homepage

# CHAPTER THREE: SYSTEM DESIGN

The system design phase marks the transition from the "what" of the system (requirements) to the "how" (technical blueprint). Following the analysis of the current manual land administration system in Bahir Dar which suffers from paper-based forgery, "double-selling" of plots, and long queues this chapter provides a comprehensive technical guide for building the **Digital Land Citizen Portal**.

The design follows Object-Oriented Design (OOD) principles, focusing on modularity, security, and scalability. This chapter establishes the architectural framework, the decomposition of system processes into specialized subsystems, and the mapping of software to hardware environments. It serves as a detailed roadmap for the development team to ensure the final product is both legally compliant and technically robust.

### *Design Goals*

Design goals describe the qualities that the developers should optimize. These goals are derived from the non-functional requirements and serve as a guide for making technical trade-offs.

#### *Functionality*

The system must provide a seamless experience for all stakeholders. For the **Citizen**, this means intuitive navigation for land searches and secure document uploads. For the **Land Officer**, it means a powerful administrative dashboard for verifying "Sened" (Title Deeds). The design ensures that all business logic such as verifying a plot’s availability before it is listed—is handled accurately by the core application engine.

#### *Performance*

Performance is critical for a web-based portal serving a large city like Bahir Dar.

- **Response Time:** The system is designed to return search results for land parcels in under 2 seconds.
- **Throughput:** The architecture supports hundreds of concurrent users, especially during peak hours when marketplace activity is high.
- **Optimization:** We utilize database indexing on geospatial data and plot numbers to ensure fast retrieval.

#### *Dependability*

Dependability is the system's ability to deliver services that can be trusted. Given that this portal manages legal land titles and financial transactions via TeleBirr/CBE Birr, the dependability requirements are significantly higher than those of a standard web application. This goal is broken down into four critical pillars: Reliability, Availability, Security, and Error Handling.

***Reliability***

Reliability is the probability that the system will perform its intended function without failure for a specified period. In the Digital Land Portal, reliability refers to the consistency of land records and the accuracy of ownership data.

- **Data Persistence:** To ensure reliability, we use **PostgreSQL with ACID (Atomicity, Consistency, Isolation, Durability) properties**. This ensures that even if the server crashes during a title transfer, the database will roll back to its previous state rather than leaving the land record in a corrupted or "half-transferred" state.
- **Validation Logic:** Reliability is further enhanced through strict backend validation. Every document uploaded for verification undergoes a checksum check to ensure that the file was not corrupted during the upload process.
- **Version Control for Records:** To track the "Reliability" of land history, the system maintains a versioned history of every parcel. Instead of overwriting data, the system creates a new entry for every transfer, allowing land officers to trace the chain of ownership back to the original manual "Sened."

***Availability***

Availability is the percentage of time the system is operational and accessible to the citizens of Bahir Dar.

- **Uptime Objective:** The design goal is **99.9% uptime**. This is achieved by deploying the application on a containerized environment (like Docker) which can be easily restarted if a service fails.
- **Redundancy:** To prevent a single point of failure, the system design includes redundant database backups. If the primary database fails, the system can switch to a "Read-Only" mode using a backup server so that citizens can still search for land plots while repairs are being made.
- **Mobile-First Accessibility:** Recognizing the infrastructure in Ethiopia, availability also means the system must be reachable on low-bandwidth networks. We utilize **Progressive Web App**

**(PWA)** technology so that users can view previously cached land details even if their internet connection is temporarily lost.

### *Security*

Security is the most vital pillar of dependability for land administration. We implement a "Defense in Depth" strategy.

- **Authentication and Authorization:** We use **JWT (JSON Web Tokens)** for session management and **Multi-Factor Authentication (MFA)**. For critical actions like "Approving a Land Sale," the Land Officer must enter a secondary OTP sent to their registered phone number.
- **Cryptographic Protection:** Sensitive data, particularly user passwords and Kebele ID numbers, are hashed using **Bcrypt**. All digital title certificates generated by the system are digitally signed with a unique cryptographic hash to prevent forgery.
- **Role-Based Access Control (RBAC):** Access is strictly segmented. A Citizen cannot access the "Verification Queue" of an Officer, and an Officer cannot modify the "Service Fee" settings managed by the Administrator.

### *Error Handling and Exception Management*

A dependable system must be "fault-tolerant." Error handling in the Digital Land Portal is designed to provide clear feedback to the user while maintaining system stability.

- **User-Centric Errors:** Instead of showing technical codes (like "Error 500"), the system provides actionable messages in Amharic and English. For example: *"The document you uploaded is too large. Please upload a file smaller than 5MB."*
- **Global Exception Handling:** On the backend (Node.js/Express), a global error-handling middleware is implemented. This ensures that even if an unexpected bug occurs, the server does not crash; it logs the error for the developer and sends a polite "Service Temporarily Unavailable" message to the user.
- **Transaction Logging:** Every error is recorded in a **System Audit Log**. This log includes the timestamp, the user ID, and the nature of the error. This is crucial for "post-mortem" analysis if a transaction fails during a payment process with TeleBirr.

### *Maintainability*

Maintainability is the ease with which a software system can be modified to correct faults, improve performance, or adapt to a changed environment. For the **Digital Land Citizen Portal**, maintainability is vital because land laws and administrative procedures in Ethiopia are subject to change.

### Comprehensive Documentation

To ensure the system can be managed by future developers at the Bahir Dar Institute of Technology or the Land Administration Bureau, we prioritize three levels of documentation:

- **Technical Specification Documentation:** This includes detailed descriptions of the system architecture, subsystem interfaces, and API endpoints (documented via tools like Swagger/OpenAPI).
- **System Administration Manual:** A guide for IT staff on how to manage the servers, perform database backups of land records, and monitor system health.
- **User Documentation:** Bilingual (Amharic and English) manuals for both Citizens and Land Officers to reduce the learning curve and minimize support requests.

### Code Readability and Standards

The system is developed using **TypeScript**, which provides static typing to catch errors during development.

- **Modular Programming:** We follow the **SOLID** principles of Object-Oriented Design. Each feature (like payment or verification) is isolated into its own module.
- **Naming Conventions:** Variable and function names follow a strict semantic naming convention (e.g., `verify LandDeed()` instead of `vld()`), ensuring that the code "tells a story" to any developer reading it.
- **Version Control:** Using **Git**, we maintain a clear history of all changes, allowing the team to "roll back" to a previous stable version if a new update introduces bugs.

### Error Monitoring and Logging

A maintainable system must be "self-aware."

- **Automated Logging:** Every transaction—especially those involving the **TeleBirr/CBE Birr** gateway—is logged with a unique transaction ID.

- **Real-time Alerts:** The system is designed to use monitoring tools that alert administrators if the server's CPU usage spikes or if the database connection is lost.
- **Audit Trails:** In land management, maintainability also includes the ability to audit data. The system maintains a "Log Table" that records every change made to a land parcel, including the timestamp and the ID of the officer who made the change.

***Scalability***

Scalability ensures the system can maintain its performance as the user base grows.

- **Horizontal Scalability:** The application tier is designed to be "stateless," meaning we can add more servers to handle increased traffic from different Weredas in the Amhara region without changing the code.
- **Database Scaling:** We utilize PostgreSQL indexing and partitioning strategies to ensure that even when the database contains hundreds of thousands of land plots, search queries remain instantaneous.

***End-User Criteria***

The system is designed for Ethiopian citizens with varying levels of digital literacy.

- **Usability:** The interface is clean and follows a "three-click rule" (users should find any information within three clicks).
- **Language Support:** The system supports both Amharic and English to ensure inclusivity.
- **Accessibility:** The marketplace is optimized for low-bandwidth mobile connections, ensuring it works even on slower 3G/4G networks.

## 3.1 Architectural Design

The Digital Land Citizen Portal adopts a **Three-Tier Architecture Model**. This separation is essential for security and system stability.

1. **Presentation Tier (Client Layer):** This is the top-most level. It manages the User Interface (UI). We use **Next.js** for the web portal and **React Native** for the mobile app. It captures user data and displays land information but contains no business logic.

2. **Application Tier (Logic Layer):** The middle tier acts as the "brain." Hosted on an Express/Node.js server, it processes requests, validates documents, calculates transaction fees, and enforces safety rules (e.g., preventing a user from buying their own land).
3. **Data Tier (Storage Layer):** The bottom layer consists of a **PostgreSQL database** for structured land data and a secure **File Storage** system for high-resolution scans of original physical land title

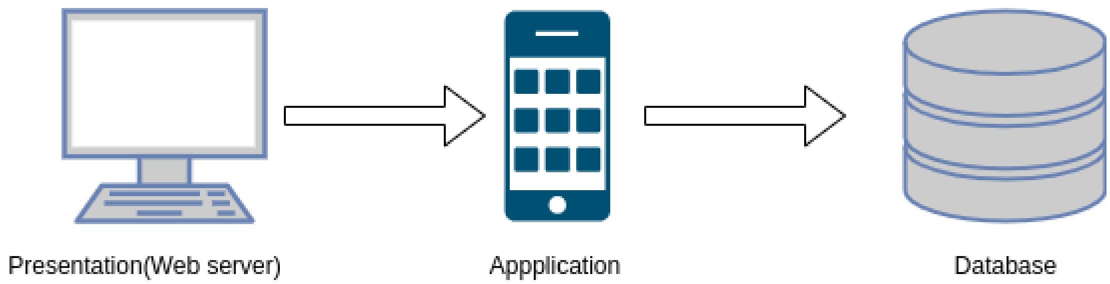



Figure 39 three-tier architecture model

## Subsystem composition

To manage complexity, the system is decomposed into specialized subsystems.

### User and Identity Subsystem

Responsible for registration, login, and Role-Based Access Control (RBAC). It integrates with KYC services to ensure every citizen is tied to a real Kebele ID.

### Ownership Verification Subsystem

This is the core engine for land officials. It manages the workflow of "Sened" verification. Citizens upload photos of their deeds, and land officers review them against the master registry. It tracks the lifecycle of an application from "Pending" to "Approved."

### Marketplace and Search Subsystem

This subsystem manages land listings. It includes a search engine with filters for location (Kebele/Wereda), land size, and grade. It ensures that only "Verified" parcels are visible to prevent fraud in the marketplace.

**Payment and Billing Subsystem**

This handles financial transactions. It provides a secure interface for **TeleBirr**, **CBE Birr**, and **EthSwitch**. It automatically generates receipts and updates the transaction status once payment is confirmed by the bank API.

**Notification Subsystem**

A background service that sends automated SMS and Email alerts to users. For example, a citizen is notified the moment their land title is verified or when a buyer makes an offer on their listed plot.

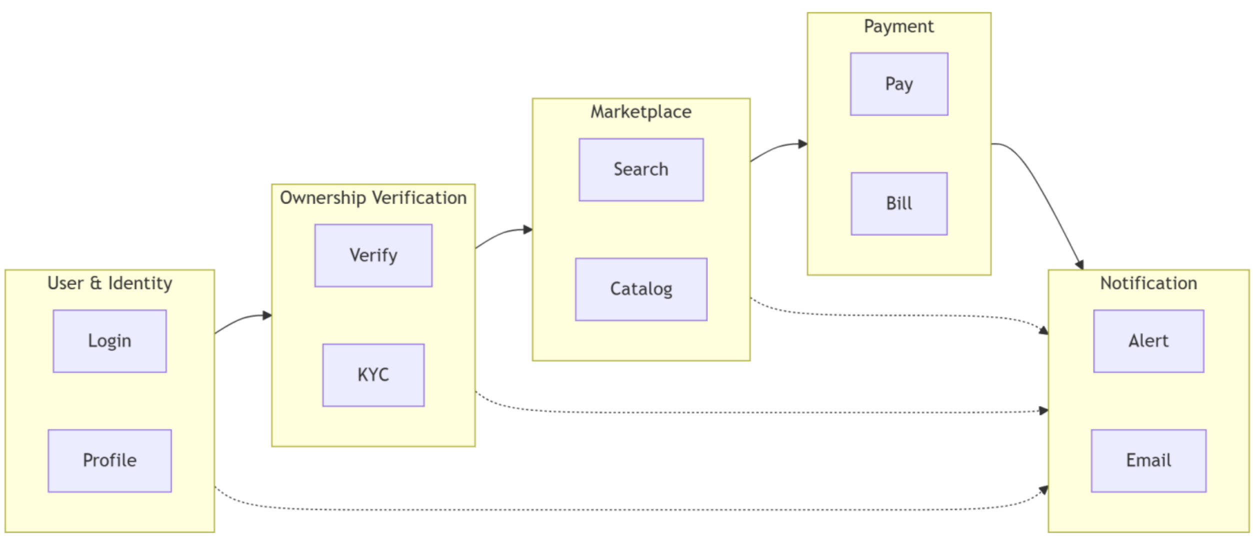



Figure 40 Subsystem Model

### *3.1.1 Component Modeling*

The component model illustrates the physical organization of the software units. For the **Digital Land Citizen Portal**, these components are designed to be modular, ensuring that the "Ownership Verification" logic is independent of the "Marketplace" services.

*User Interface (UI) Components*

- **Authentication UI:** Manages the specialized login and registration views, including the **OTP (One-Time Password) Verification** interface required for secure access.
- **Land Registry Dashboard:** A complex UI component that allows Citizens to view their verified land parcels and track the status of pending applications in real-time.
- **Marketplace Search Component:** Provides the "Geospatial Search" filters (Kebele, size, grade) and property recommendation interfaces.
- **Officer Verification Portal:** A dedicated UI for land officers to review "Sened" documents, validate supporting materials, and digitally sign approvals.

*Backend Logic Components*

- **Blockchain Ledger Wrapper:** A critical component that ensures the immutability of land titles and transaction histories to prevent fraud.
- **Payment Gateway Wrapper:** Handles communication with **TeleBirr, CBE Birr, and EthSwitch**, standardizing the transaction flow for land service fees and marketplace payments.
- **Geospatial Processor:** Manages the retrieval of parcel boundaries and location coordinates from the **PostgreSQL** database.
- **Notification Service:** An automated component triggered by status changes (e.g., from "Pending" to "Approved") to send SMS/Email updates to users.

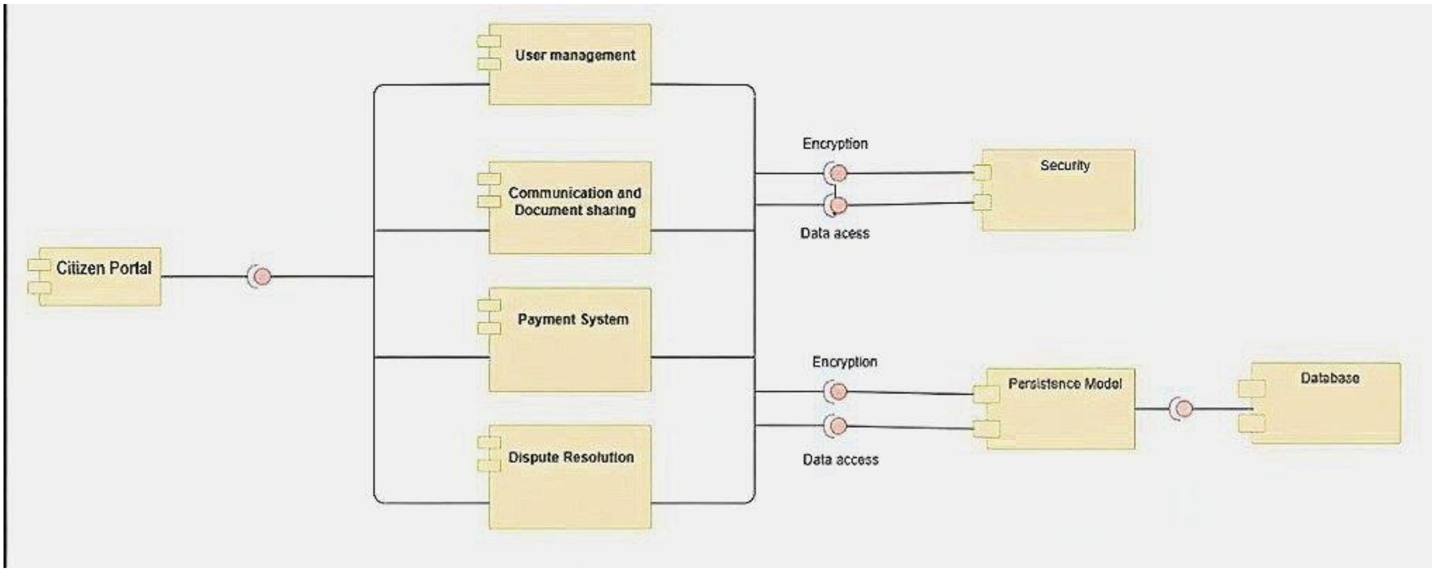



Figure 41Component Diagram

### *3.1.2 Deployment Modeling*

The deployment model maps these software components to the physical hardware. This system utilizes a distributed approach to ensure high availability despite local internet constraints.

***A. Client Layer (Client Devices)***

- **Client Devices:** The portal is deployed to **Smartphones (Android/iOS)** via React Native and **Personal Computers** via web browsers (Firefox, Chrome).
- **Hosting Environment:** UI components are executed locally on user devices, communicating with the server through secured **HTTPS/TLS** channels.

*B. Application Layer (Web and Application Server)*

- **Web/Application Server:** Hosted on a Linux-based environment (Ubuntu) running the **Node.js** runtime.
- **Hosting Environment:** This tier manages the **Prisma ORM** for efficient database operations and enforces the "Two-Officer Verification" rules for high-value transactions.

*C. Data Layer (Database Server & External Integrations)*

- **Database Server:** A secure **PostgreSQL** instance isolated from public access. It stores all structured land records and geospatial data.
- **Blockchain Node:** An immutable ledger specifically for ownership verification to ensure records are tamper-proof.
- **External Integrations:** These include the API nodes of the **National Bank of Ethiopia (TeleBirr/CBE Birr)** and the government's **National ID (Fayda)** system for user identity verification.

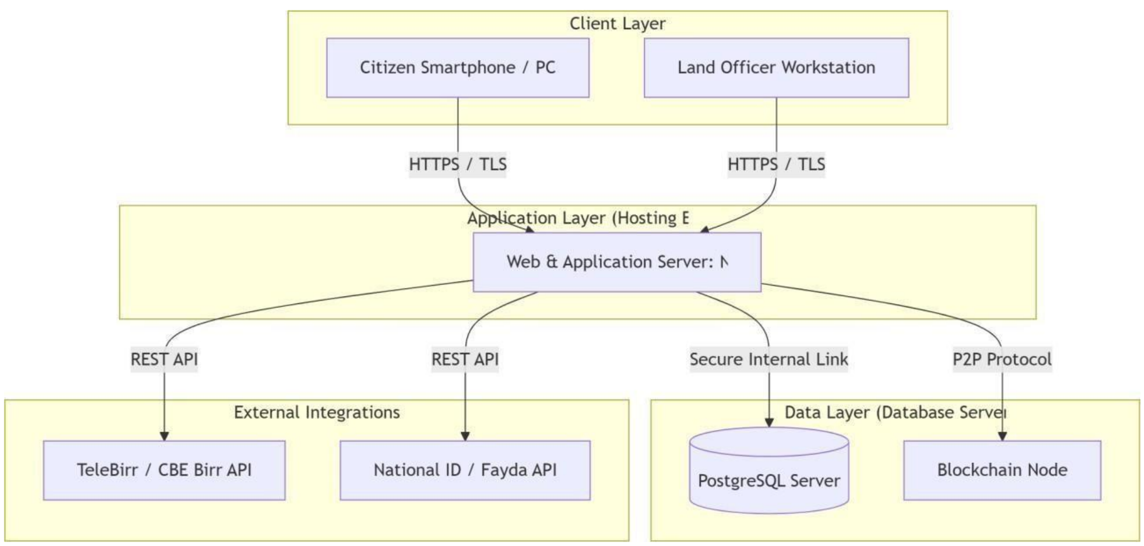



Figure 42 Deployment Diagram

## Communication Flow

The system utilizes a structured communication flow to maintain data integrity:

1. **Request Phase:** The user triggers an action (e.g., "Request Ownership Transfer") from the **Client Layer**.
2. **Processing Phase:** The **Application Layer** validates the business rules (e.g., checking for full payment or existing legal encumbrances).
3. **Data Phase:** The **Data Layer** updates the registry, and the **Blockchain component** locks the record.
4. **Feedback Phase:** The **Notification Subsystem** confirms the action via SMS/Email, and the updated status is pushed back to the **Client Layer**.

**Justification for the Architecture**

The **Three-Tier Architecture** is selected for the following reasons based on your requirements analysis:

- **Security & Encryption:** Critical for protecting sensitive land records and Kebele ID data through encapsulation and TLS/SSL.
- **Scalability for National Deployment:** The modularity allows for the system to be expanded from Bahir Dar to all 11 regions of Ethiopia without a complete redesign.
- **Reliability Under Constraints:** By separating the layers, the system can provide "Read-Only" access to cached land data even when high-bandwidth features (like map loading) are slow due to internet issues.
- **Transparency:** The architecture supports a centralized "Single Source of Truth," which is the primary objective of moving away from fragmented, paper-based workflows.

## 3.2 Detail Design

### 3.2.1 Design class model

The Design Class Model provides a structured, object-oriented blueprint for implementing the Digital Land Registry and Ownership Verification System. It ensures secure citizen access, transparent land ownership management, automated service requests, and verified digital payments. By clearly defining entities, relationships, and responsibilities, the model supports scalability, security, maintainability, and integration with existing land administration and banking systems.

A **Design Class Model** (Design Class Diagram in UML) shows:

- **Classes** used in the system

- **Attributes** (data stored in each class)
- **Methods/operations** (what each class can do)
- **Relationships** between classes (how they interact)

It is created **after requirements analysis** and is used to:

- Guide **implementation (coding)**
- Define **responsibilities of classes**
- Show **system structure at code level**

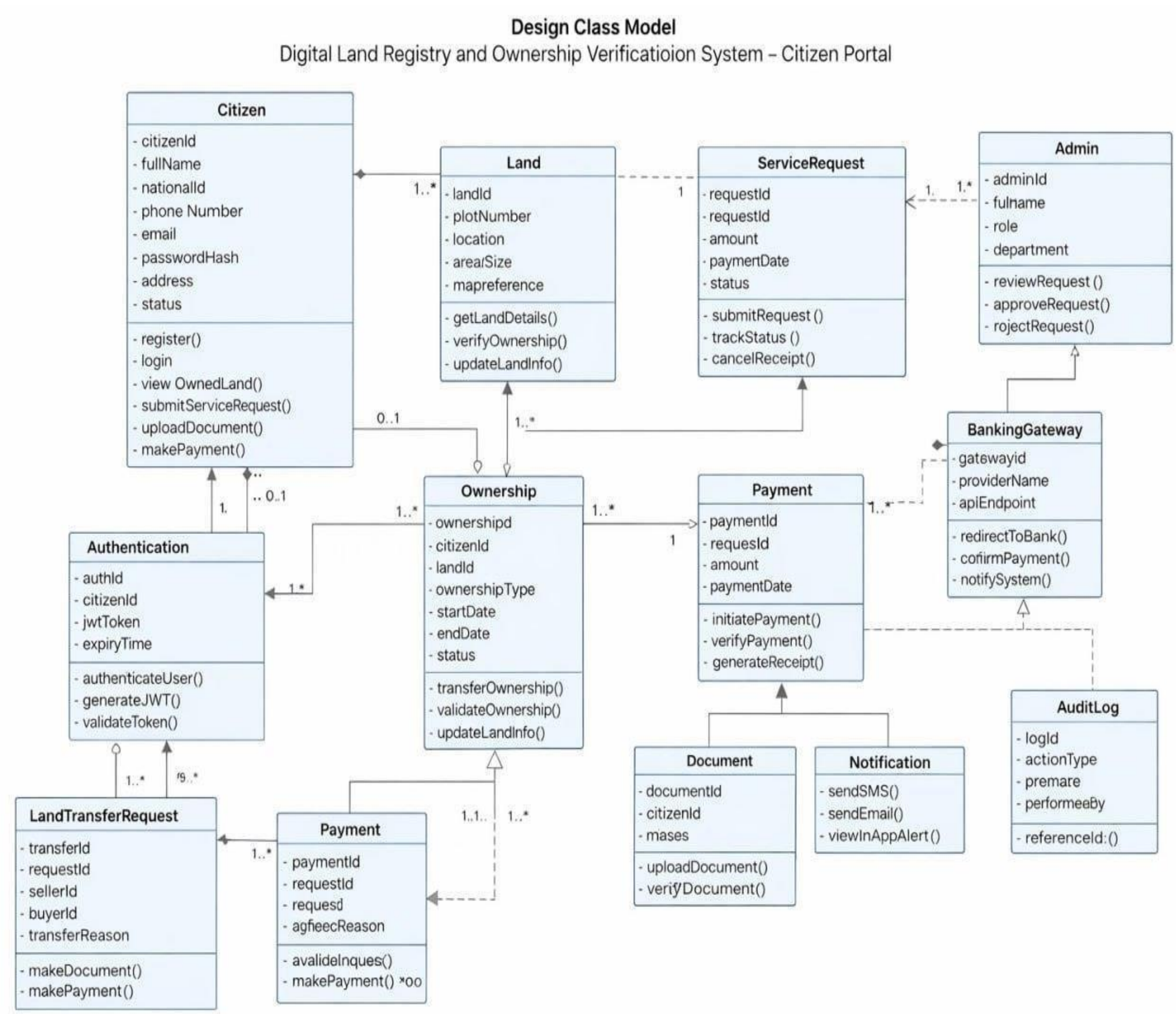


Figure 43 Class Diagram

1. Citizen Entity

| Attribute Name | Data Type | Length | Constraints |
|---|---|---|---|
| citizen_id | INT | - | Primary Key, Auto Increment |
| full_name | VARCHAR | 100 | NOT NULL |
| national_id | VARCHAR | 20 | UNIQUE, NOT NULL |
| phone_number | VARCHAR | 15 | UNIQUE, NOT NULL |
| Email | VARCHAR | 100 | UNIQUE |
| password_hash | VARCHAR | 255 | NOT NULL |
| Address | VARCHAR | 200 | NOT NULL |
| Status | VARCHAR | 20 | DEFAULT 'Active' |
| created_at | TIMESTAMP | - | DEFAULT CURRENT_TIMESTAMP |

Table 28 Citizen Entitiy

2. Authentication Entity

| Attribute Name | Data Type | Length | Constraints |
|---|---|---|---|
| auth_id | INT | - | Primary Key |
| citizen_id | INT | - | Foreign Key → Citizen(citizen_id) |
| jwt_token | VARCHAR | 500 | NOT NULL |
| expiry_time | TIMESTAMP | - | NOT NULL |
| login_time | TIMESTAMP | - | DEFAULT CURRENT_TIMESTAMP |

Table 29 Authenticiy Entity

3. Land Entity

| Attribute Name | Data Type | Length | Constraints |
|---|---|---|---|
| land_id | INT | - | Primary Key |
| plot_number | VARCHAR | 50 | UNIQUE, NOT NULL |
| Location | VARCHAR | 150 | NOT NULL |
| area_size | DECIMAL | 10.2 | NOT NULL |
| land_use_type | VARCHAR | 50 | NOT NULL |
| map_reference | VARCHAR | 100 | NOT NULL |
| registration_date | DATE | - | NOT NULL |

| | | | |
|---|---|---|---|
| Status | VARCHAR | 20 | DEFAULT 'Registered' |

Table 30 Land Entitiy

5. Ownership Entity

| Attribute Name | Data Type | Length | Constraints |
|---|---|---|---|
| ownership_id | INT | - | Primary Key |
| citizen_id | INT | - | Foreign Key → Citizen |
| land_id | INT | - | Foreign Key → Land |
| ownership_type | VARCHAR | 30 | NOT NULL |
| start_date | DATE | - | NOT NULL |
| end_date | DATE | - | NULL |
| Status | VARCHAR | 20 | DEFAULT 'Active' |

Table 31 Ownership Entity

5. ServiceRequest Entity

| Attribute Name | Data Type | Length | Constraints |
|---|---|---|---|
| request_id | INT | - | Primary Key |
| citizen_id | INT | - | Foreign Key → Citizen |
| request_type | VARCHAR | 50 | NOT NULL |
| submission_date | TIMESTAMP | - | DEFAULT CURRENT_TIMESTAMP |
| Status | VARCHAR | 30 | DEFAULT 'Pending' |
| Remarks | VARCHAR | 255 | NULL |

Table 32 Service request entity

6. Land Transfer Request Entity (Subclass)

| Attribute Name | Data Type | Length | Constraints |
|---|---|---|---|
| transfer_id | INT | - | Primary Key |
| request_id | INT | - | Foreign Key → ServiceRequest |
| seller_id | INT | - | Foreign Key → Citizen |
| buyer_id | INT | - | Foreign Key → Citizen |
| transfer_reason | VARCHAR | 100 | NOT NULL |
| agreement_document | VARCHAR | 255 | NOT NULL |

Table 33 Land Transfer Request Entity

7. Payment Entity

| Attribute Name | Data Type | Length | Constraints |
| --- | --- | --- | --- |
| payment_id | INT | - | Primary Key |
| request_id | INT | - | Foreign Key → ServiceRequest |
| Amount | DECIMAL | 10.2 | NOT NULL |
| payment_date | TIMESTAMP | - | DEFAULT CURRENT_TIME STAMP |
| payment_status | VARCHAR | 20 | NOT NULL |
| transaction_reference | VARCHAR | 100 | UNIQUE, NOT NULL |

**Table 34 Payment Gateway**

8. BankingGateway Entity

| Attribute Name | Data Type | Length | Constraints |
| --- | --- | --- | --- |
| gateway_id | INT | - | Primary Key |
| provider_name | VARCHAR | 50 | NOT NULL |
| api_endpoint | VARCHAR | 255 | NOT NULL |
| status | VARCHAR | 20 | DEFAULT 'Active' |

9. Document Entity

| Attribute Name | Data Type | Length | Constraints |
| --- | --- | --- | --- |
| document_id | INT | - | Primary Key |
| citizen_id | INT | - | Foreign Key → Citizen |
| request_id | INT | - | Foreign Key → ServiceRequest |
| document_type | VARCHAR | 50 | NOT NULL |
| file_path | VARCHAR | 255 | NOT NULL |
| upload_date | TIMESTAMP | - | DEFAULT CURRENT_TIMES TAMP |
| verification_status | VARCHAR | 20 | DEFAULT 'Pending' |

**Table 35 Document Entity**

10. Notification Entity

| Attribute Name | Data Type | Length | Constraints |
| --- | --- | --- | --- |
| notification_id | INT | - | Primary Key |
| citizen_id | INT | - | Foreign Key → Citizen |
| Message | VARCHAR | 255 | NOT NULL |
| notification_type | VARCHAR | 20 | NOT NULL |

| | | | |
|---|---|---|---|
| sent_date | TIMESTAMP | - | DEFAULT CURRENT_TIMESTAMP |

Table 36 Notification Entity

11. Admin Entity

| Attribute Name | Data Type | Length | Constraints |
|---|---|---|---|
| admin_id | INT | - | Primary Key |
| full_name | VARCHAR | 100 | NOT NULL |
| Role | VARCHAR | 50 | NOT NULL |
| Department | VARCHAR | 50 | NOT NULL |
| email | VARCHAR | 100 | UNIQUE |

Table 37 Admin Entity

12. AuditLog Entity

| Attribute Name | Data Type | Length | Constraints |
|---|---|---|---|
| log_id | INT | - | Primary Key |
| action_type | VARCHAR | 100 | NOT NULL |
| action_date | TIMESTAMP | - | DEFAULT CURRENT_TIMESTAMP |
| performed_by | VARCHAR | 50 | NOT NULL |
| reference_id | INT | - | NOT NULL |

Table 38 Audit Log

### 3.2.2 Entity–Relationship (E-R) Model

The E-R model accurately represents the data structure of the Digital Land Registry and Ownership Verification System. It ensures proper handling of land ownership, service requests, digital payments, document management, and administrative control while maintaining transparency, security, and data integrity.

The E-R model represents:

- Entities (real-world objects)
- Attributes (properties of entities)
- Relationships (how entities are connected)
- Cardinality & participation

This model focuses on data structure, unlike the design class model which focuses on behavior.

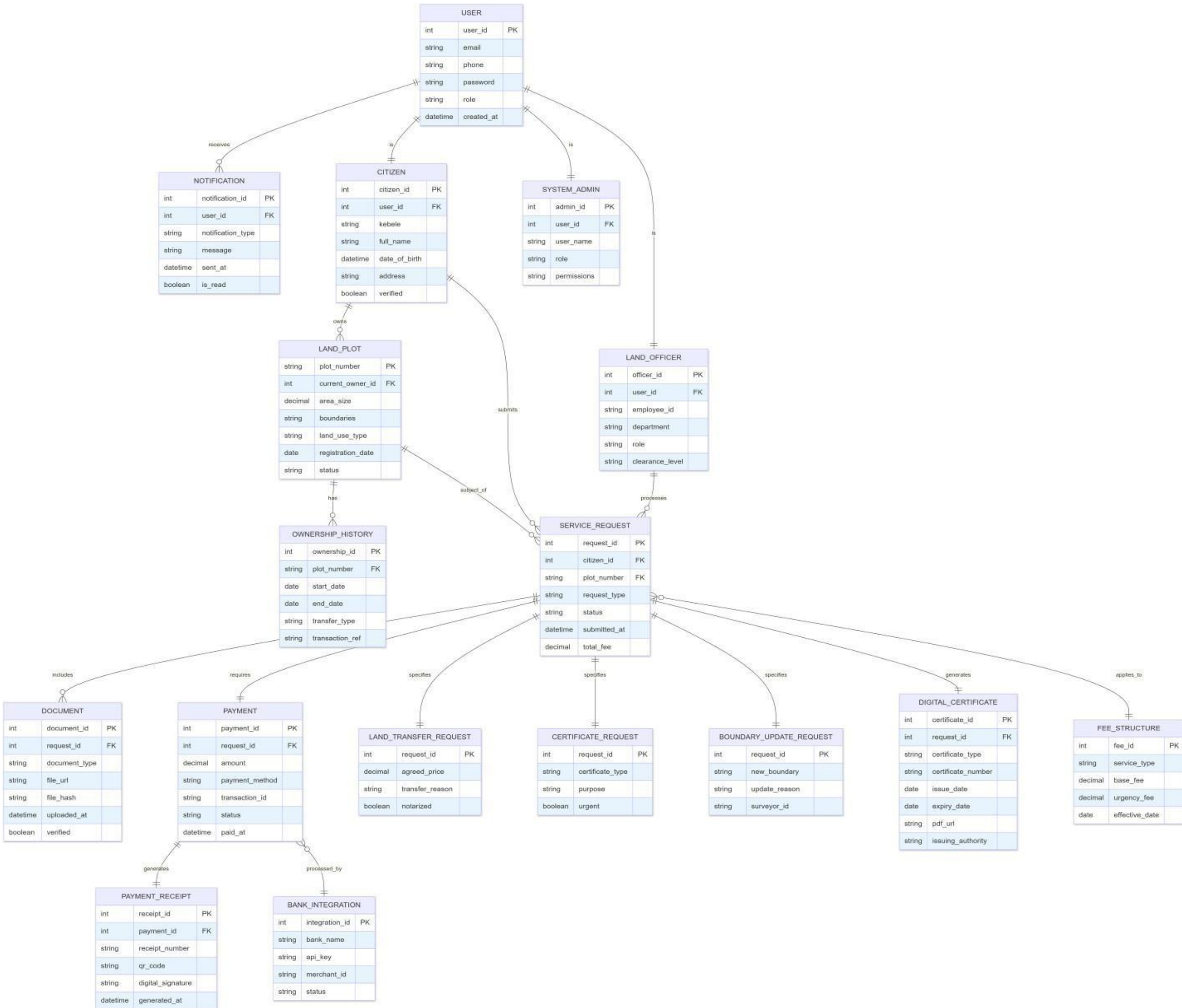
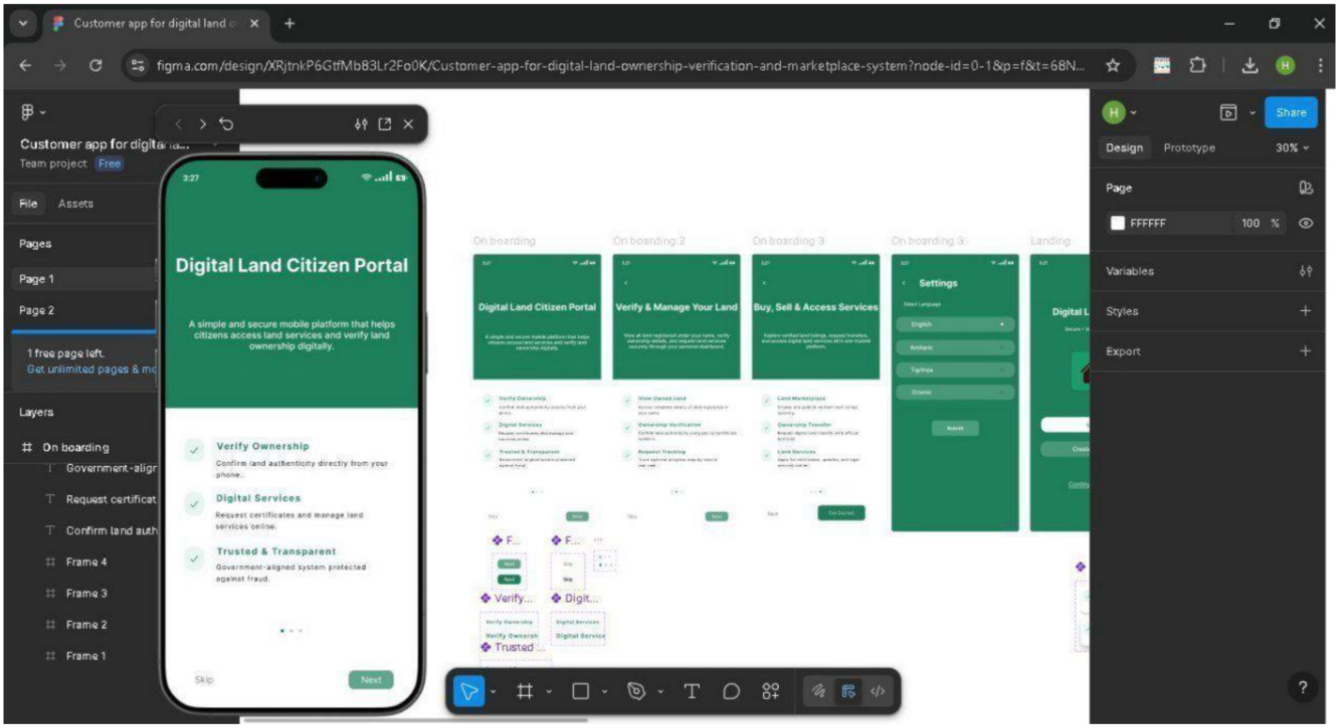
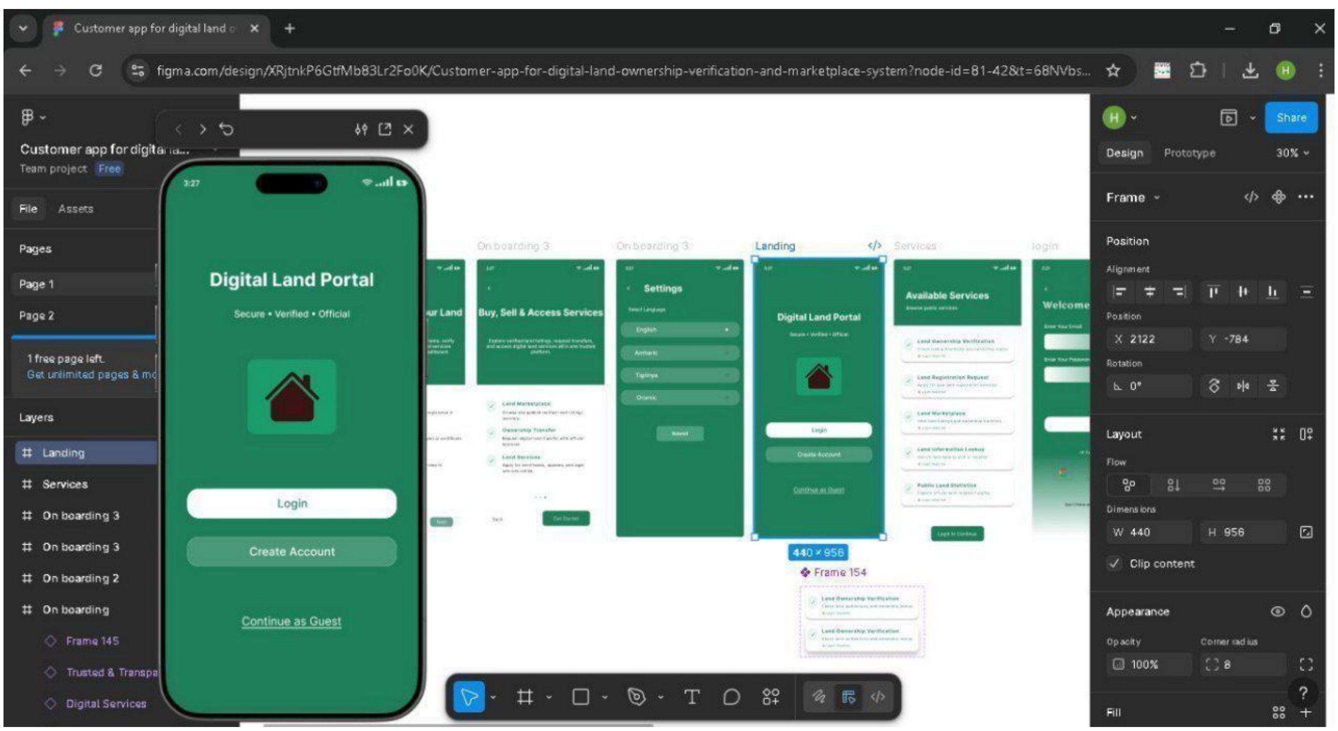



**Figure 44 E-R Diagram**

## 3.3 User Interface Design




## 3.4 Access Control and Security

The system implements **Role-Based Access Control (RBAC)** to ensure that only authorized users can access specific functionalities. Each user is assigned a role, and each role is granted permissions to perform specific tasks within the system.

The access control and security mechanism ensures that citizens can only access their own land records and service requests, while administrators manage verification and approval processes. Automated system services handle payments, notifications, and logging to ensure transparency, data integrity, and secure land administration.

| **Task \ Role** | Admin | Citizen | System |
|---|---|---|---|
| Register Account | | ⍰ | |
| Login | ⍰ | ⍰ | |
| View Land Records | ⍰ | ⍰ | |
| Verify Ownership | ⍰ | ⍰ | |
| Submit Service Request | | ⍰ | |
| Upload Documents | | ⍰ | |
| Make Payment | | ⍰ | |
| Track Request Status | | ⍰ | |
| Approve Request | ⍰ | | |
| Reject Request | ⍰ | | |
| Update Land Records | ⍰ | | |
| Generate Certificate | ⍰ | | |
| Send Notification | | | ⍰ |
| View Audit Log | ⍰ | | ⍰ |

## References

Based on your project documentation for the **Digital Land Citizen Portal**, here is a list of academic and professional references formatted in **APA style**. I have selected these to cover the specific technical stack you used (Next.js, Blockchain, PostgreSQL) and the geographical/legal context of your project (Ethiopia).


**1. Land Administration & Governance (Contextual)**

- Deininger, K., & Feder, G. (2009). Land Registration, Governance, and Development: Evidence and Implications for Policy. *The World Bank Research Observer*, 24(2), 233-266.
- Federal Democratic Republic of Ethiopia. (2011). *Urban Lands Lease Holding Proclamation No. 721/2011*. Negarit Gazeta.
- Tura, H. A. (2018). Land rights and land grabbing in Oromia, Ethiopia. *Land Use Policy*, 70, 247-255.
- World Bank. (2022). *Digital Governance Projects in Sub-Saharan Africa: Best Practices and Challenges*. World Bank Publications.

**2. Technology Stack & Frameworks**

- **Next.js & React:**
    - Cholakov, S. (2021). *Next.js Projects: Build scalable, high-performance, and modern web applications with Next.js and React*. Packt Publishing.
- **PostgreSQL & Prisma:**
    - Bantz, A. (2022). *Prisma Express: A guide to the next-generation ORM for Node.js and TypeScript*. O'Reilly Media.
- **Blockchain for Land Registry:**
    - Vos, J., Lemmen, C., & Beentjes, B. (2017). Blockchain-based Land Administration: Feasible, Illusory or a Panacea? *Proceedings of the 2017 World Bank Conference on Land and Poverty*.
    - Thakur, V., Doja, M. N., Dwivedi, Y. K., Ahmad, T., & Ganesh, G. (2020). Land records management system: A deployment of blockchain and smart contracts for retrievable and tamper-proof records. *Information Systems Frontiers*, 22, 1131-1140.

**3. Systems Analysis & Design (Methodology)**

- Larman, C. (2015). *Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development*. Pearson Education.
- Nielsen, J. (1994). *Usability Engineering*. Morgan Kaufmann (for your UI/UX and Prototyping sections).

**4. Ethiopian Digital Infrastructure**

- Ministry of Innovation and Technology (MinT). (2020). *Digital Ethiopia 2025: A Digital Strategy for Ethiopia's Inclusive Prosperity*.
- National Bank of Ethiopia. (2020). *Licensing and Authorization of Payment Instrument Issuers (Directive No. ONPS/01/2020)* (For your TeleBirr/CBE Birr integration).

# Appendices

### Appendix A: Sample Working Forms

These are the physical papers currently used by the Bahir Dar land administration. Including these proves you understood the "as-is" system before building the digital portal.

- **A.1: Land Title Deed (Sened):** A scanned copy of a manual ownership certificate.
- **A.2: Application for Title Transfer:** The paper form citizens currently fill out to sell land.
- **A.3: Payment Receipts:** Samples of manual revenue receipts from the district (Wereda) finance office.

### Appendix B: Requirement Gathering Tools

This section validates your methodology by showing the exact questions you used to gather data in Bahir Dar.

### B.1: Interview Guide (For Officials)

**Target:** Land Administration Office Managers.

1. What are the most common reasons for delays in land verification requests?
2. How do you currently verify if a land title is not forged?
3. What are the biggest security risks in the current paper-based record-keeping?

### B.2: Citizen Questionnaire

**Target:** Residents and Land Buyers.

1. On a scale of 1–5, how transparent do you find the current land buying process?
2. Have you ever experienced "double-selling" or fraud during a transaction?
3. Would you feel comfortable paying land service fees via TeleBirr or CBE Birr?